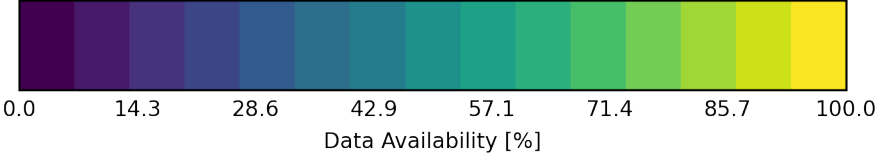


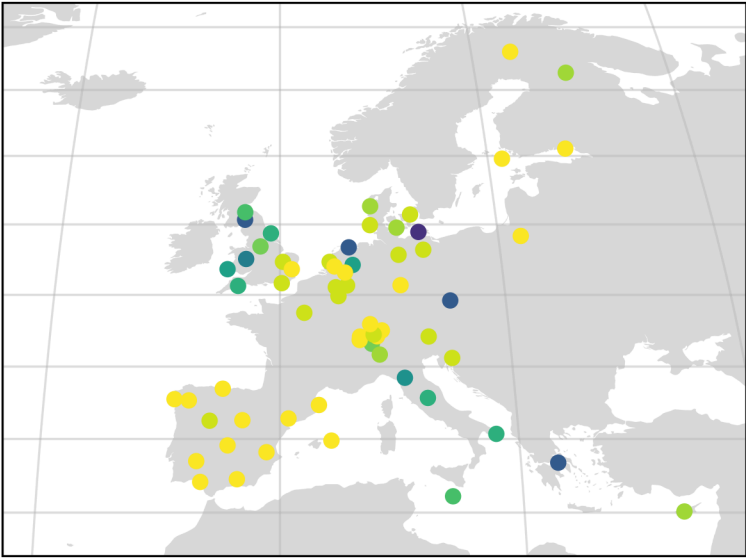
EVALUATION REPORT

Network: EBAS
Species: O3, NO2
Temporal resolution: Hourly
Date range: 01/01/2018 - 01/04/2018
Models: MONARCH
Subsections: LOCAL·All, LOCAL·France, LOCAL·Germany, LOCAL
·Spain, LOCAL·UK

Map Data Availability (Summary)
EBAS|sconco3



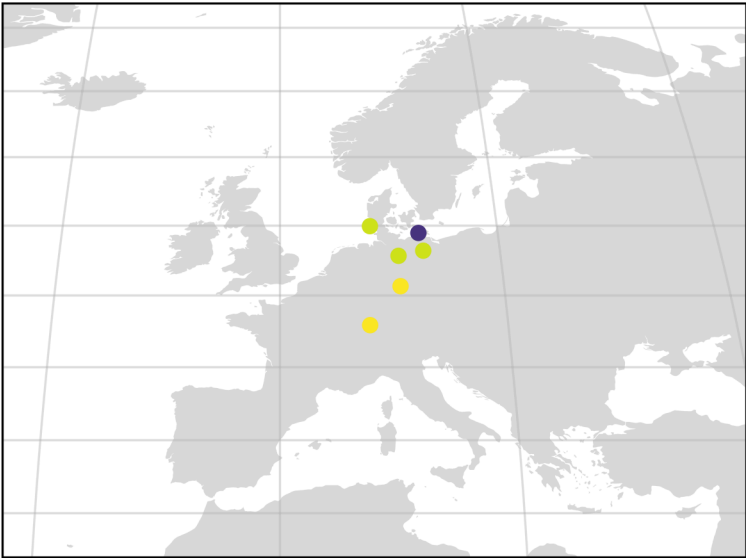
observations
LOCAL·All (62 stations)



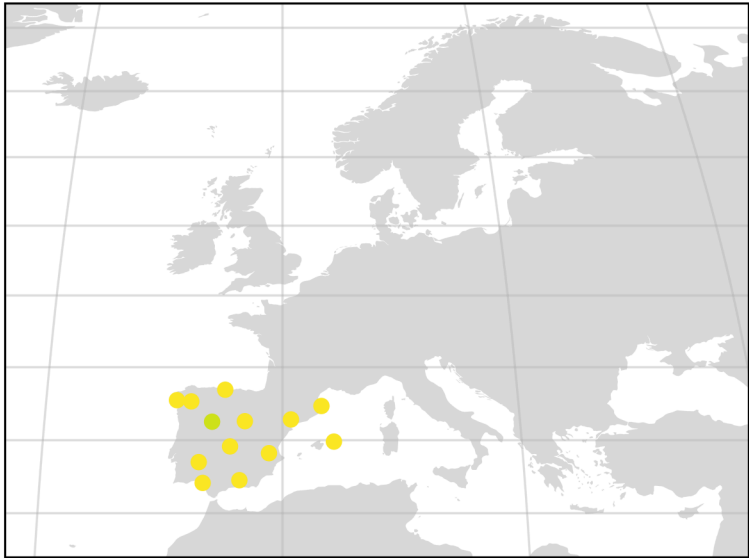
observations
LOCAL·France (1 station)



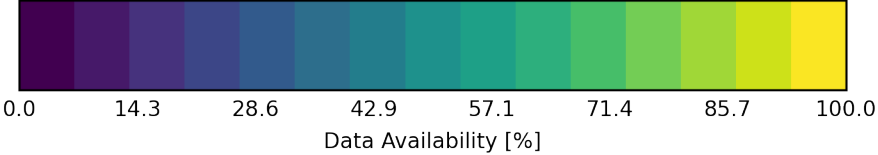
observations
LOCAL·Germany (6 stations)



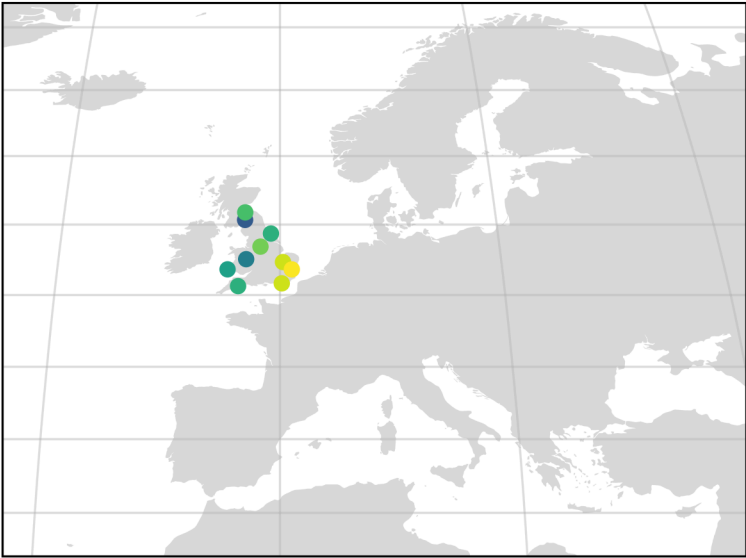
observations
LOCAL·Spain (13 stations)



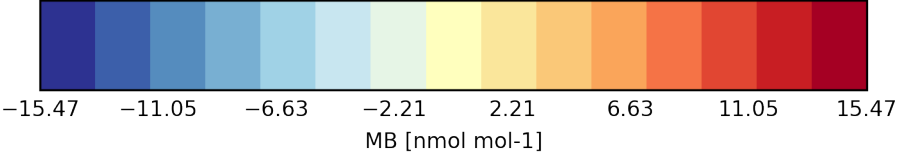
Map Data Availability (Summary)
EBAS|sconco3



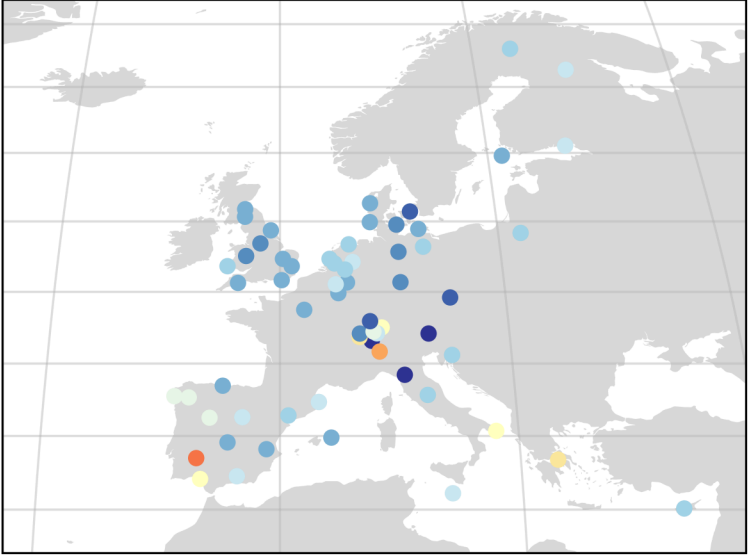
observations
LOCAL·UK (10 stations)



Map MB (Summary)
EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)

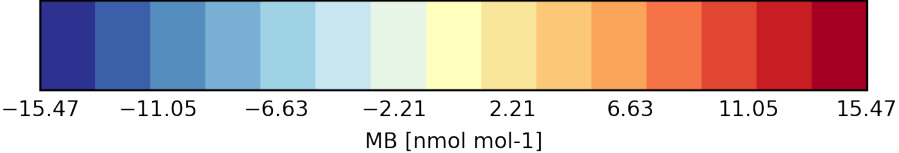


MONARCH
LOCAL·Germany (6 stations)

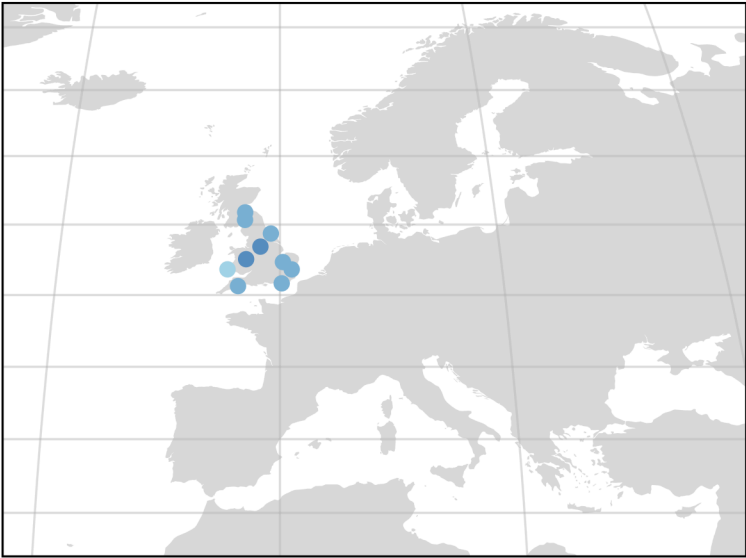


MONARCH
LOCAL·Spain (13 stations)

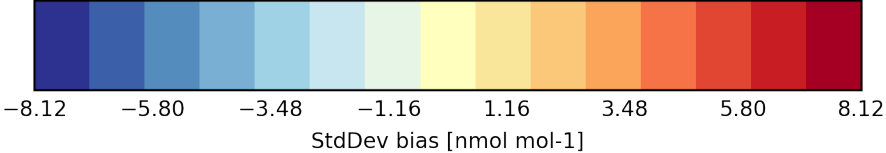




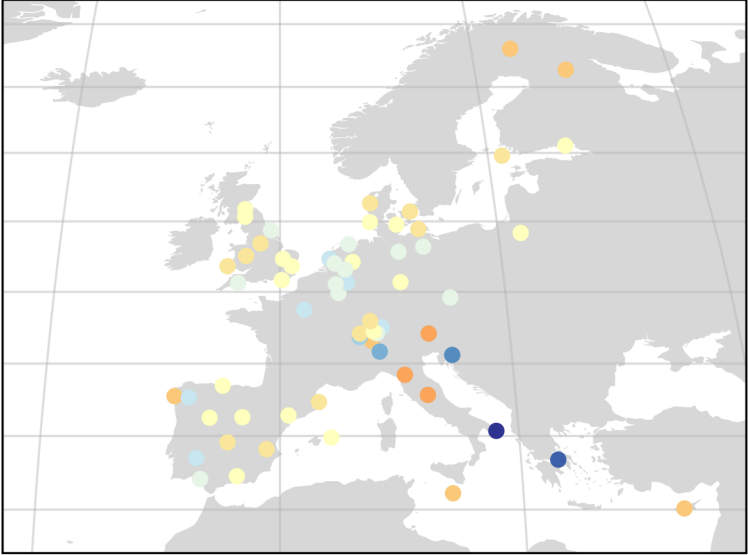
MONARCH
LOCAL·UK (10 stations)



Map StdDev bias (Summary)
EBAS|sconco3



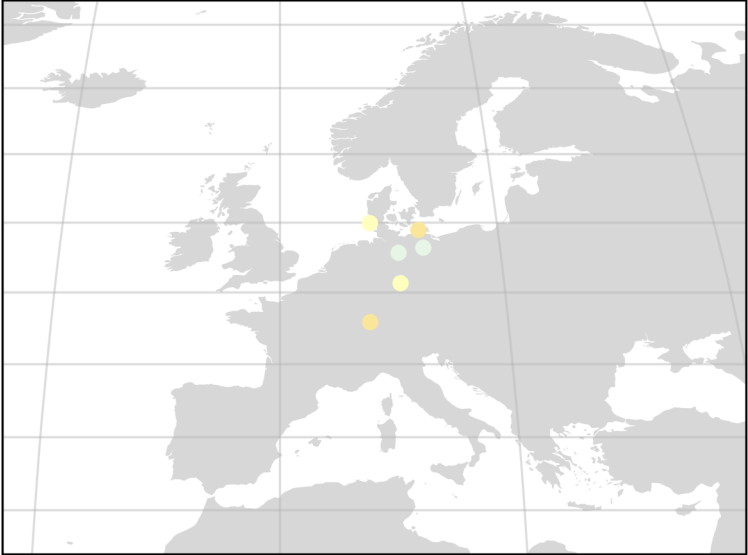
MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



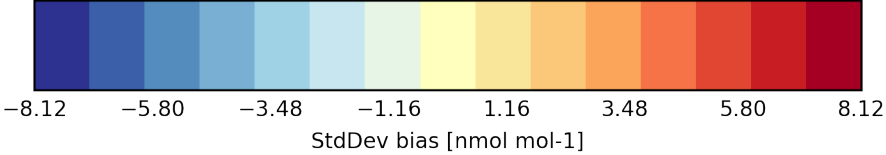
MONARCH
LOCAL·Germany (6 stations)



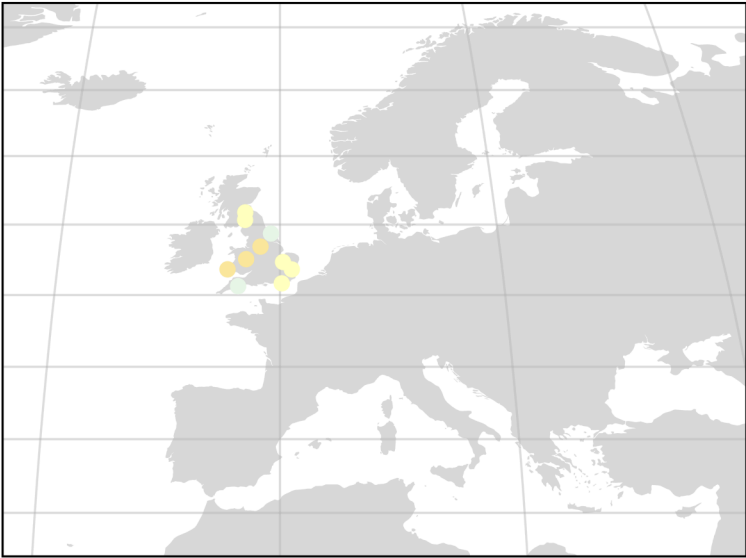
MONARCH
LOCAL·Spain (13 stations)



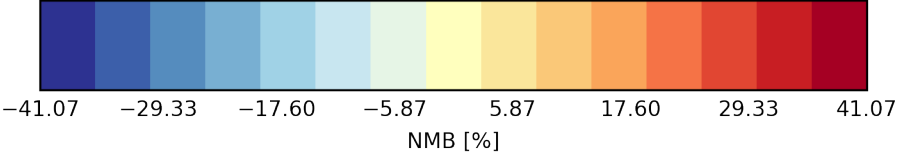
Map StdDev bias (Summary)
EBAS|sconco3



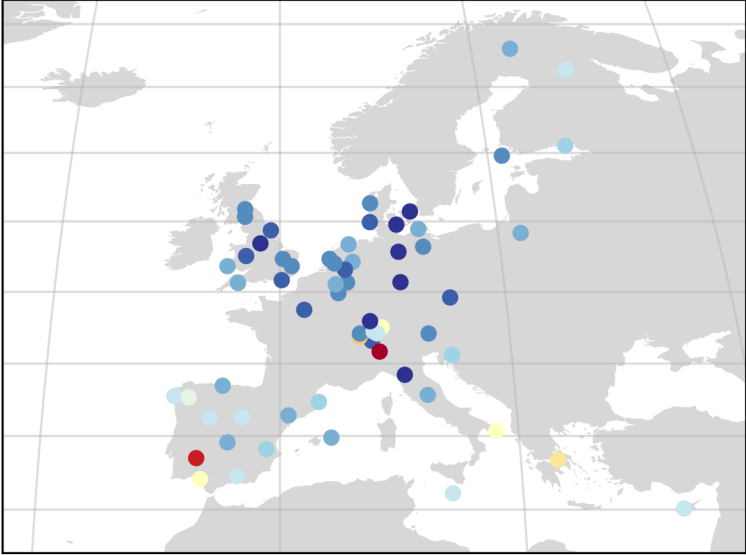
MONARCH
LOCAL·UK (10 stations)



Map NMB (Summary)
EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



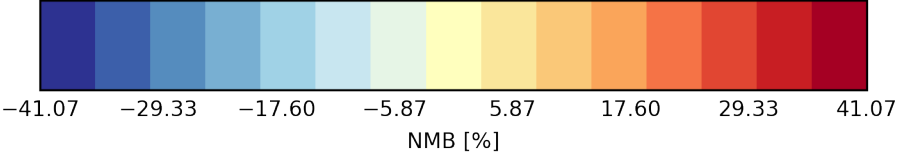
MONARCH
LOCAL·Germany (6 stations)



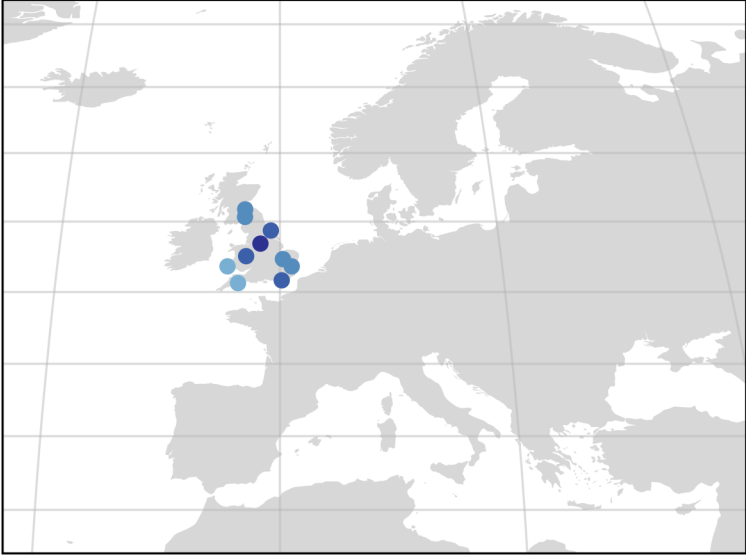
MONARCH
LOCAL·Spain (13 stations)



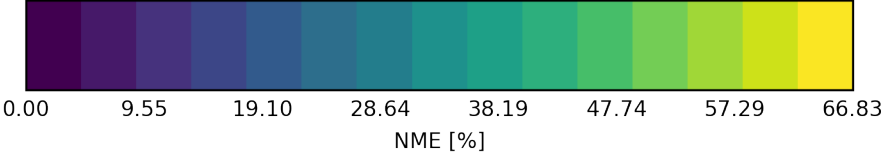
Map NMB (Summary)
EBAS|sconco3



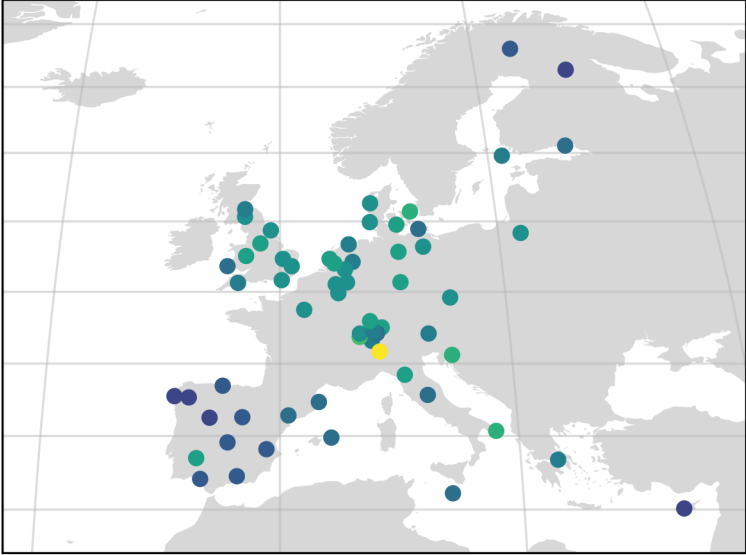
MONARCH
LOCAL·UK (10 stations)



Map NME (Summary)
EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



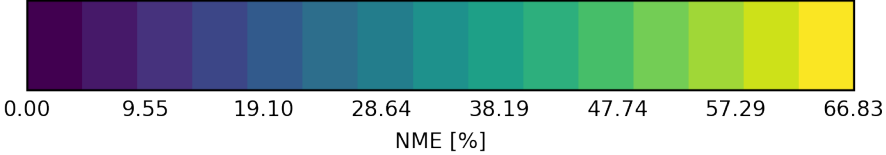
MONARCH
LOCAL·Germany (6 stations)



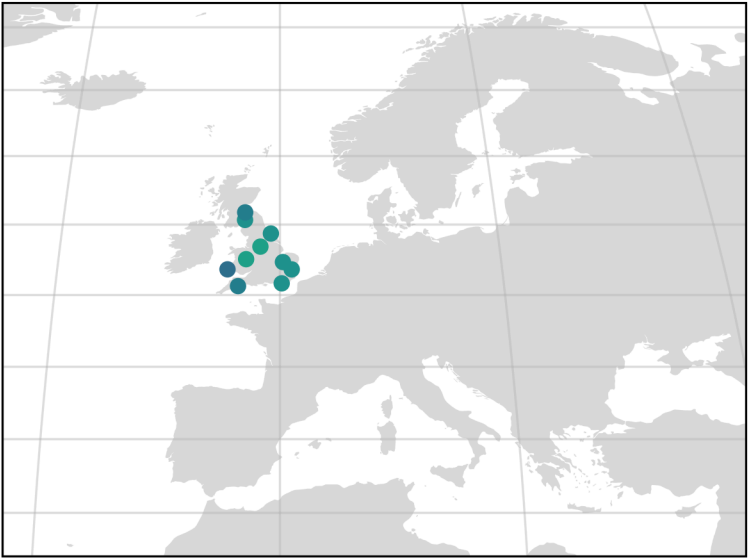
MONARCH
LOCAL·Spain (13 stations)



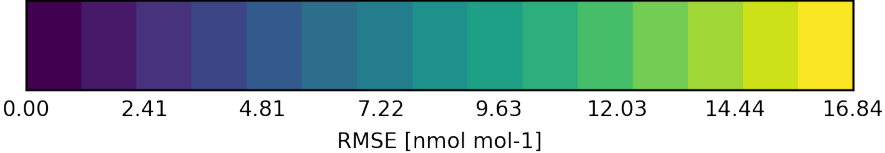
Map NME (Summary)
EBAS|sconco3



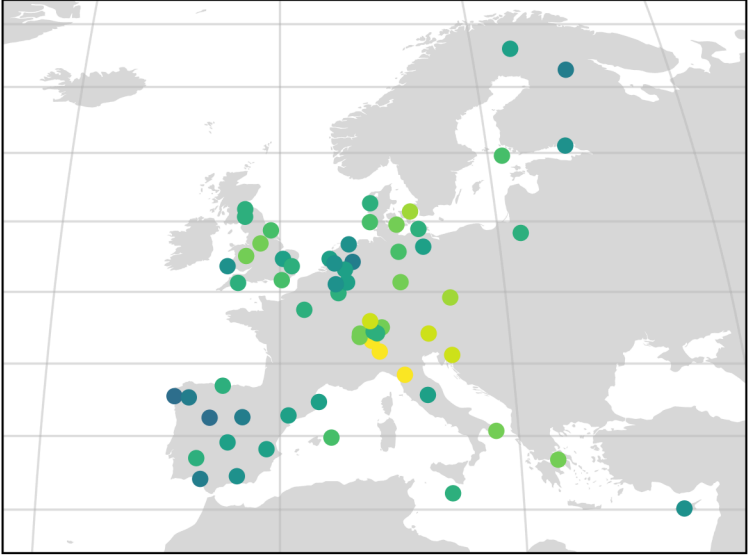
MONARCH
LOCAL-UK (10 stations)



Map RMSE (Summary)
EBAS|sconco3



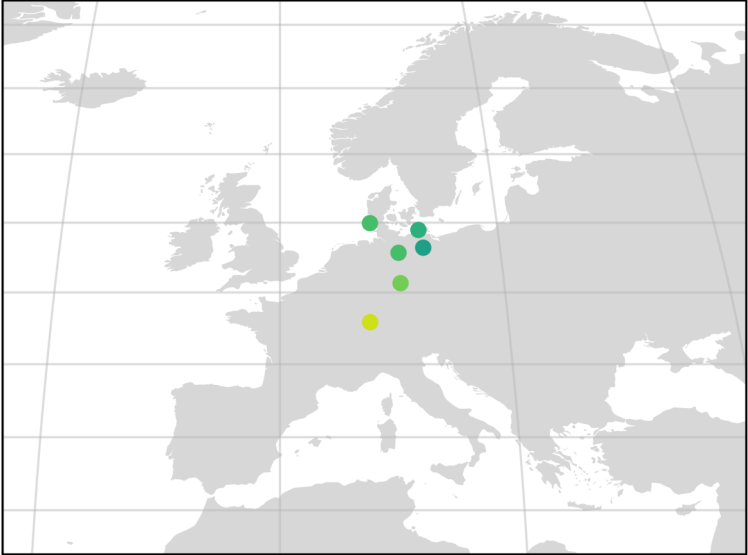
MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



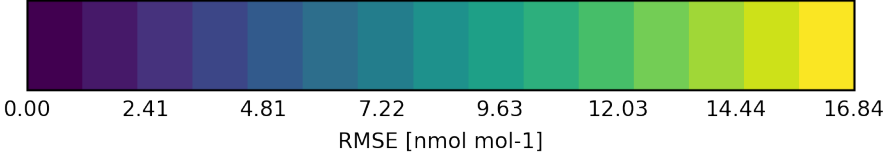
MONARCH
LOCAL·Germany (6 stations)



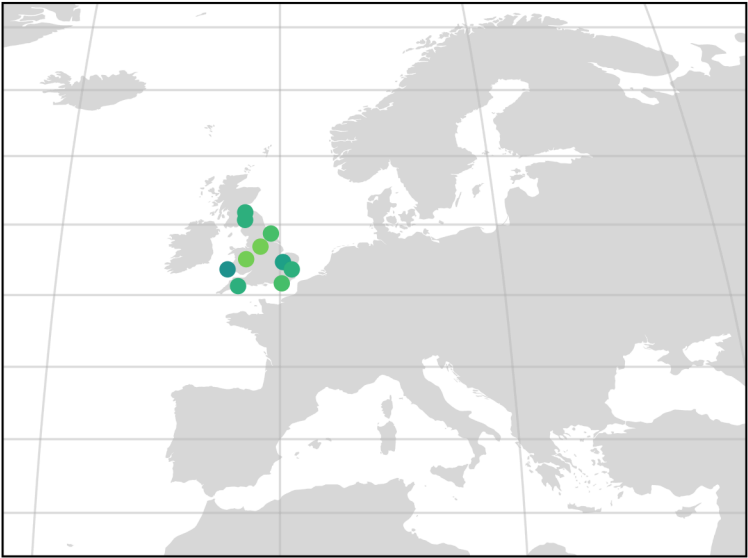
MONARCH
LOCAL·Spain (13 stations)



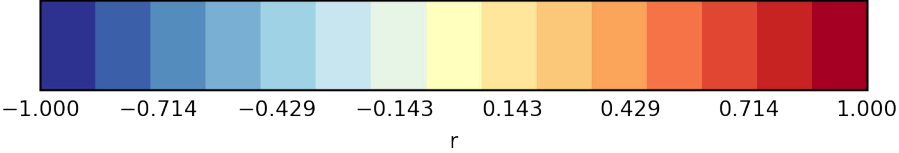
Map RMSE (Summary)
EBAS|sconco3



MONARCH
LOCAL·UK (10 stations)



Map r (Summary)
EBAS|sconco3



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



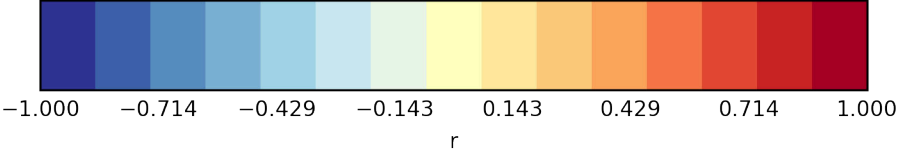
MONARCH
LOCAL·Germany (6 stations)



MONARCH
LOCAL·Spain (13 stations)



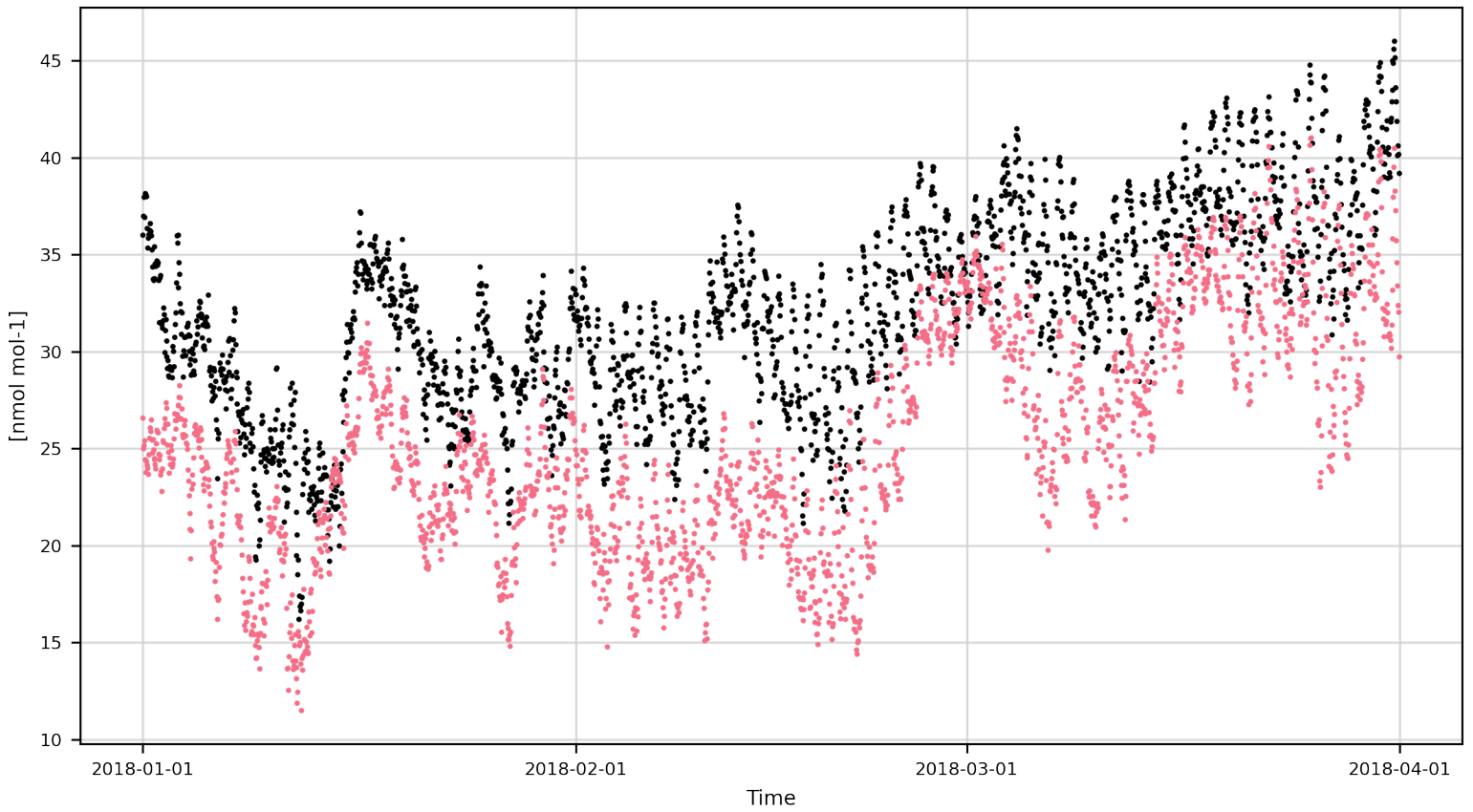
Map r (Summary)
EBAS|sconco3



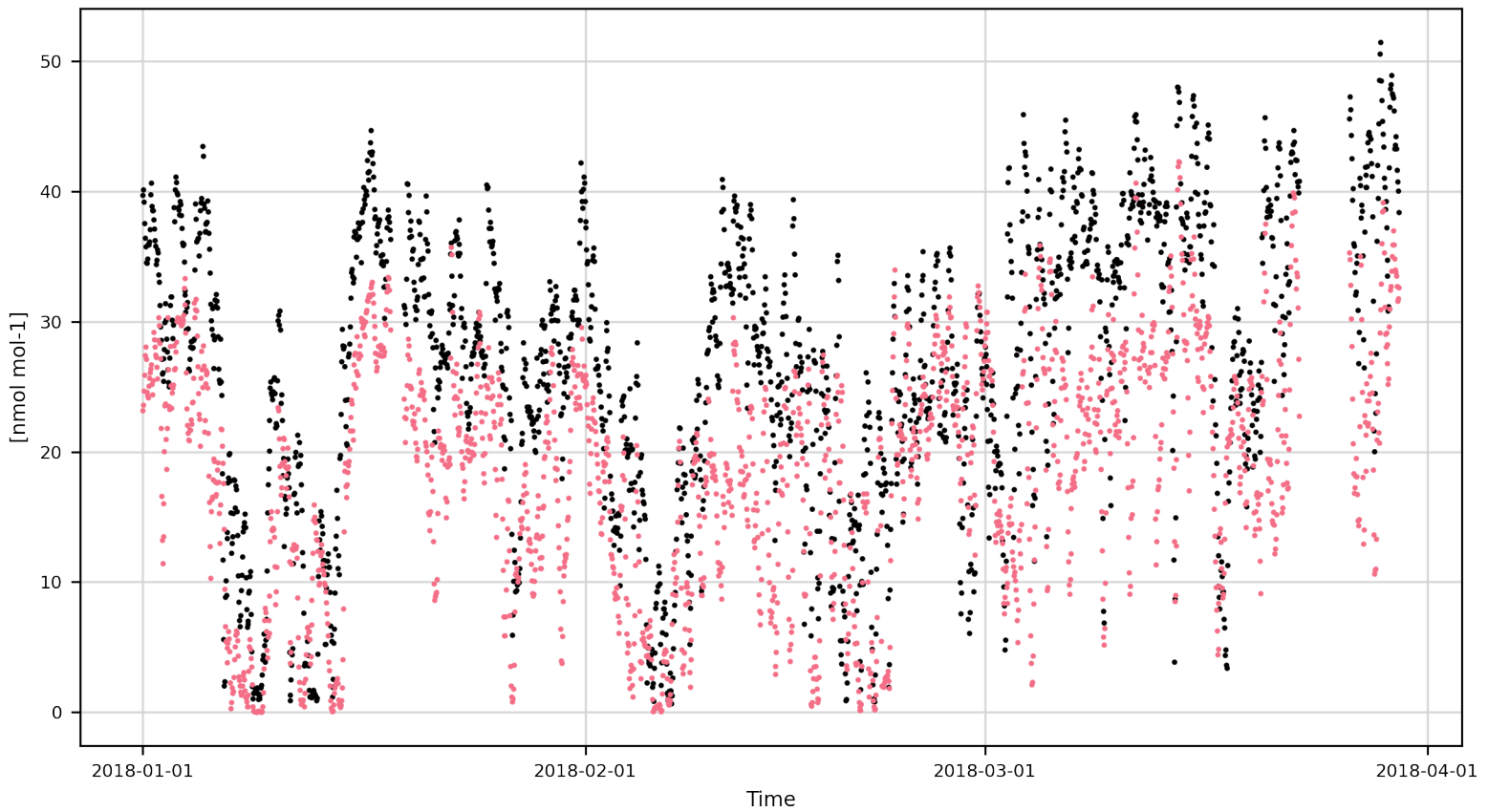
MONARCH
LOCAL·UK (10 stations)



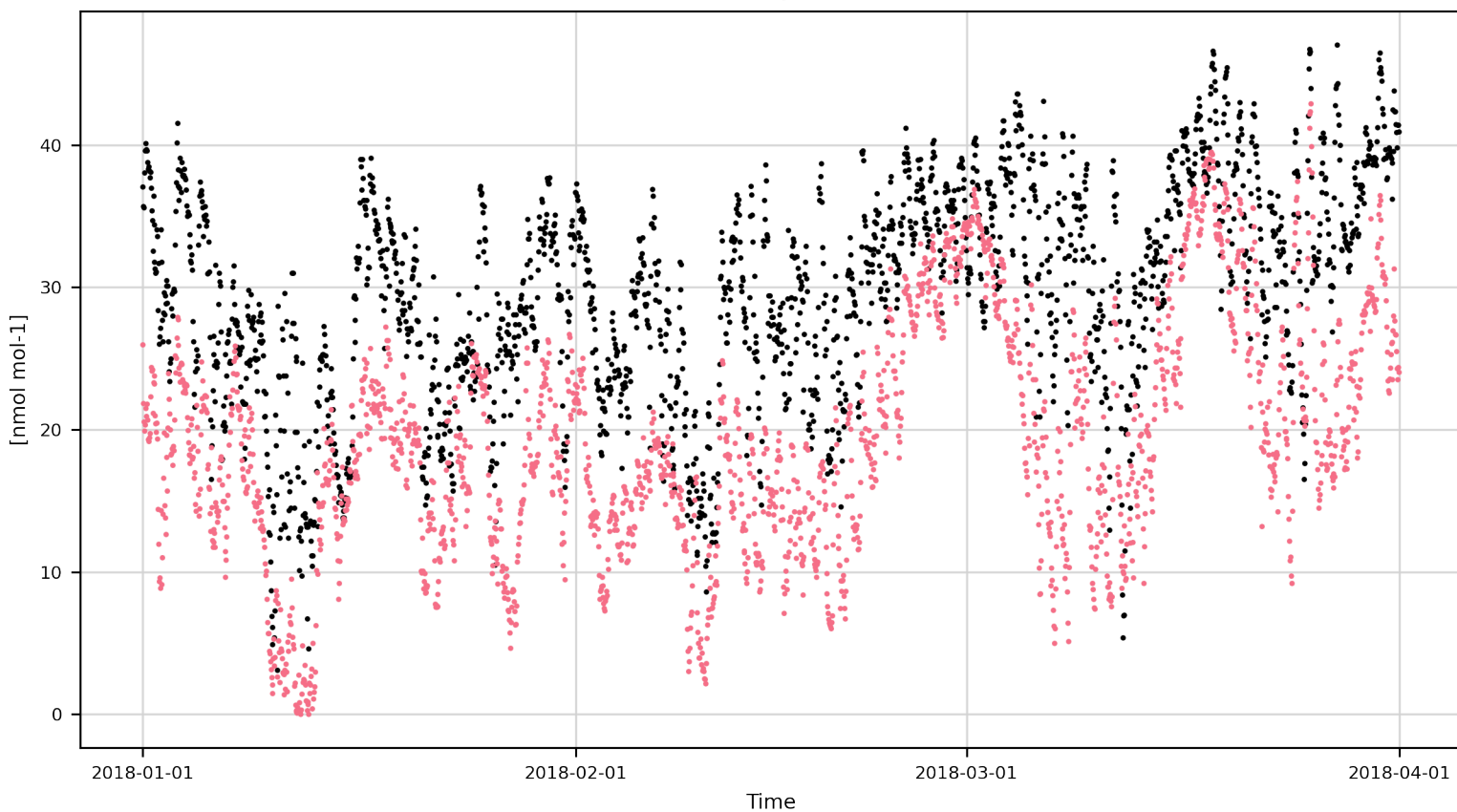
LOCAL·All (62 stations)



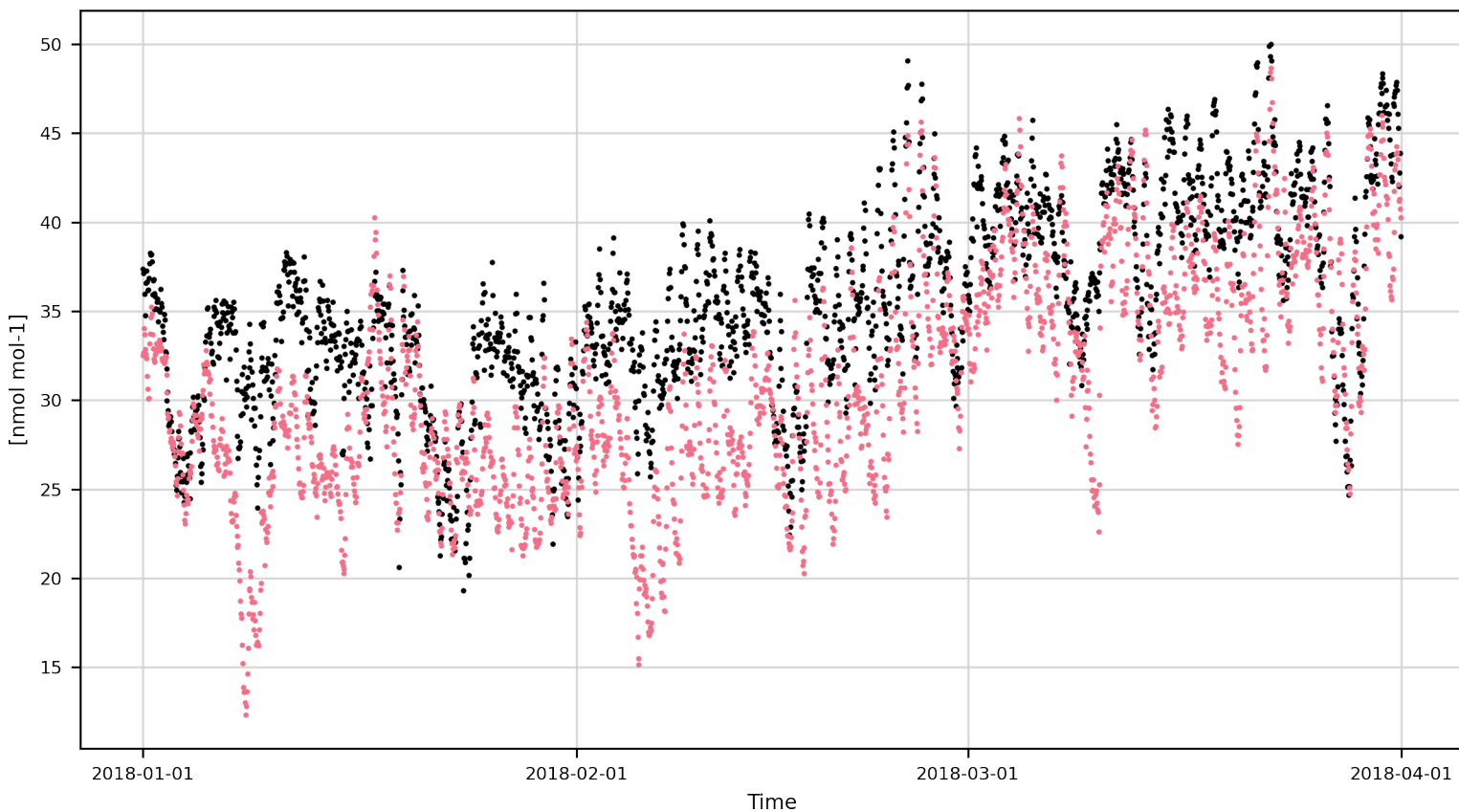
LOCAL·France (1 station)

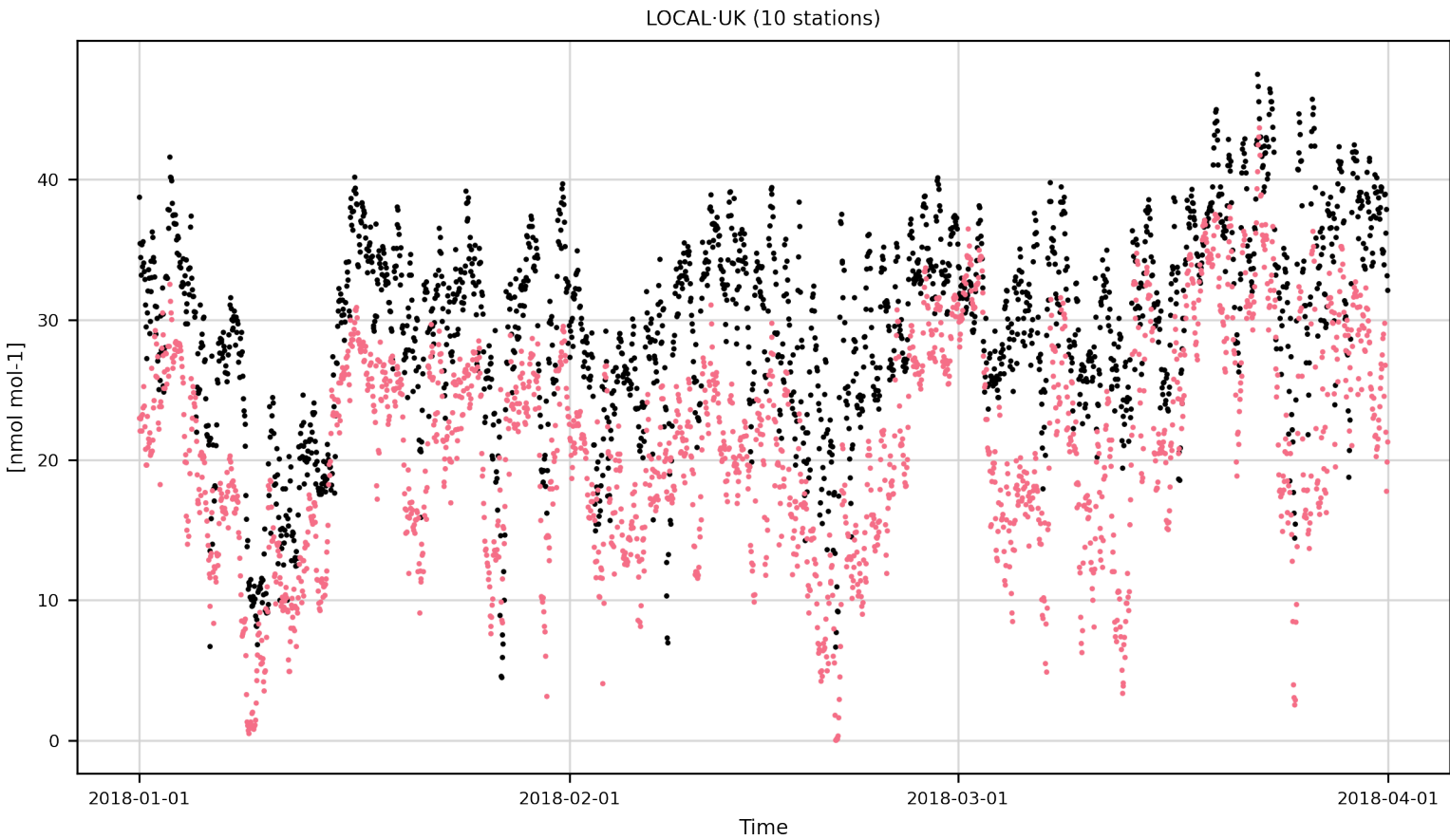


LOCAL-Germany (6 stations)



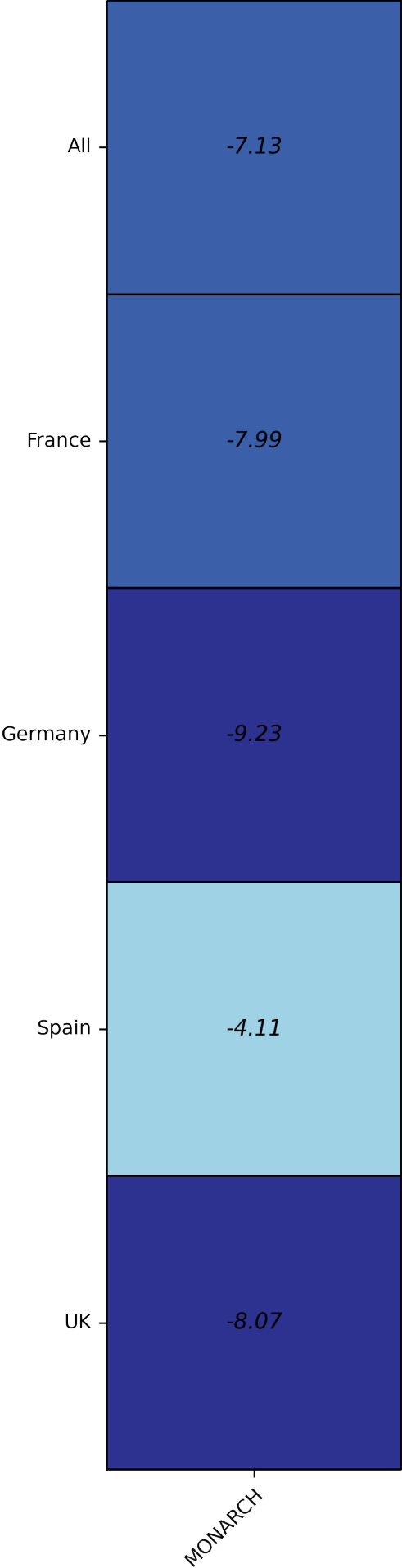
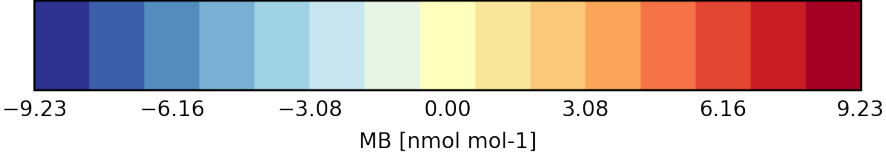
LOCAL-Spain (13 stations)



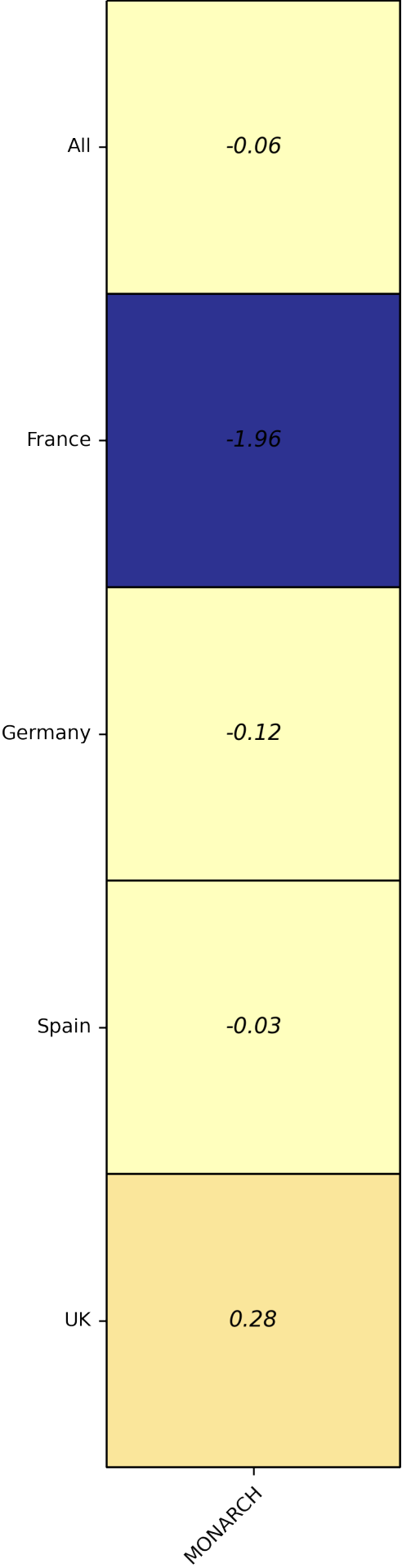
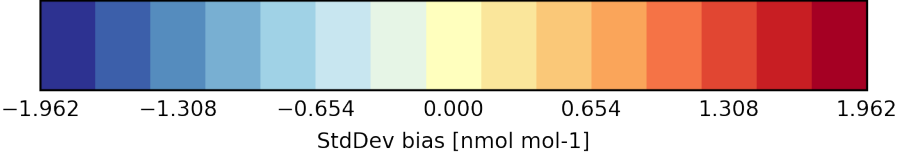


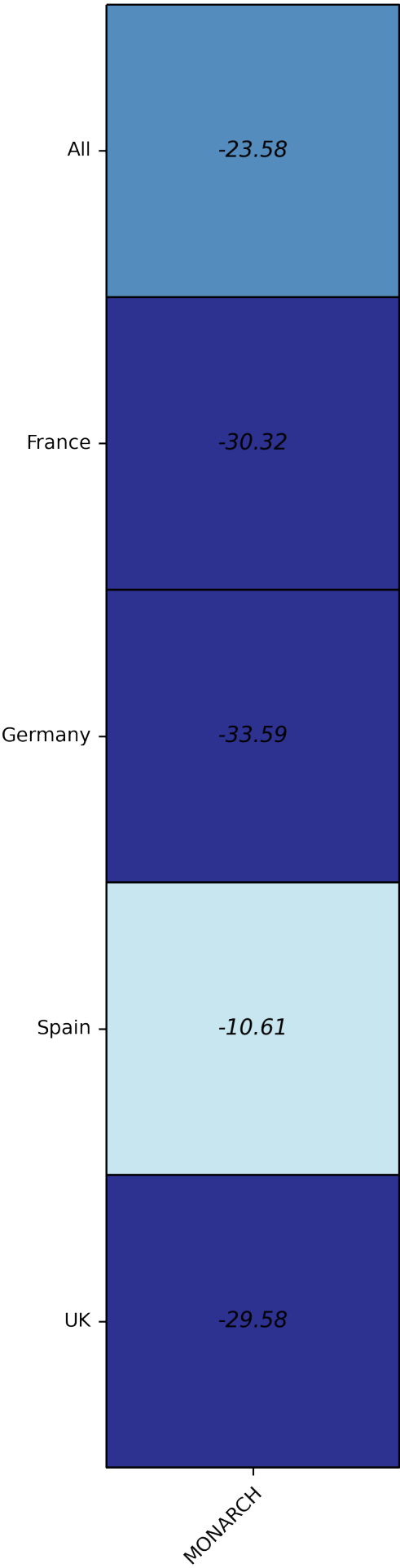
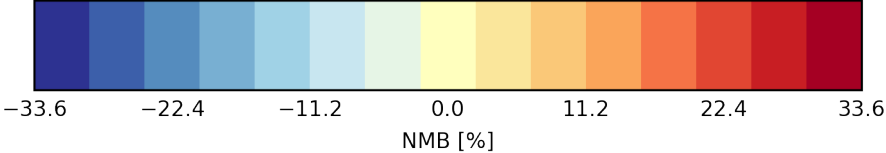
		Mean	StdDev	p5	Median	p95
All	observations	30.15	8.78	13.67	31.07	42.90
	MONARCH	25.25	8.84	9.70	25.48	38.21
France	observations	26.37	11.18	4.67	27.52	41.94
	MONARCH	18.37	9.22	1.79	19.54	31.94
Germany	observations	29.61	9.02	12.20	30.65	42.80
	MONARCH	19.55	8.86	4.86	19.65	35.43
Spain	observations	34.07	7.06	21.57	34.40	46.15
	MONARCH	31.00	7.66	19.42	30.79	43.19
UK	observations	29.19	8.52	13.80	30.34	41.01
	MONARCH	19.72	8.93	5.41	19.85	34.37

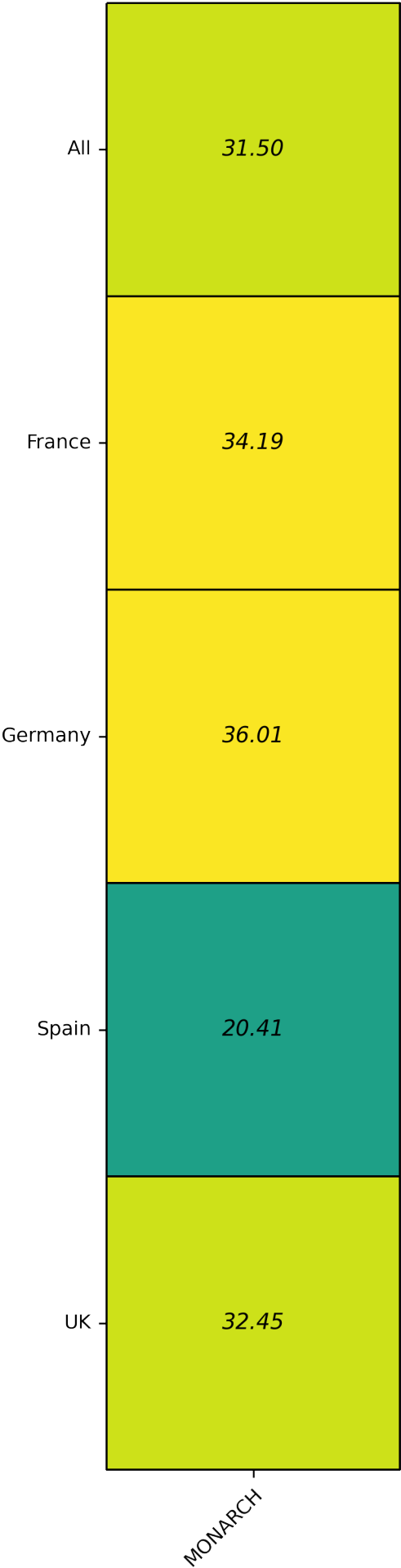
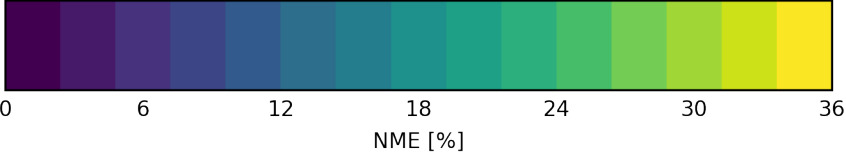
		Mean_bias	StdDev_bias	NMB	NME	RMSE	r
All	MONARCH	-7.13	-0.06	-23.58	31.50	10.54	0.66
France	MONARCH	-7.99	-1.96	-30.32	34.19	10.73	0.77
Germany	MONARCH	-9.23	-0.12	-33.59	36.01	11.80	0.72
Spain	MONARCH	-4.11	-0.03	-10.61	20.41	9.14	0.65
UK	MONARCH	-8.07	0.28	-29.58	32.45	10.63	0.70



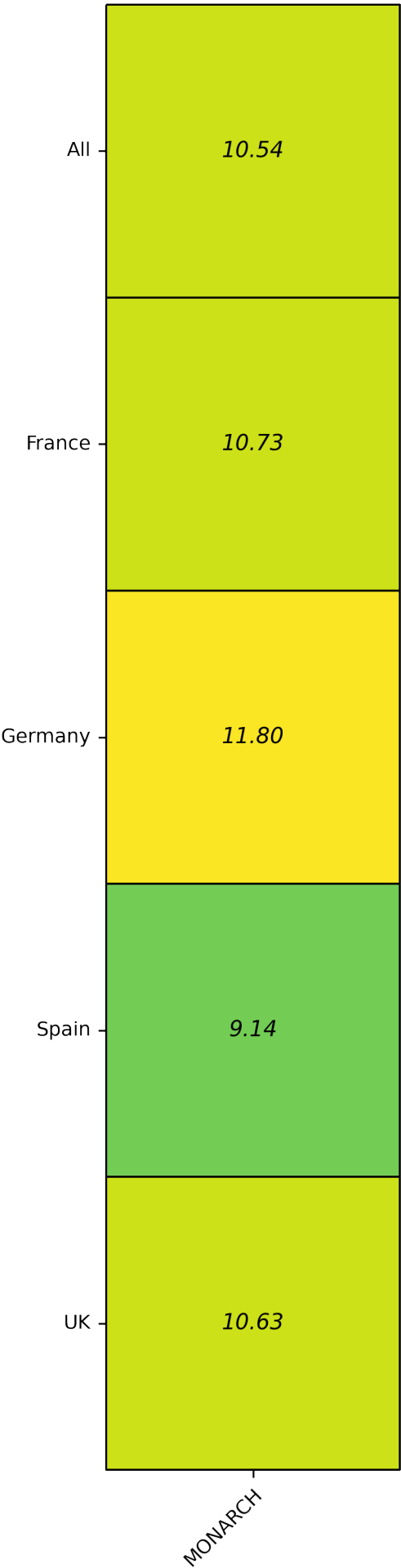
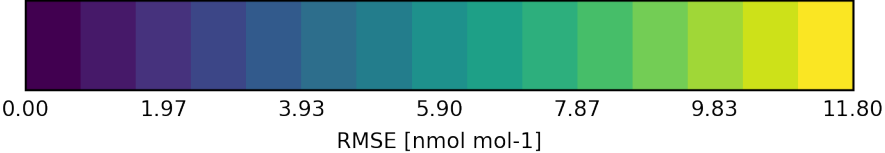
Heatmap StdDev bias (Summary)
EBAS|sconco3



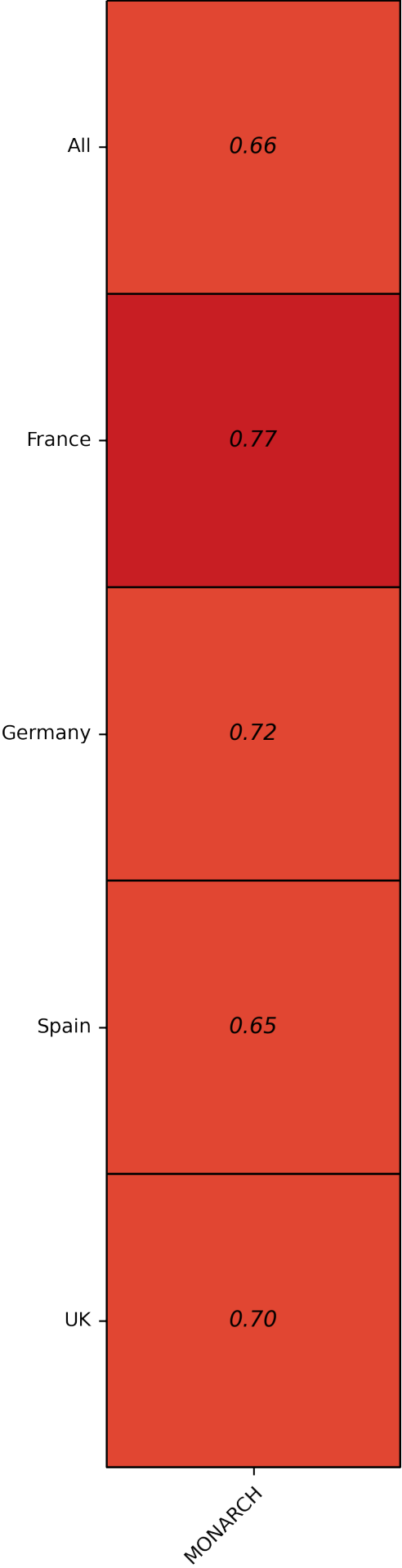
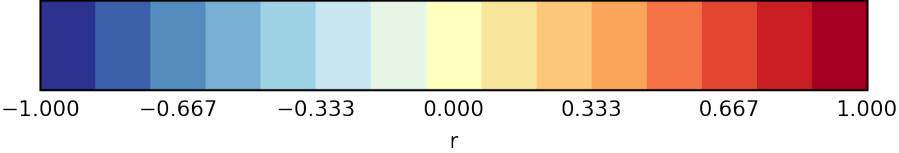


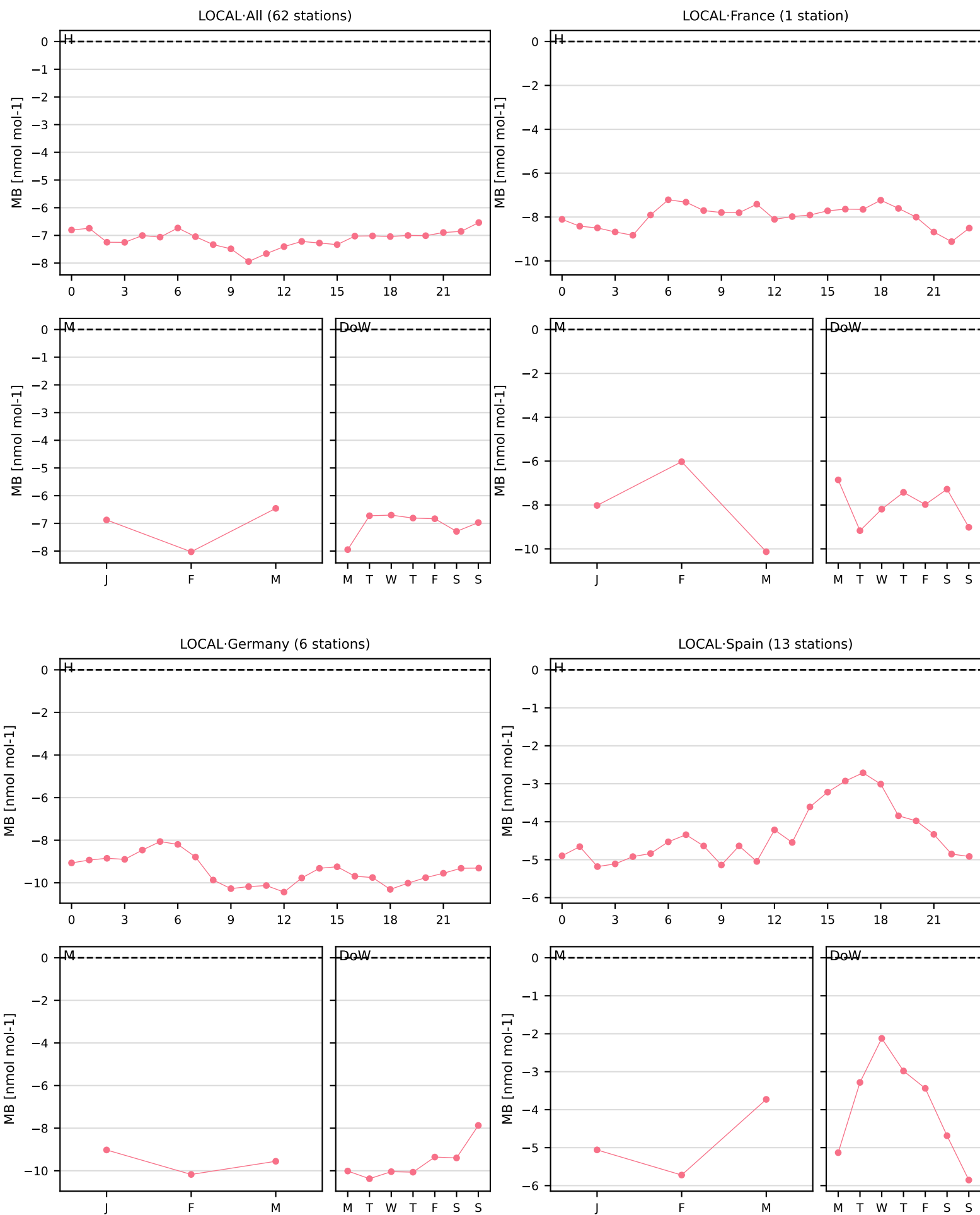


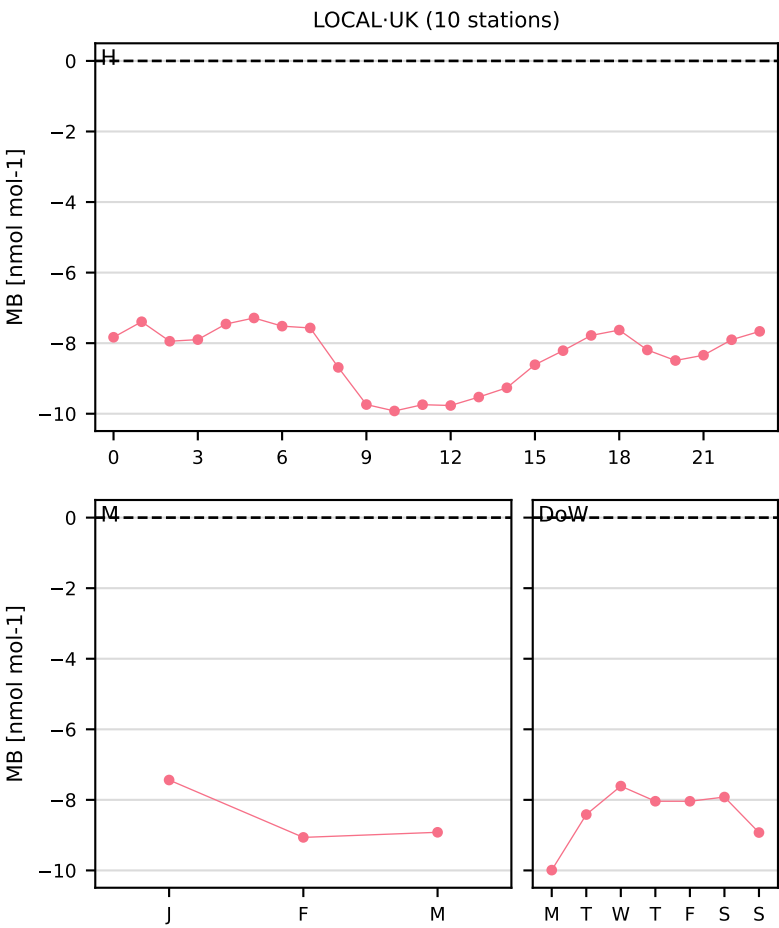
Heatmap RMSE (Summary)
EBAS|sconco3



Heatmap r (Summary)
EBAS|sconco3

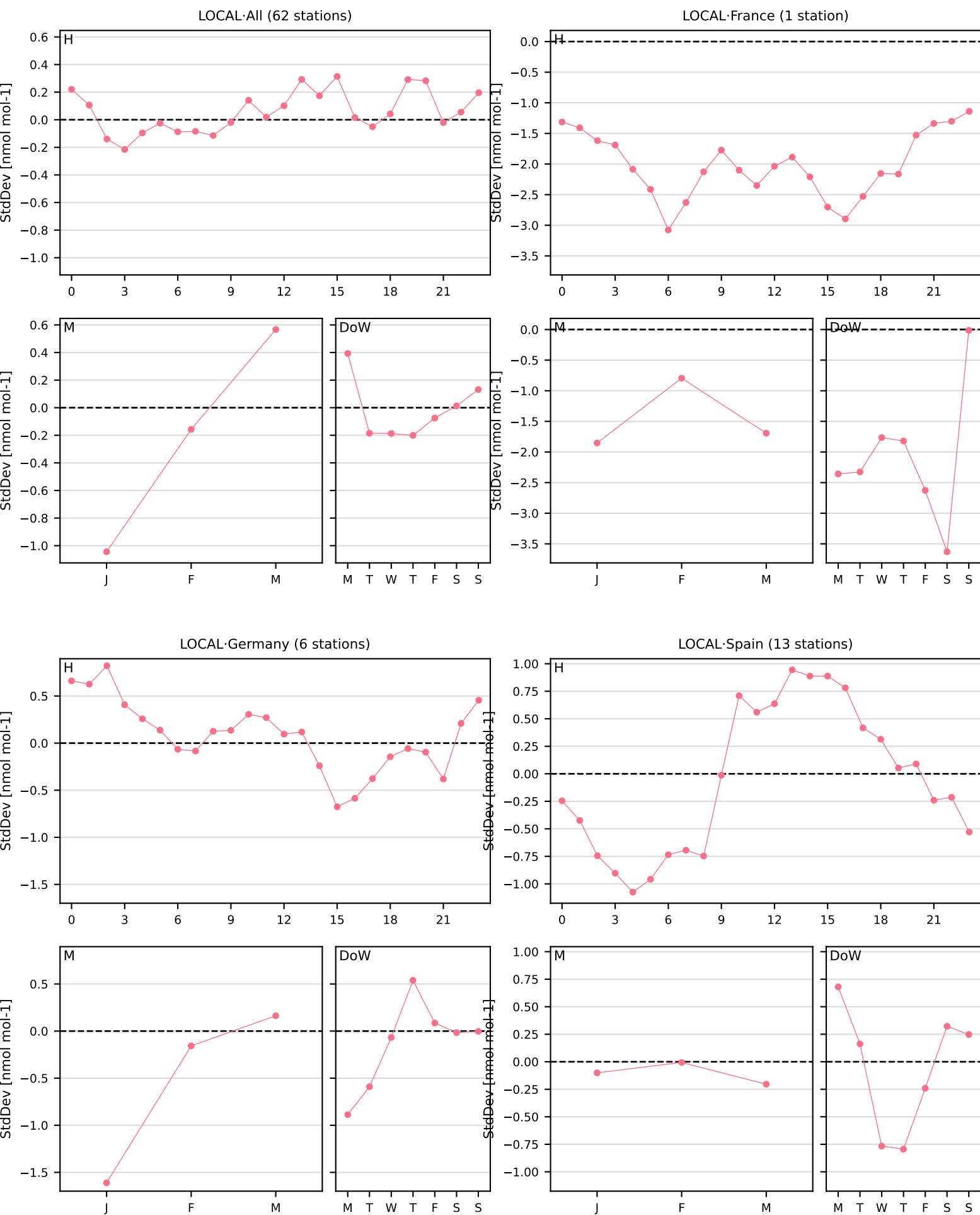


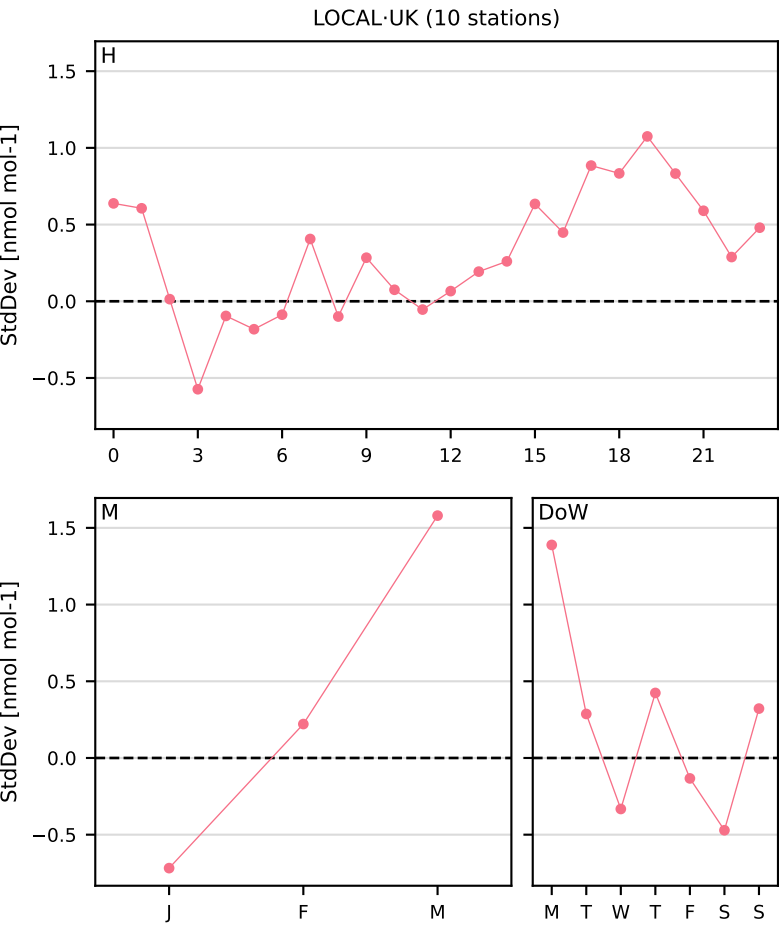


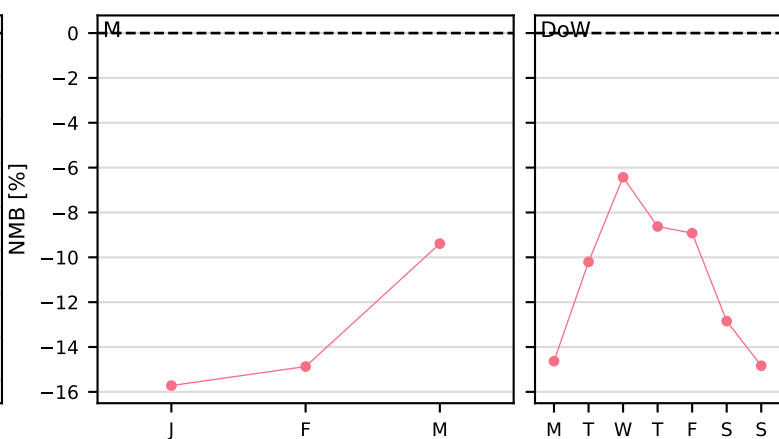
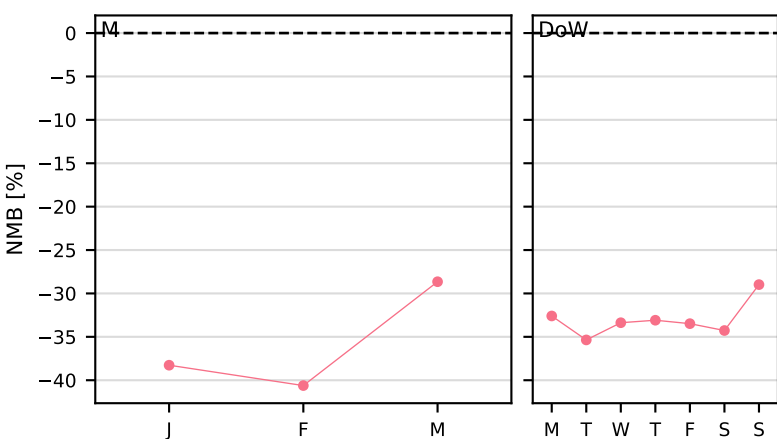
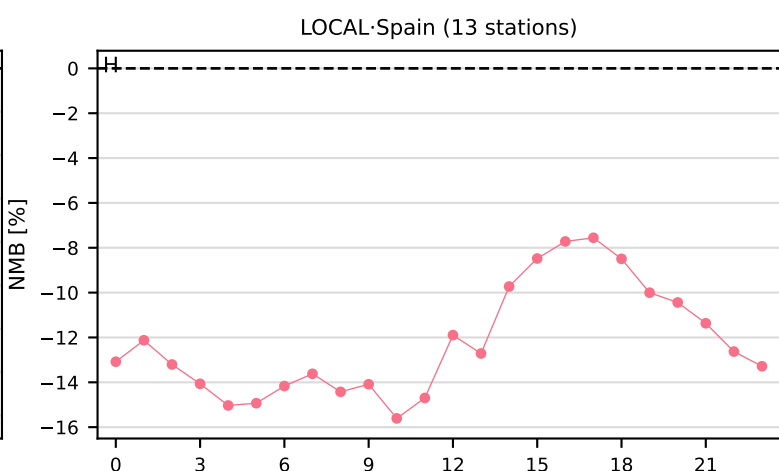
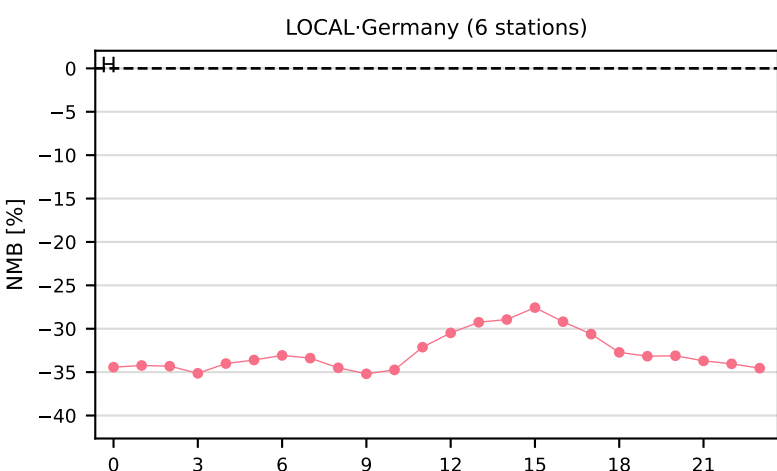
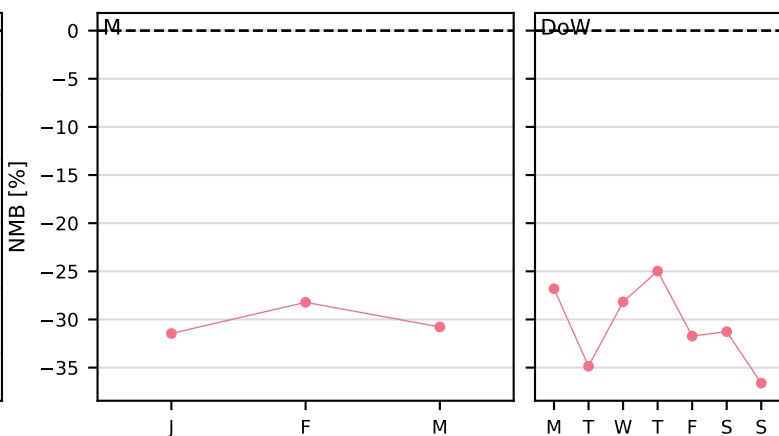
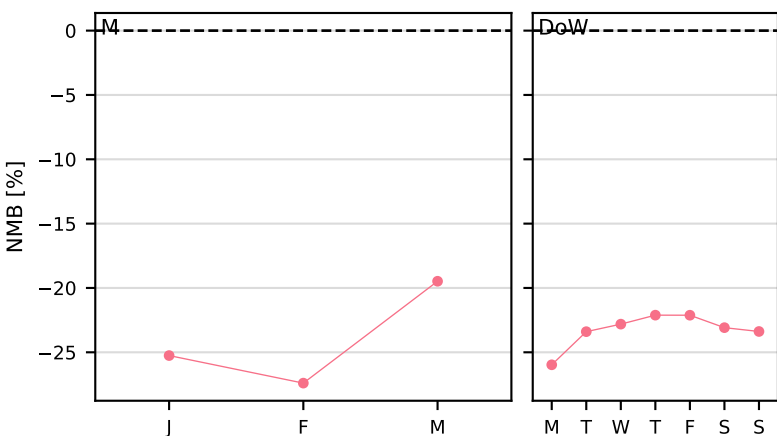
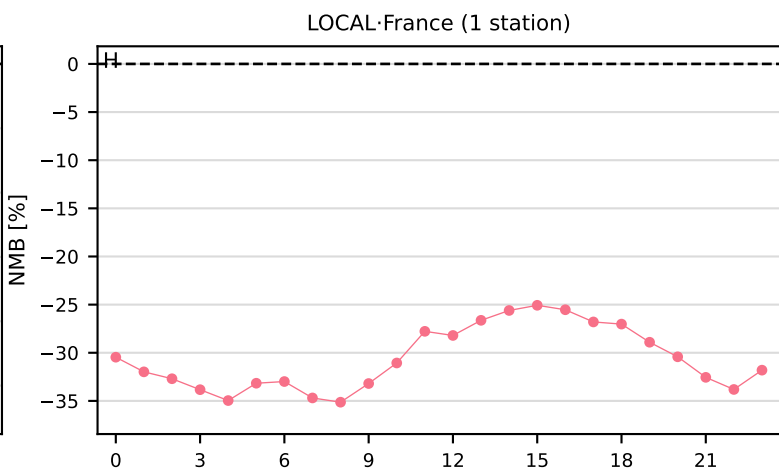
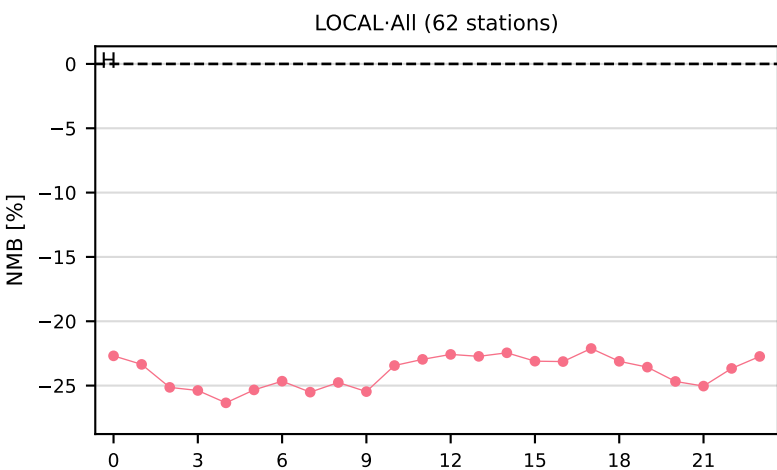


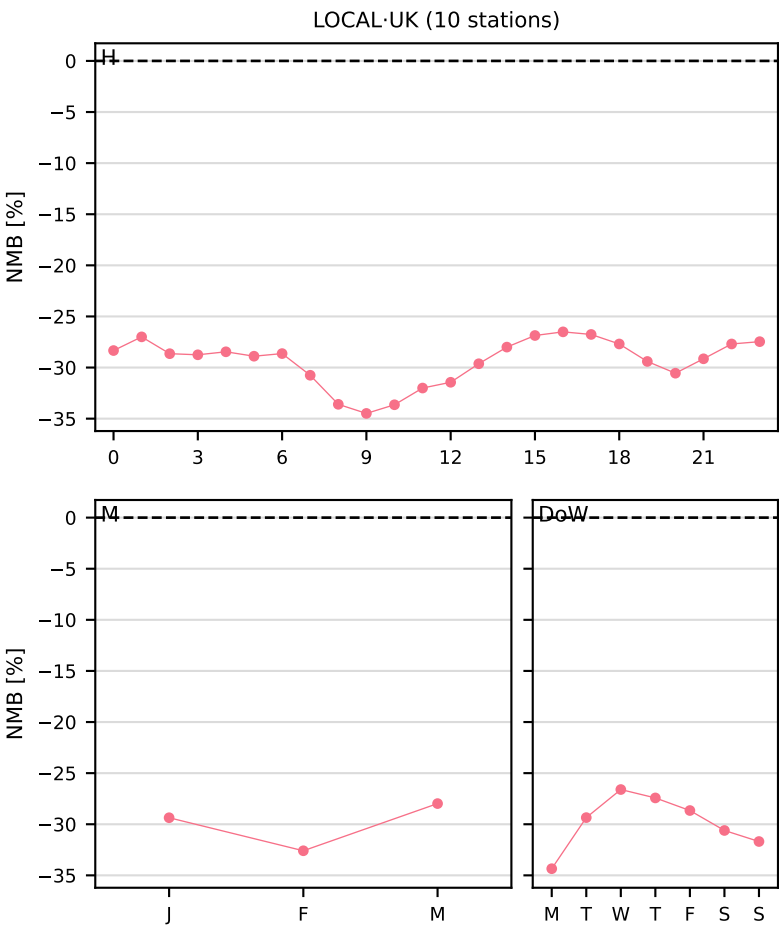
Periodic StdDev bias (Summary)
EBAS|sconco3

MONARCH

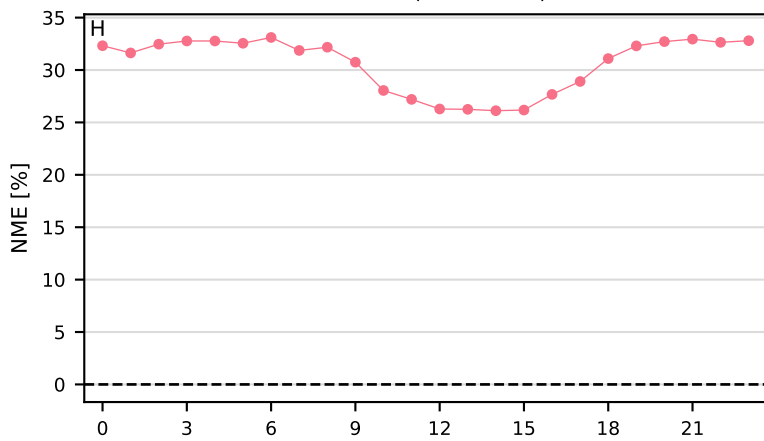




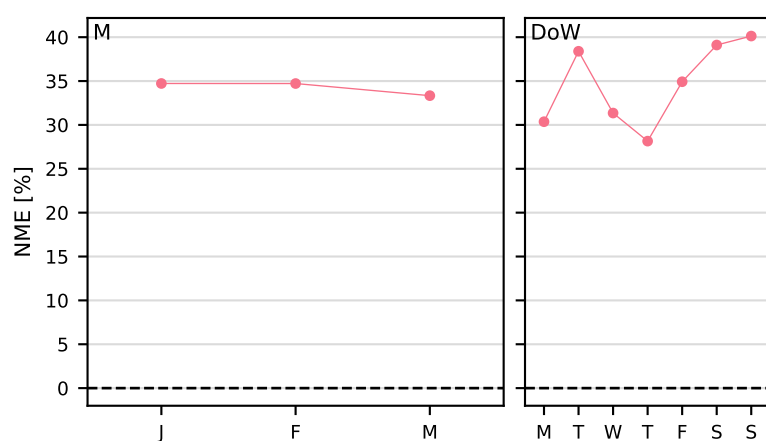
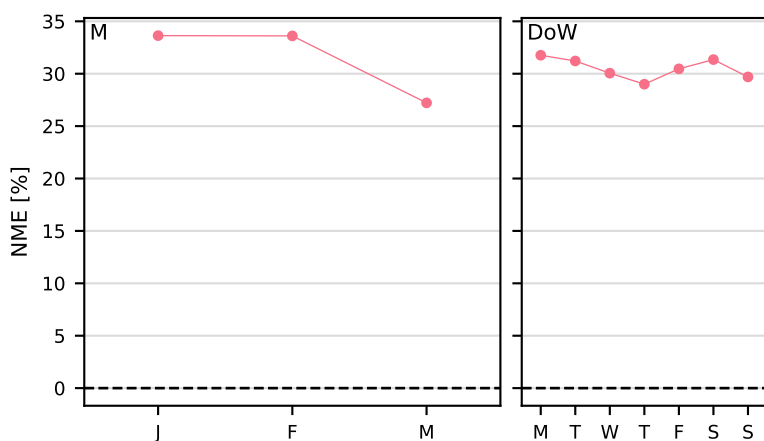
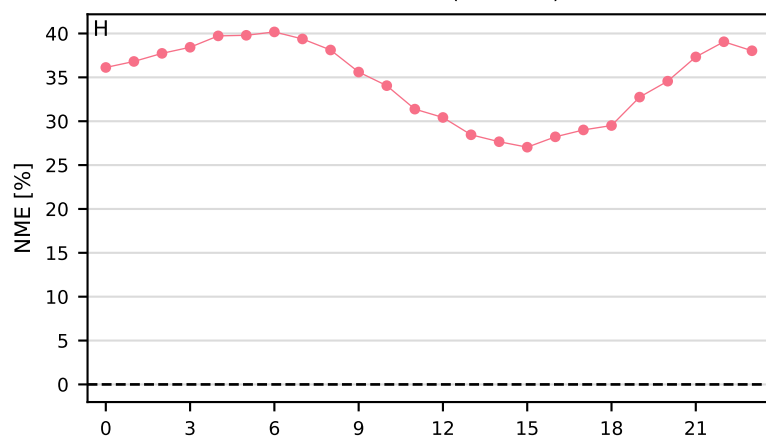




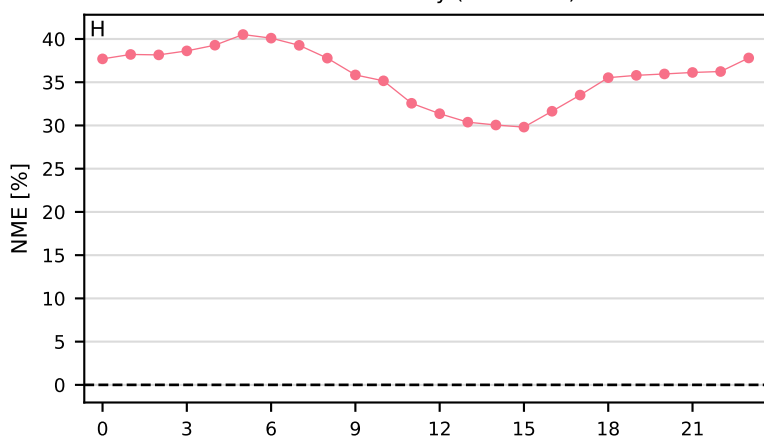
LOCAL·All (62 stations)



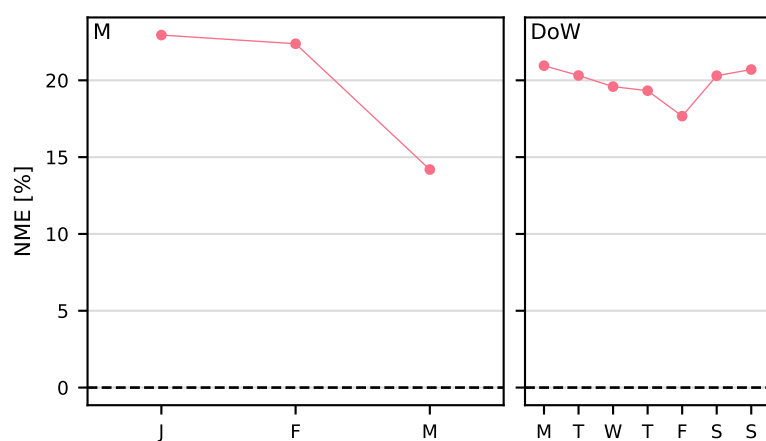
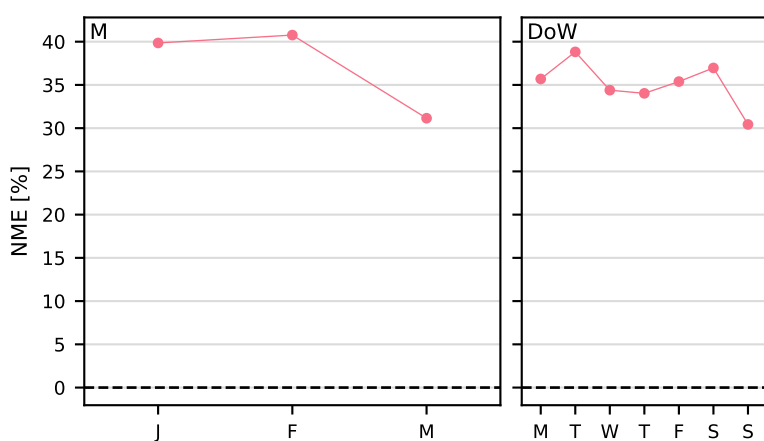
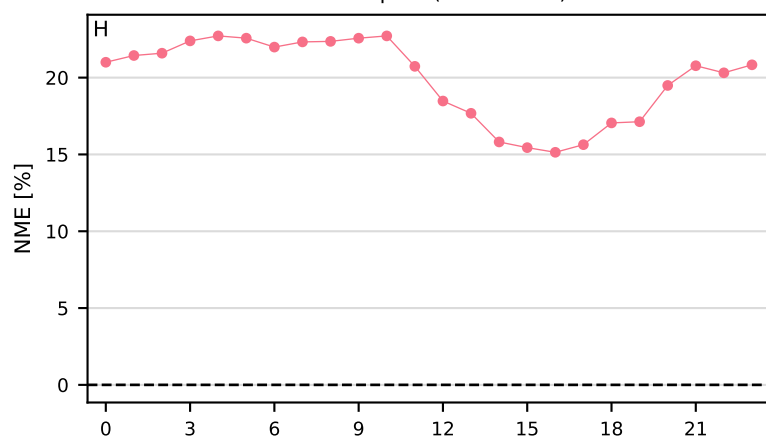
LOCAL·France (1 station)

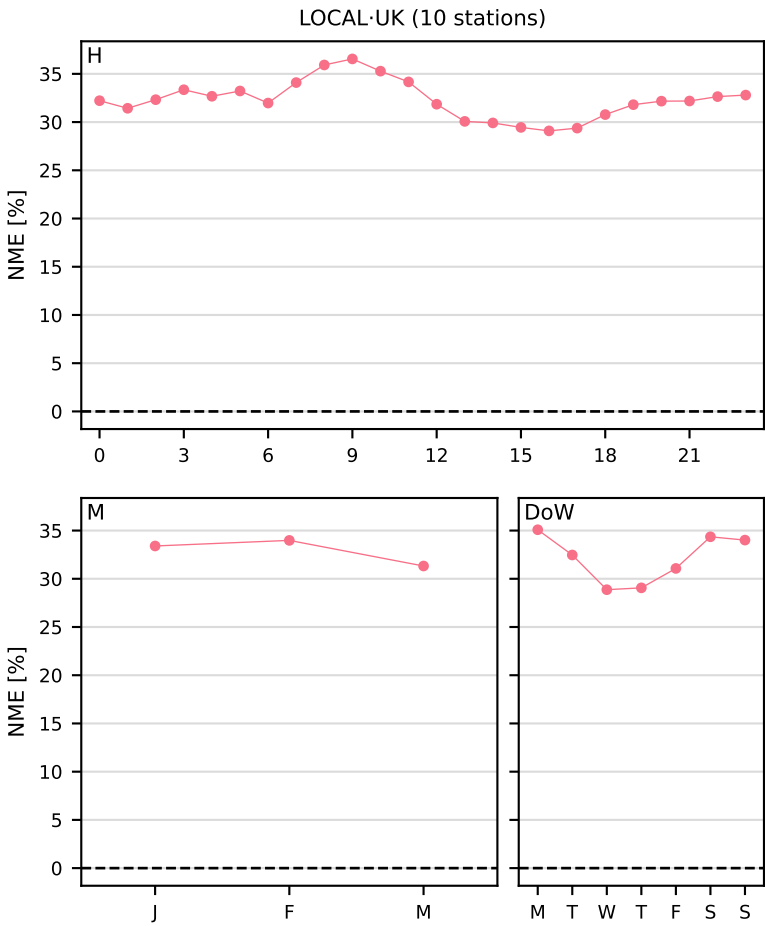


LOCAL·Germany (6 stations)

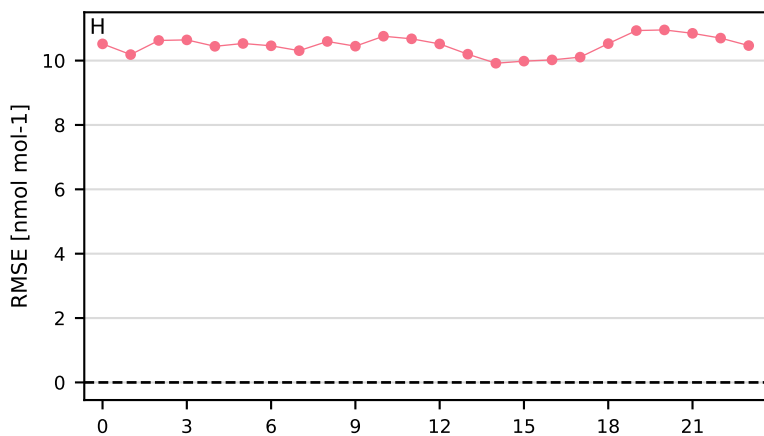


LOCAL·Spain (13 stations)

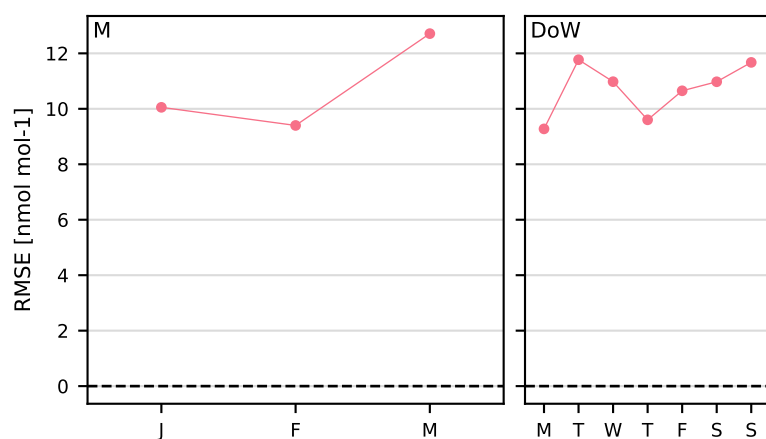
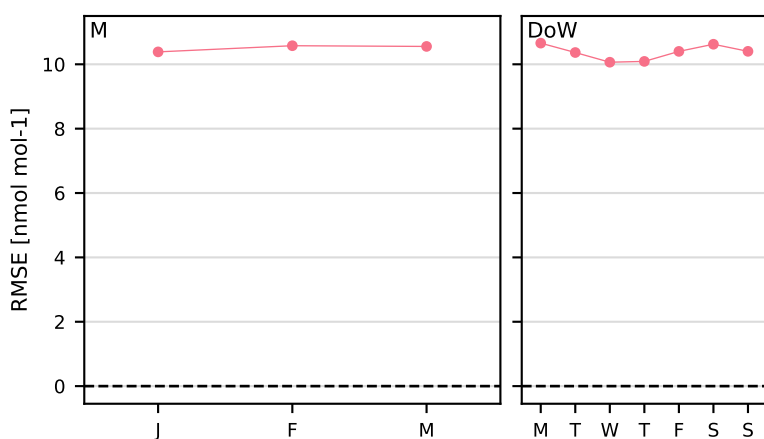
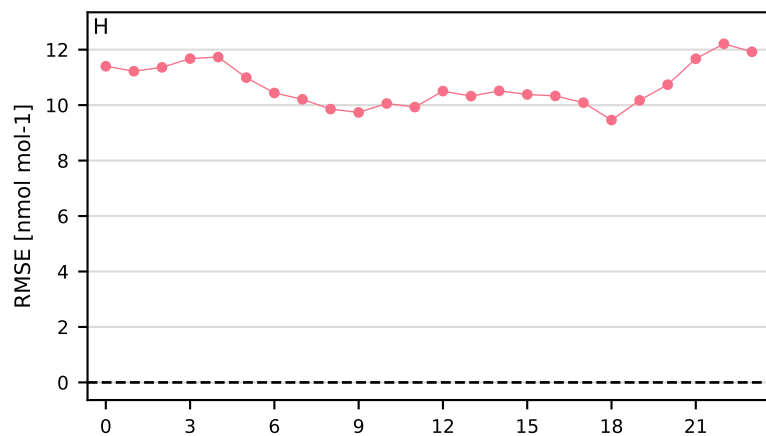




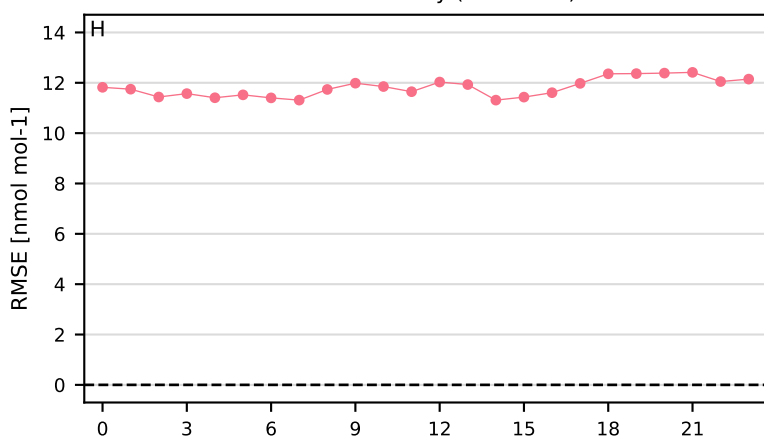
LOCAL·All (62 stations)



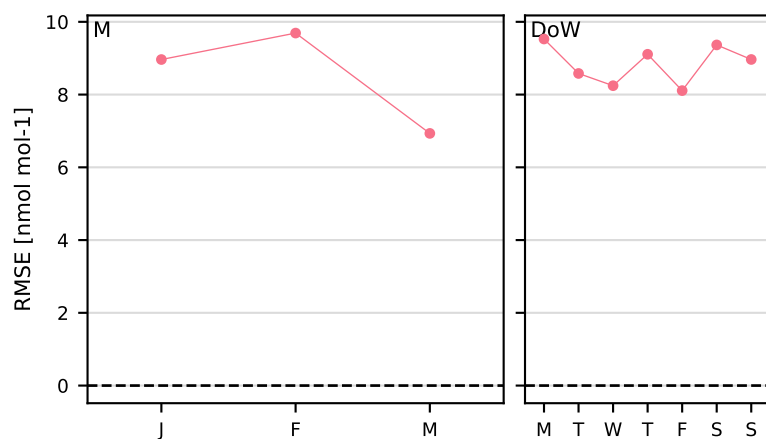
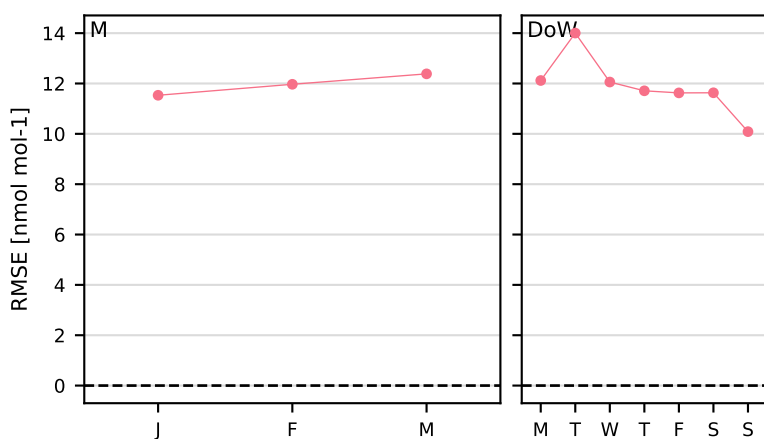
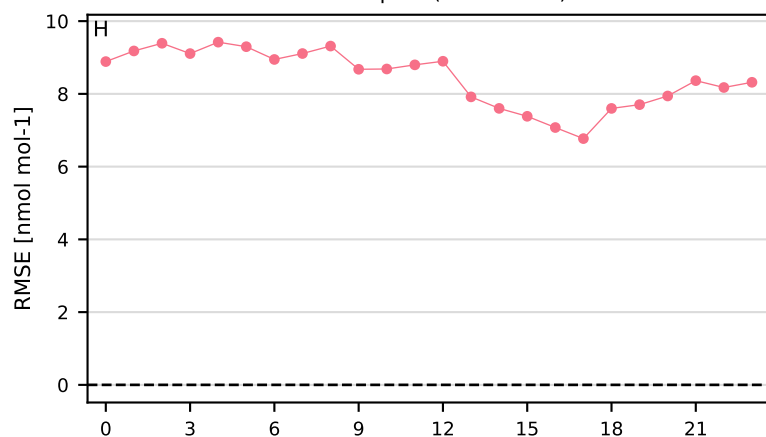
LOCAL·France (1 station)

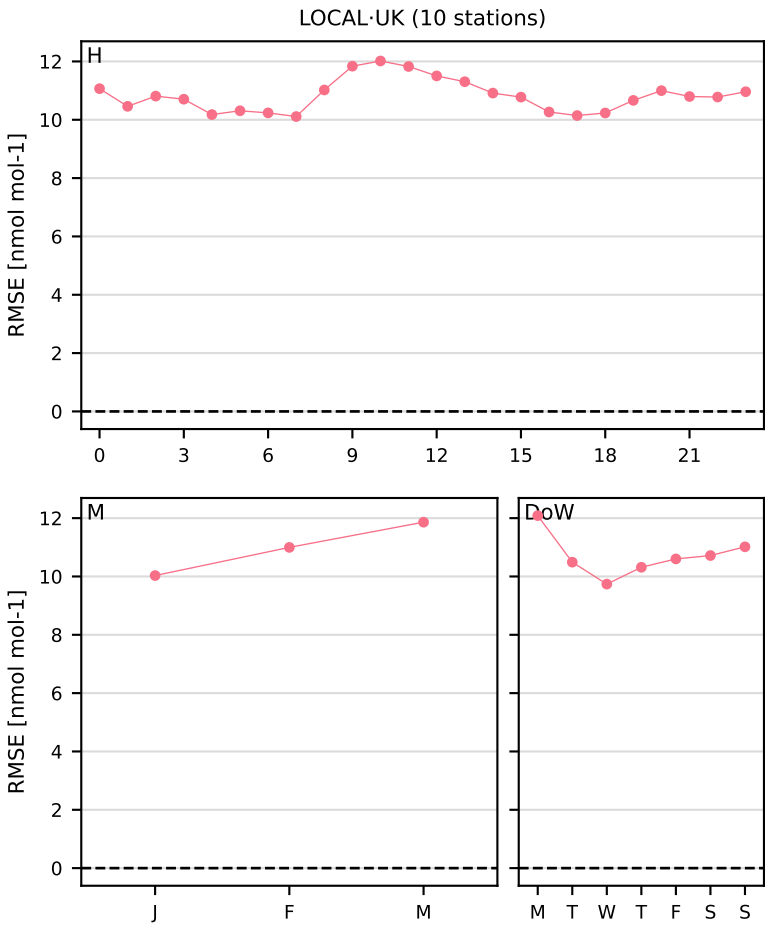


LOCAL·Germany (6 stations)

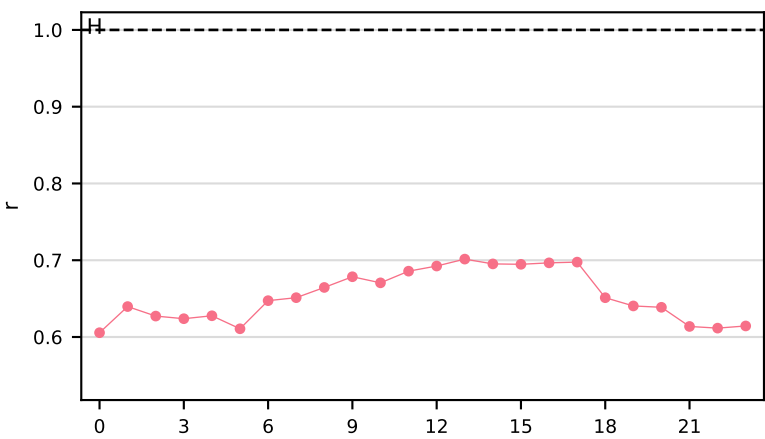


LOCAL·Spain (13 stations)

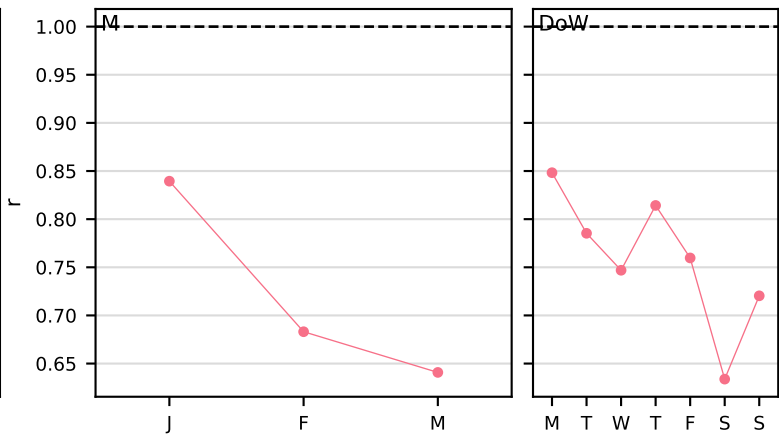
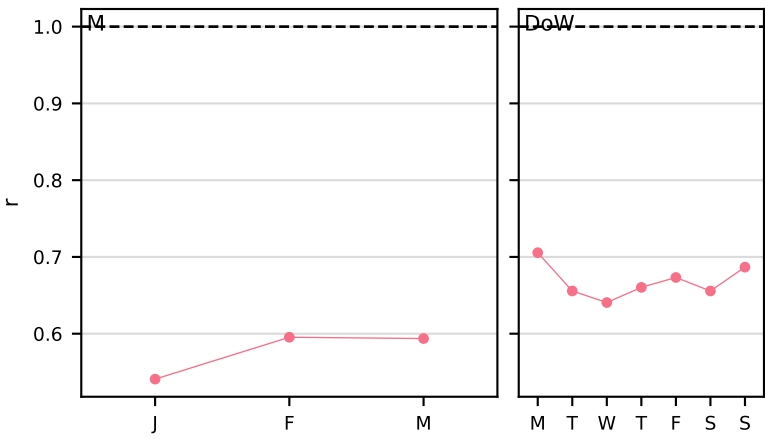
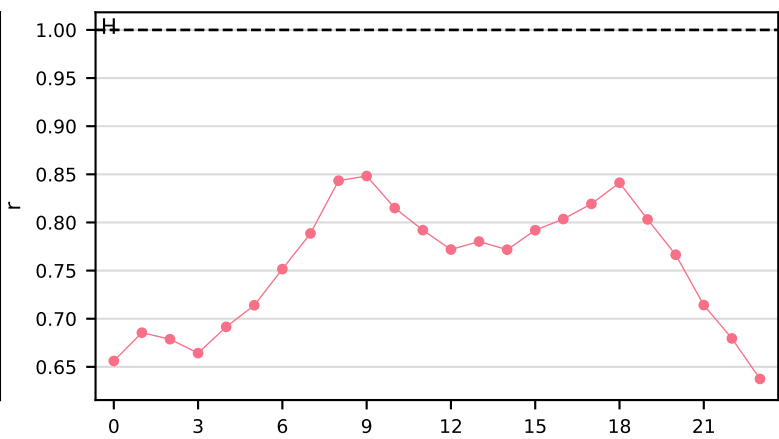




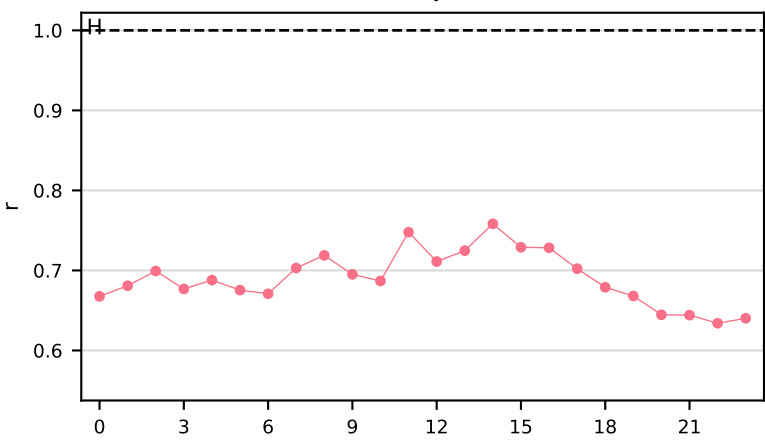
LOCAL·All (62 stations)



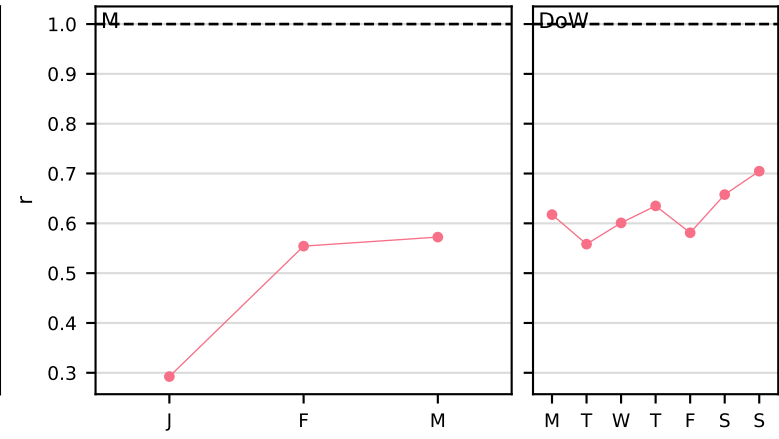
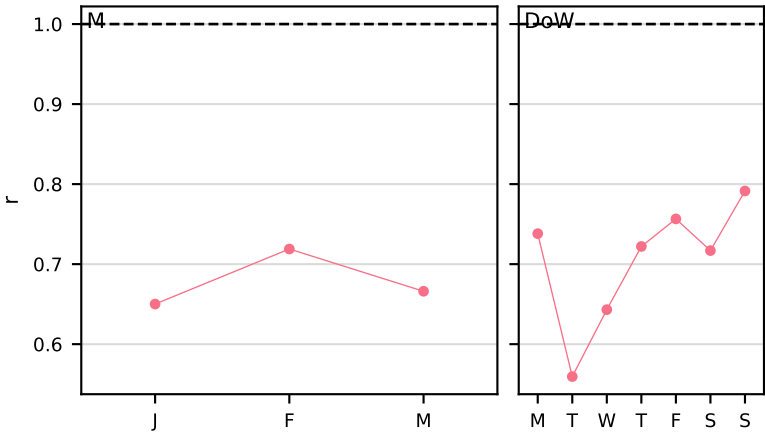
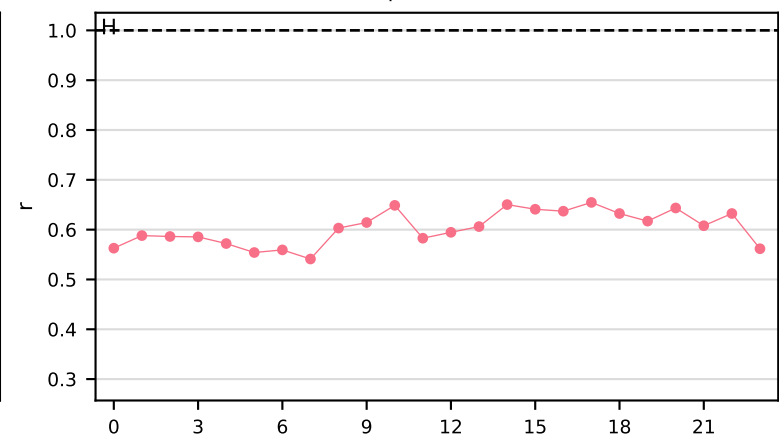
LOCAL·France (1 station)

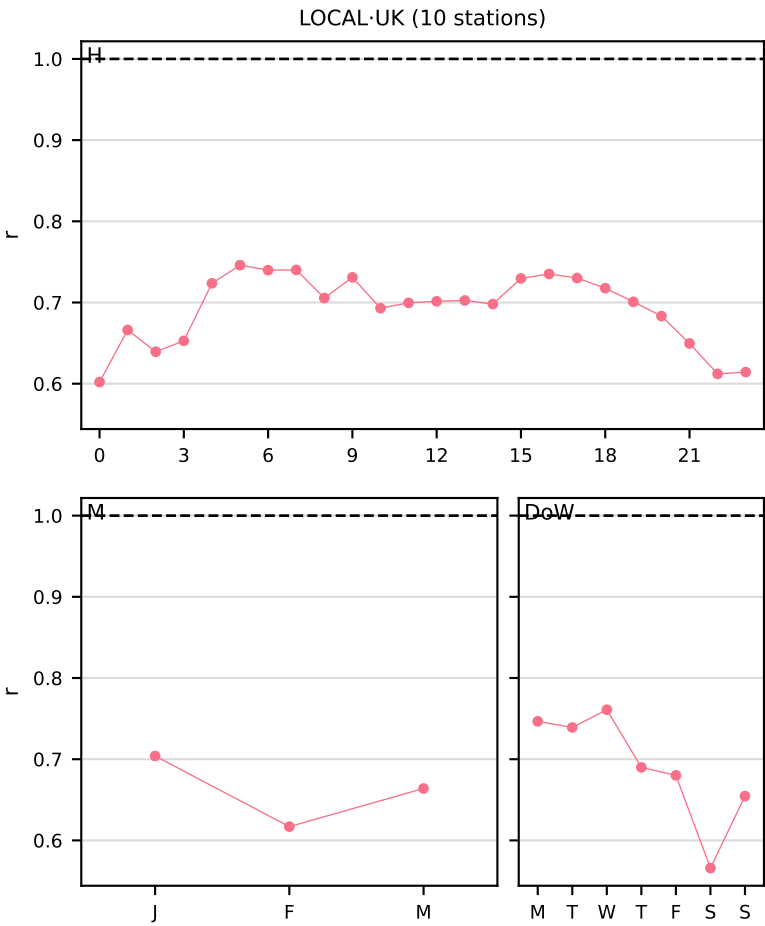


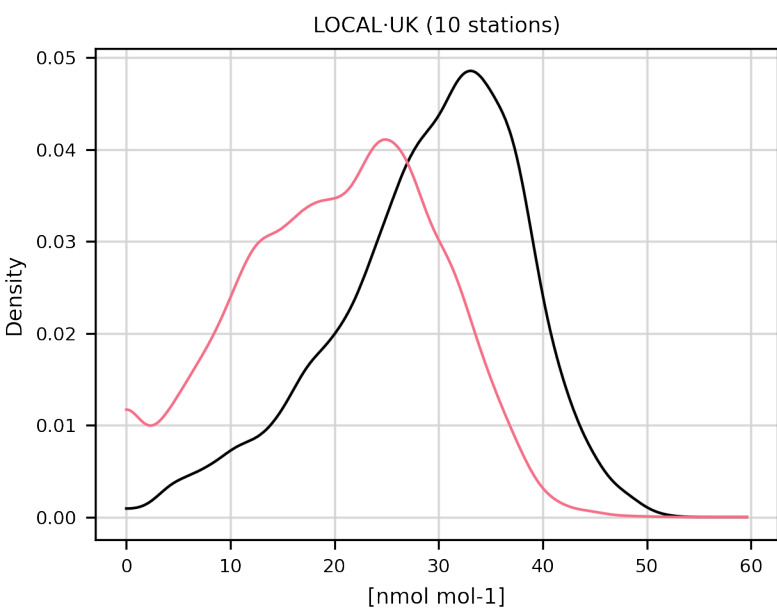
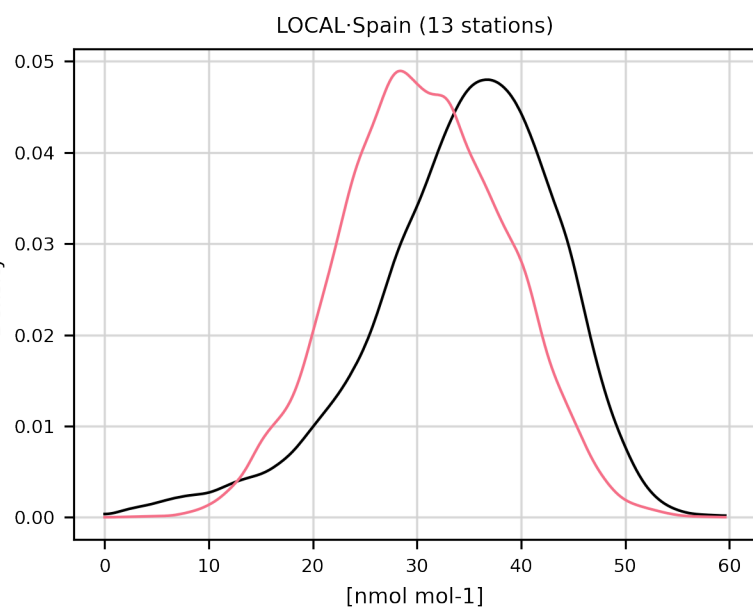
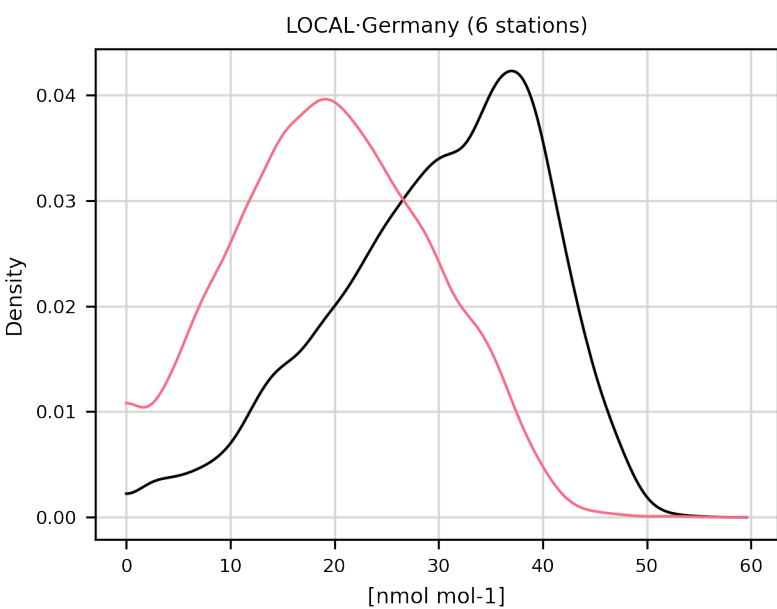
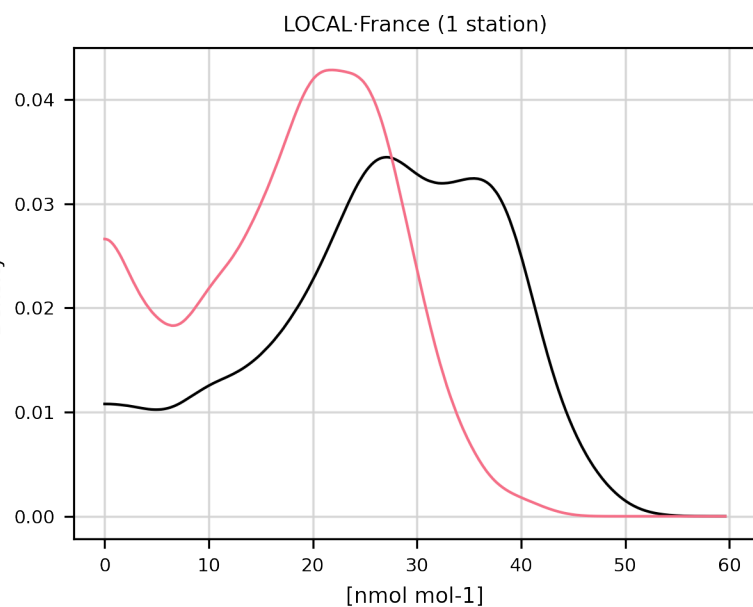
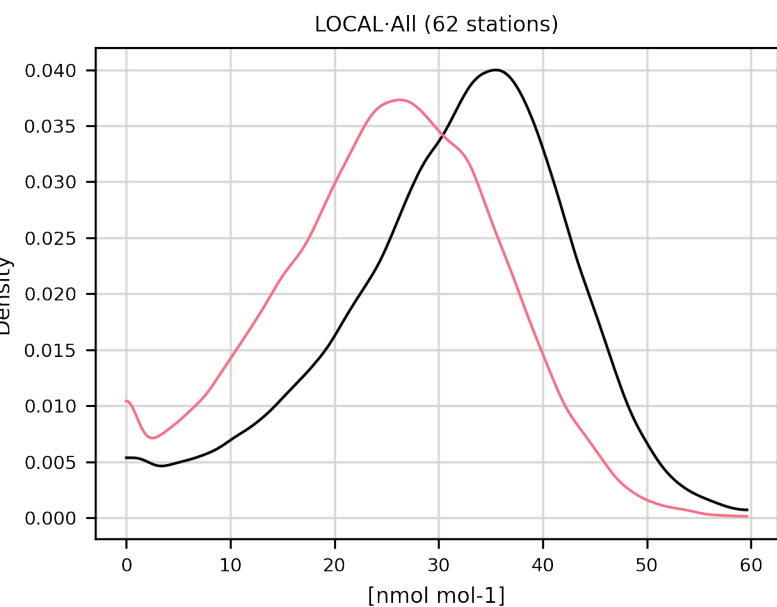
LOCAL·Germany (6 stations)

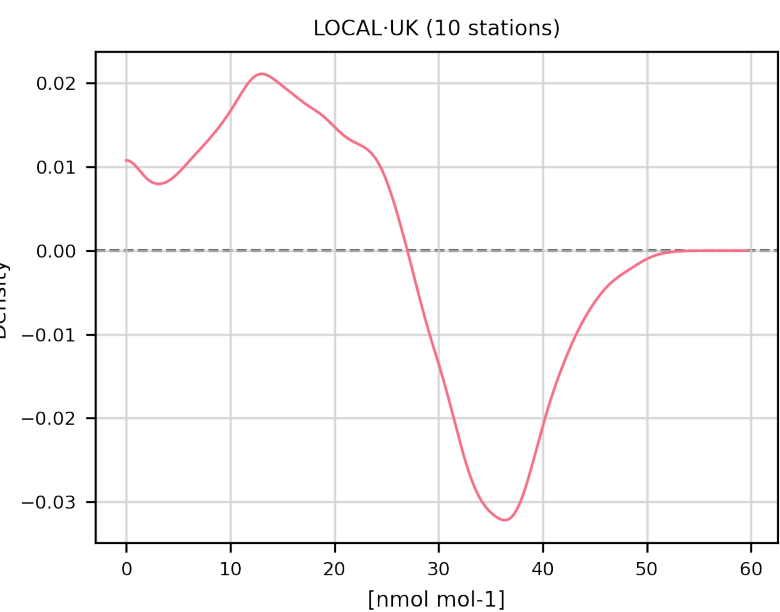
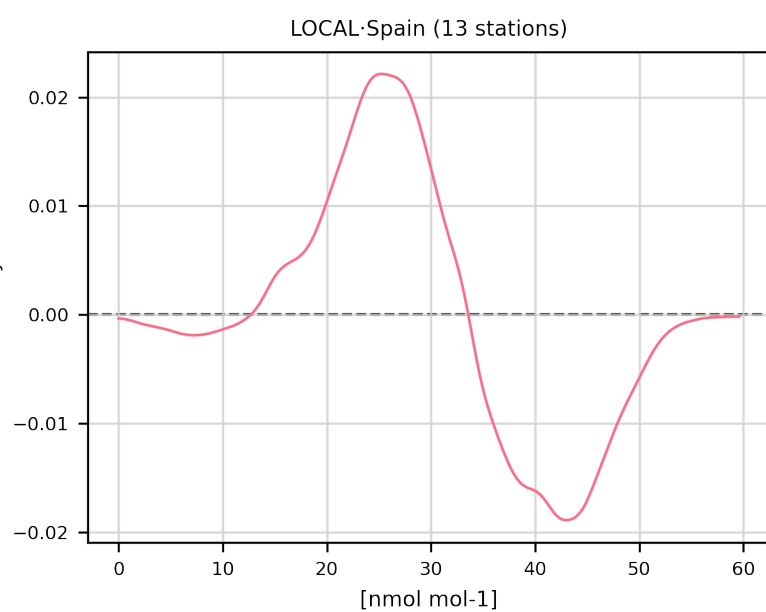
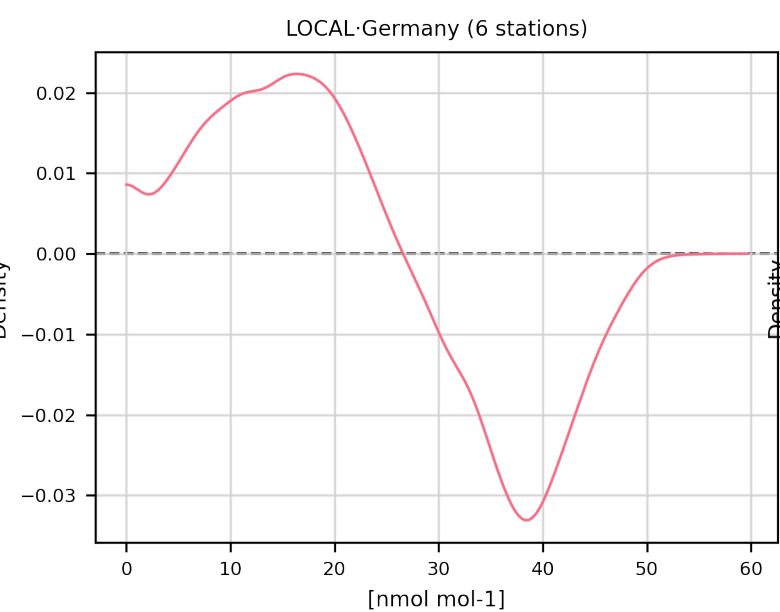
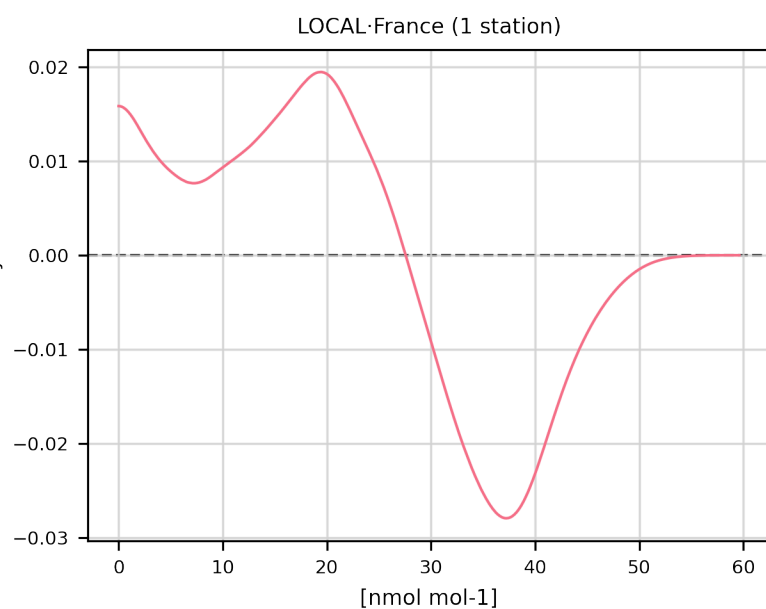
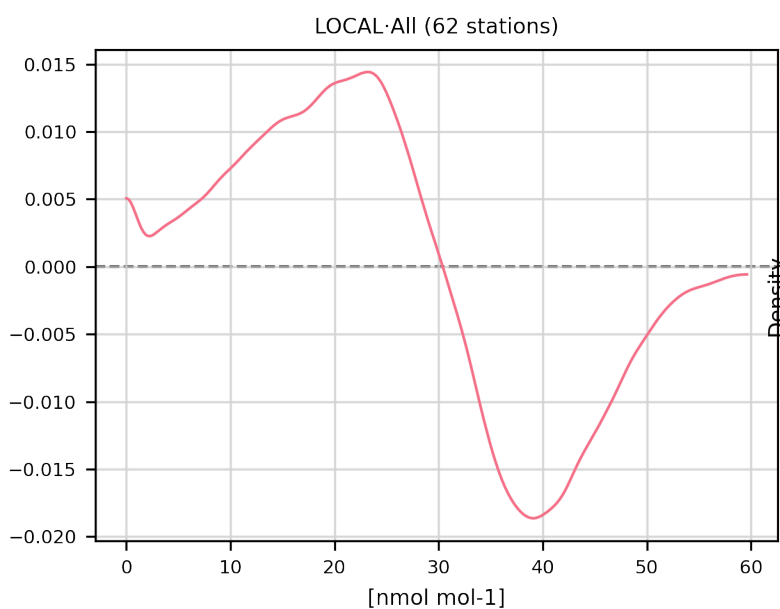


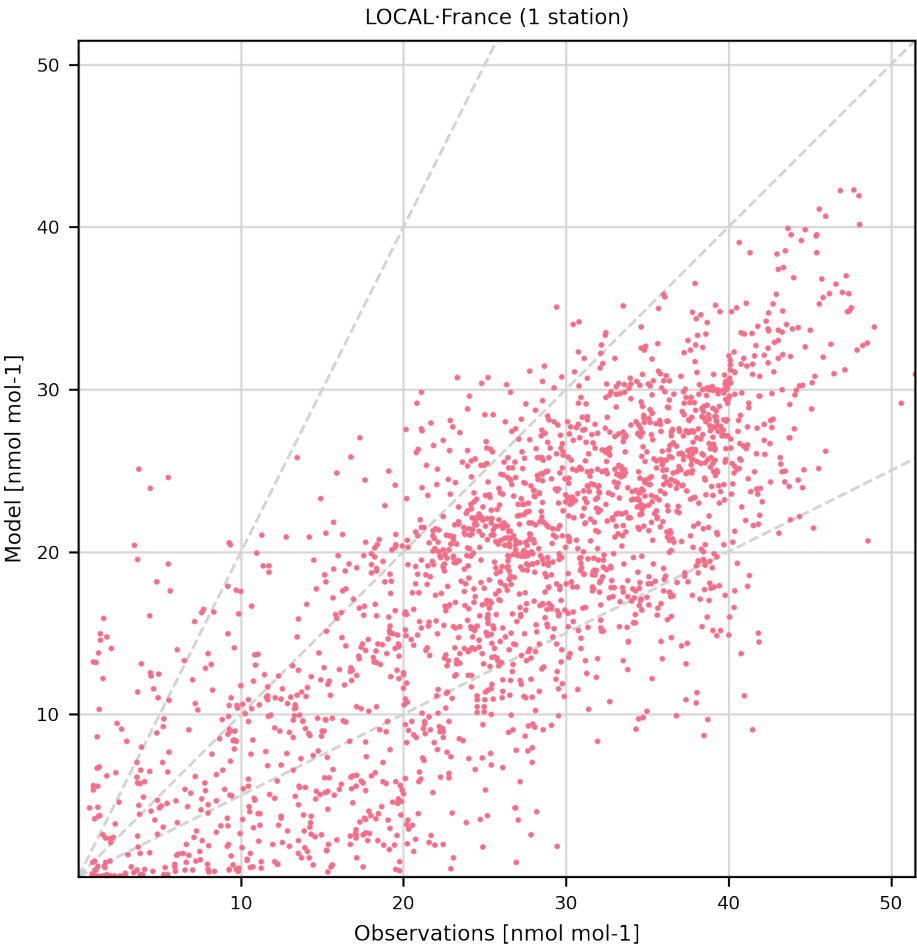
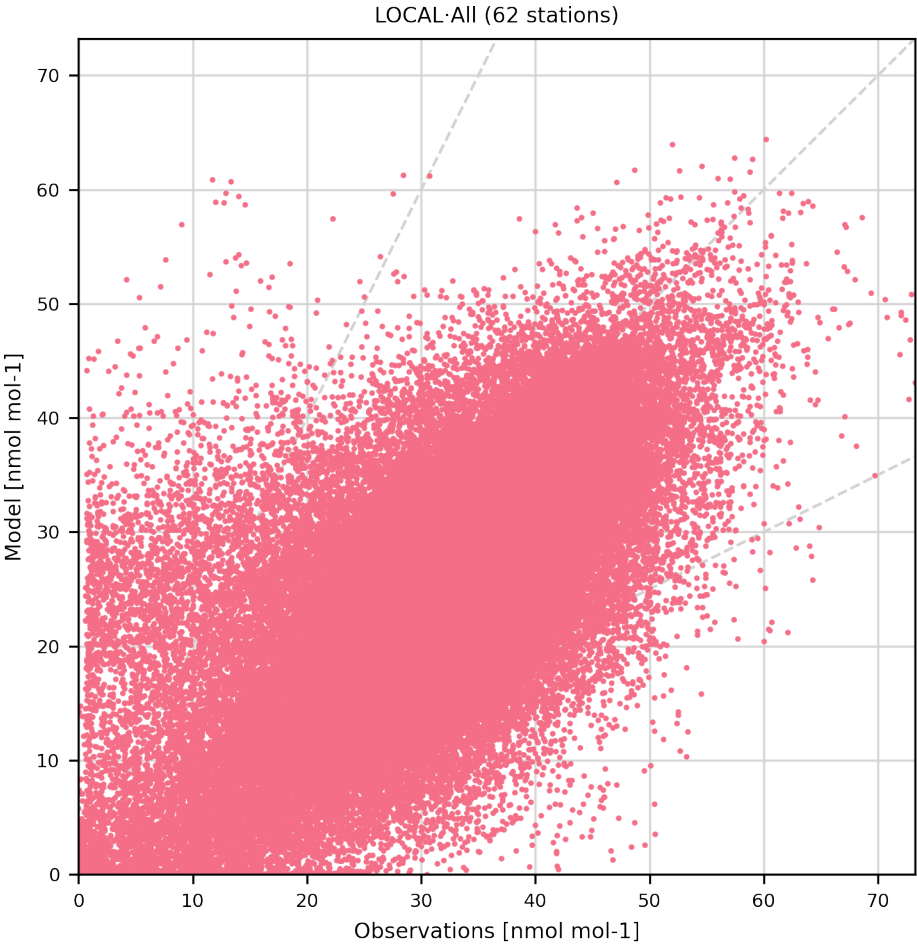
LOCAL·Spain (13 stations)

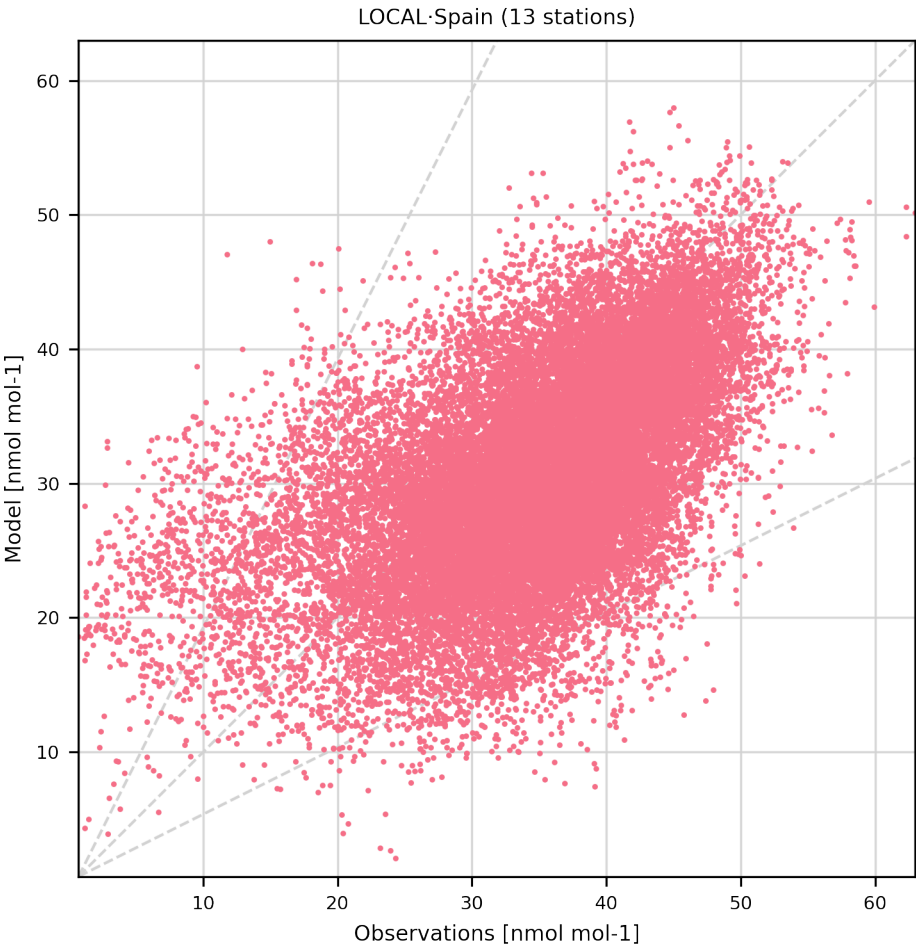
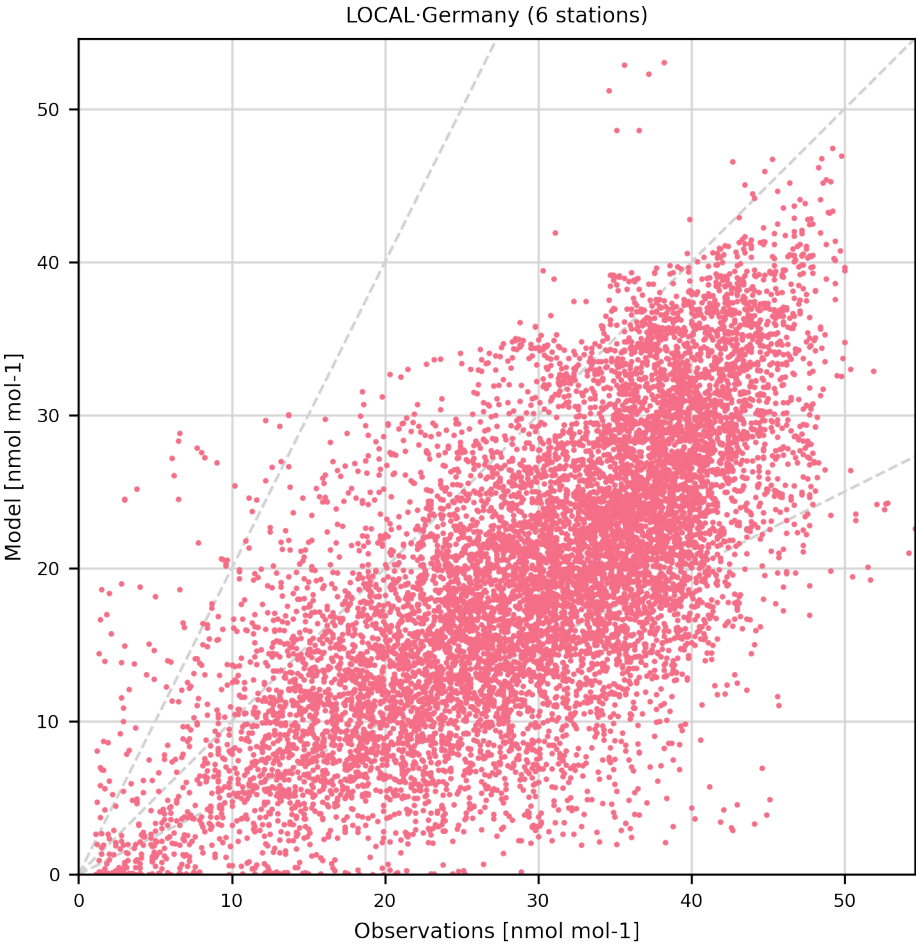


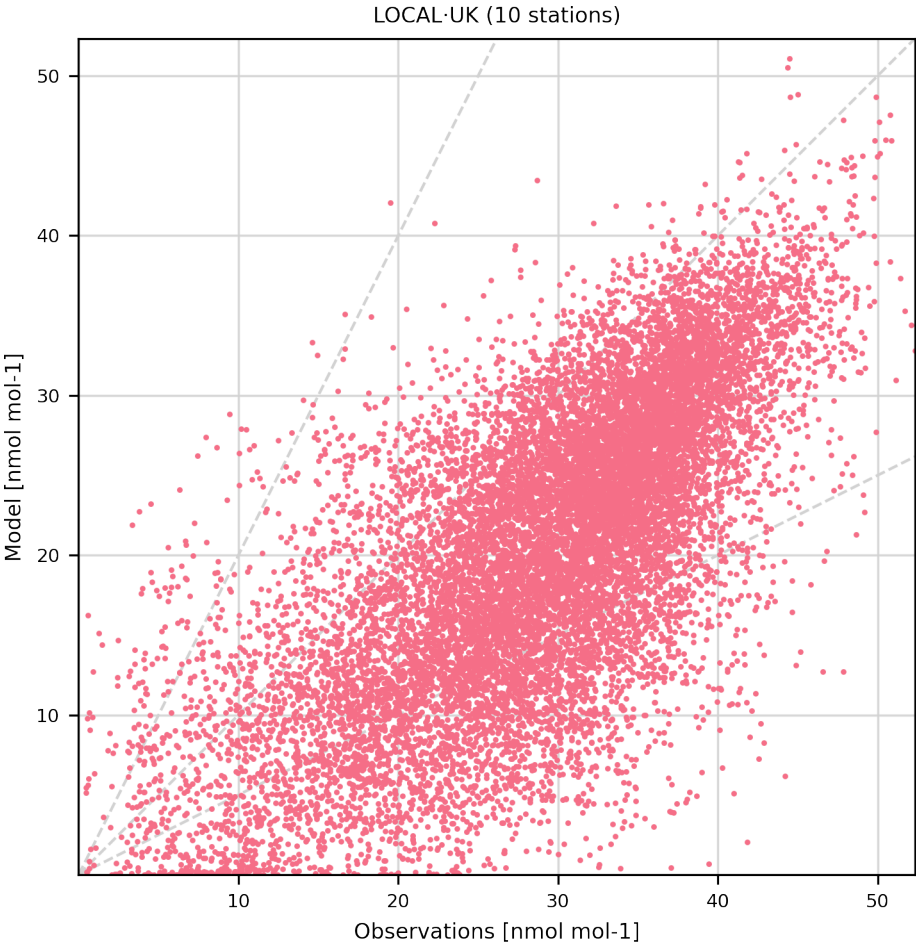




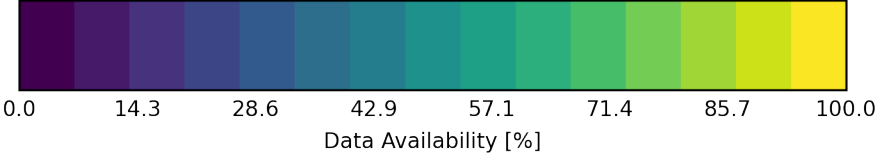




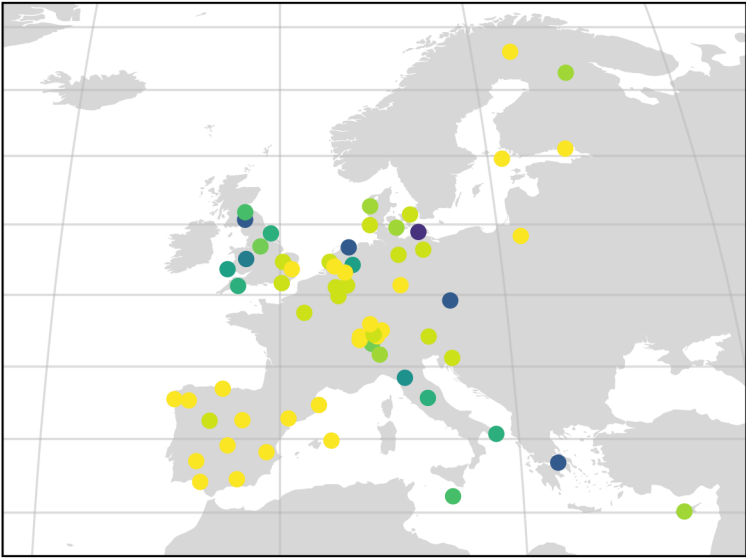




Map Data Availability (Summary)
EBAS|sconcn02



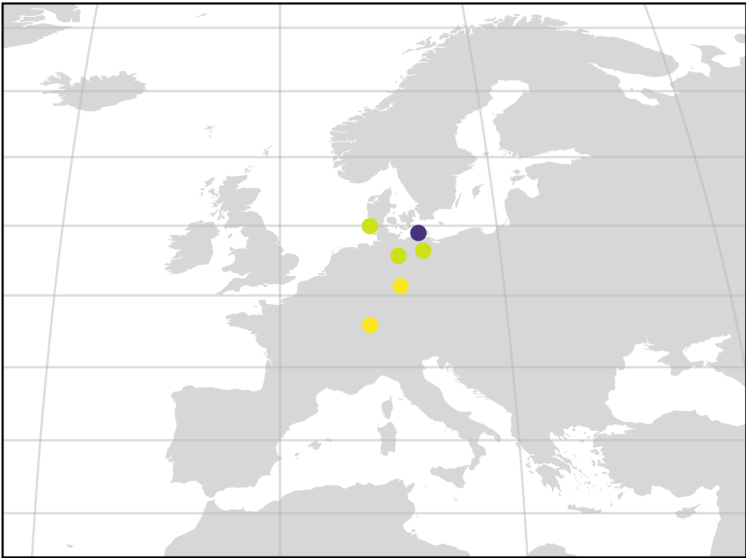
observations
LOCAL·All (62 stations)



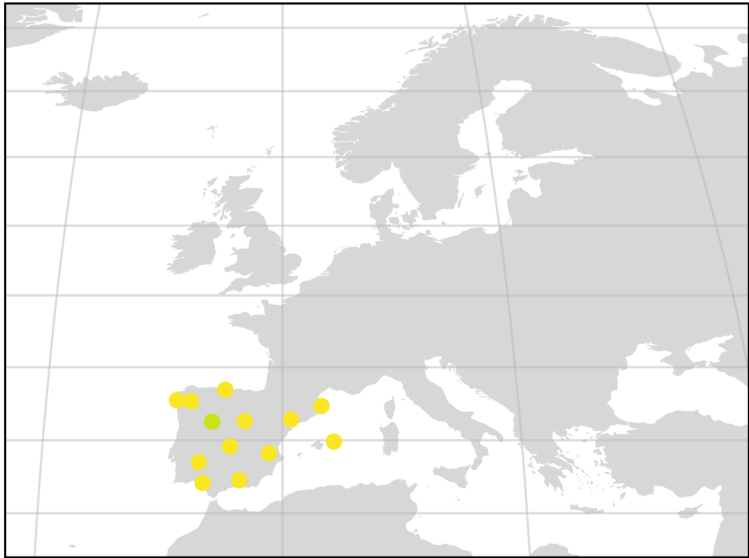
observations
LOCAL·France (1 station)



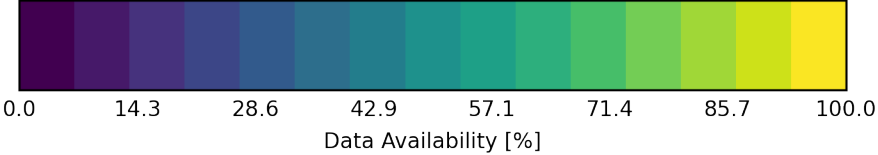
observations
LOCAL·Germany (6 stations)



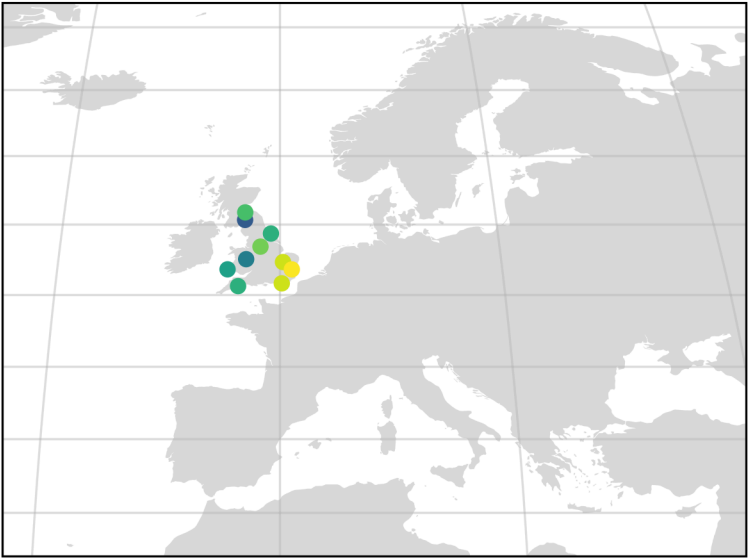
observations
LOCAL·Spain (13 stations)



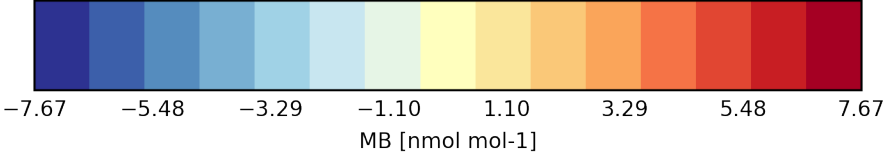
Map Data Availability (Summary)
EBAS|sconcn02



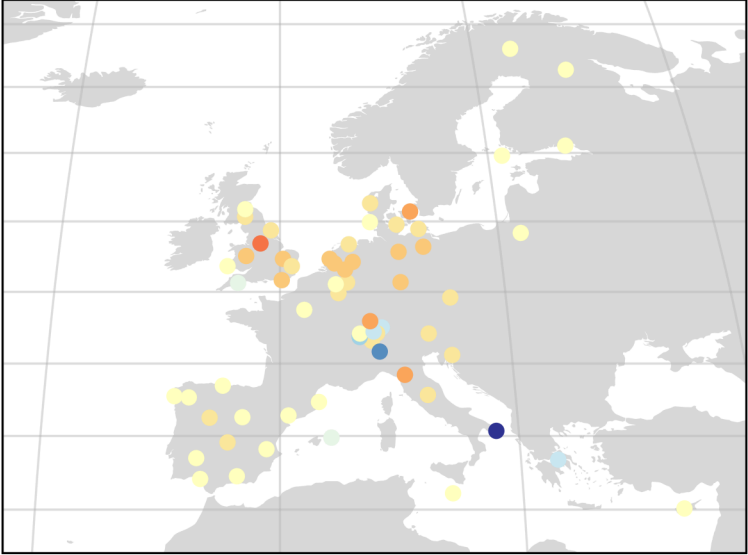
observations
LOCAL·UK (10 stations)



Map MB (Summary)
EBAS|sconchno2



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



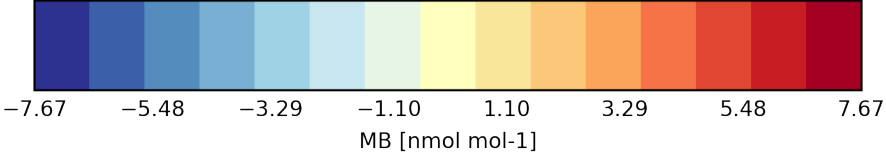
MONARCH
LOCAL·Germany (6 stations)



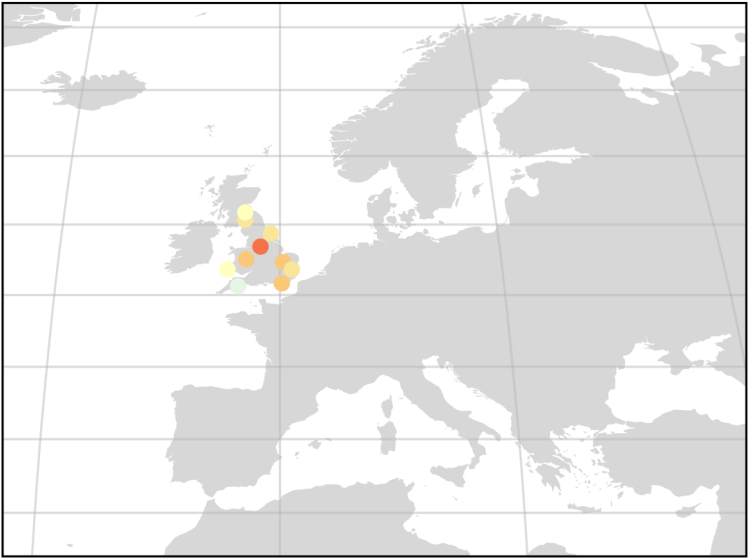
MONARCH
LOCAL·Spain (13 stations)



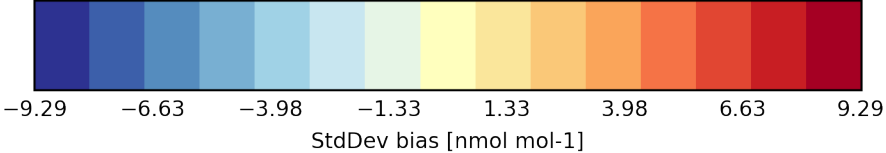
Map MB (Summary)
EBAS|sconchno2



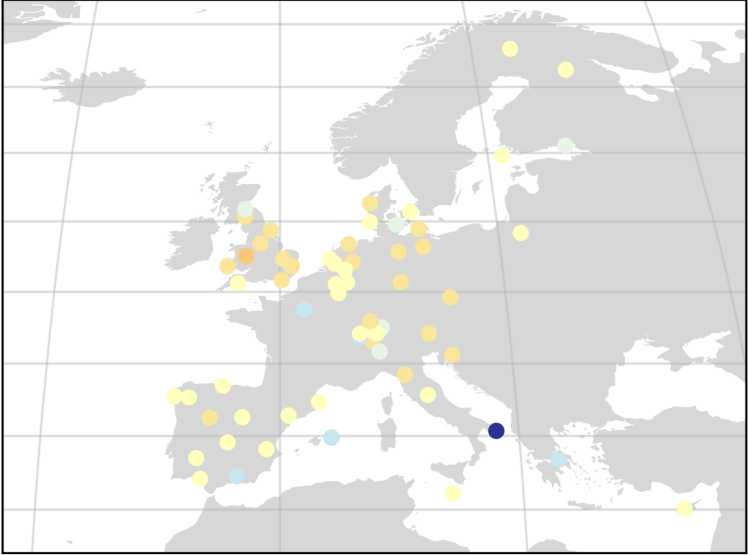
MONARCH
LOCAL·UK (10 stations)



Map StdDev bias (Summary)
EBAS|sconcnO2



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



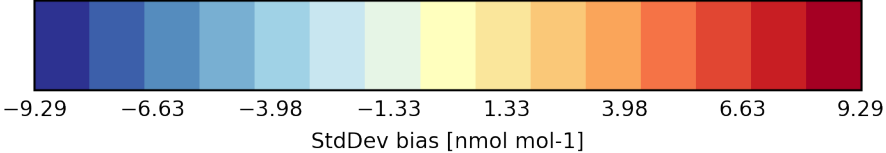
MONARCH
LOCAL·Germany (6 stations)



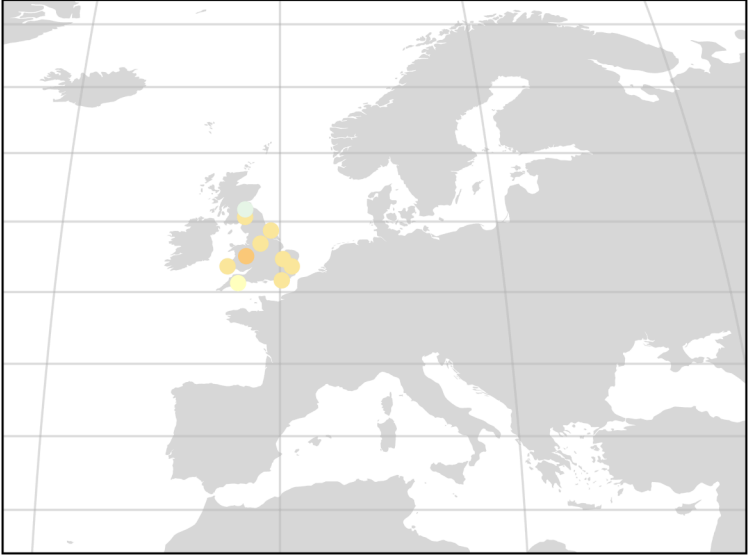
MONARCH
LOCAL·Spain (13 stations)



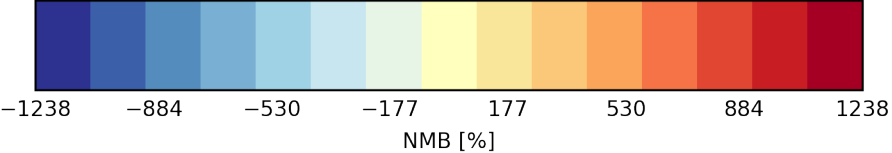
Map StdDev bias (Summary)
EBAS|sconcnO2



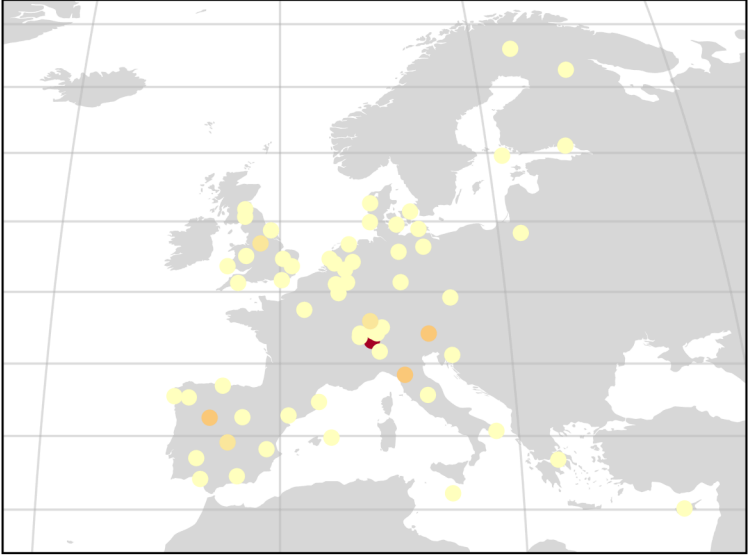
MONARCH
LOCAL·UK (10 stations)



Map NMB (Summary)
EBAS|sconcn02



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



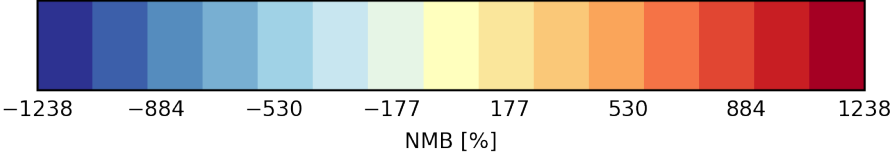
MONARCH
LOCAL·Germany (6 stations)



MONARCH
LOCAL·Spain (13 stations)



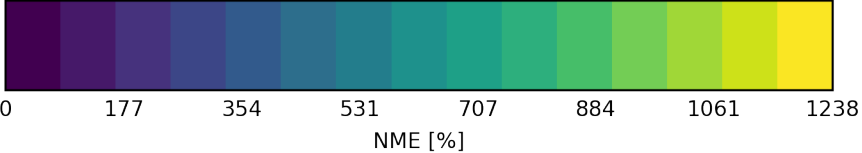
Map NMB (Summary)
EBAS|sconcn02



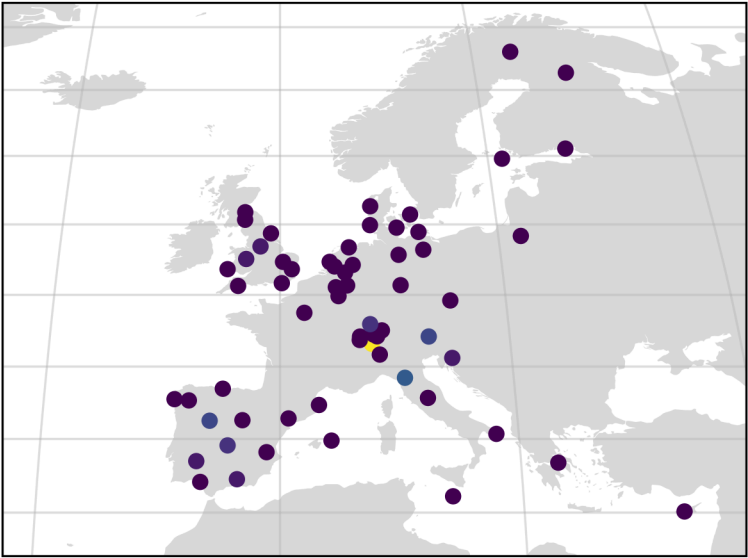
MONARCH
LOCAL-UK (10 stations)



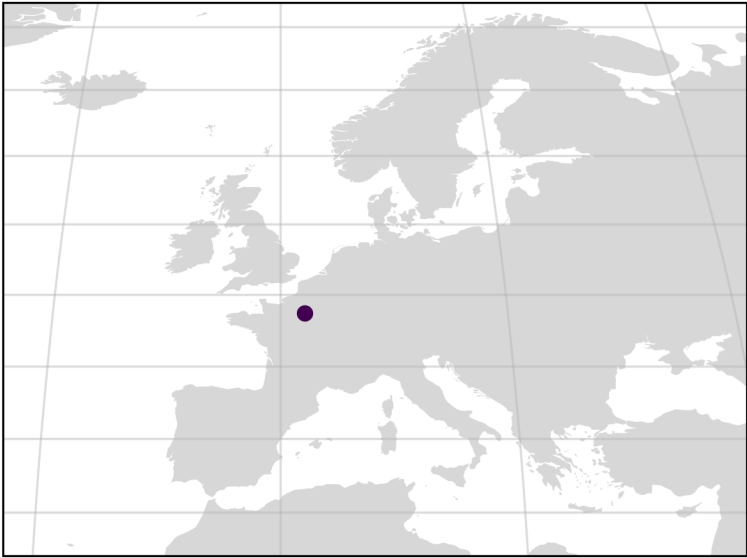
Map NME (Summary)
EBAS|sconchno2



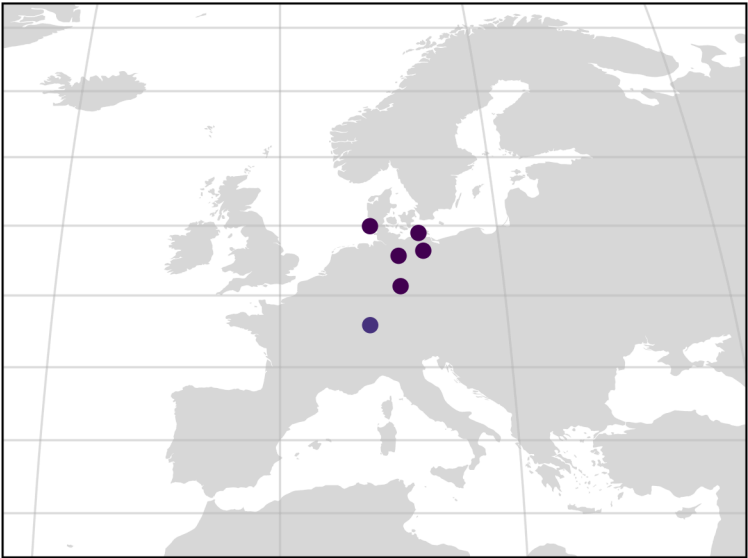
MONARCH
LOCAL·All (62 stations)



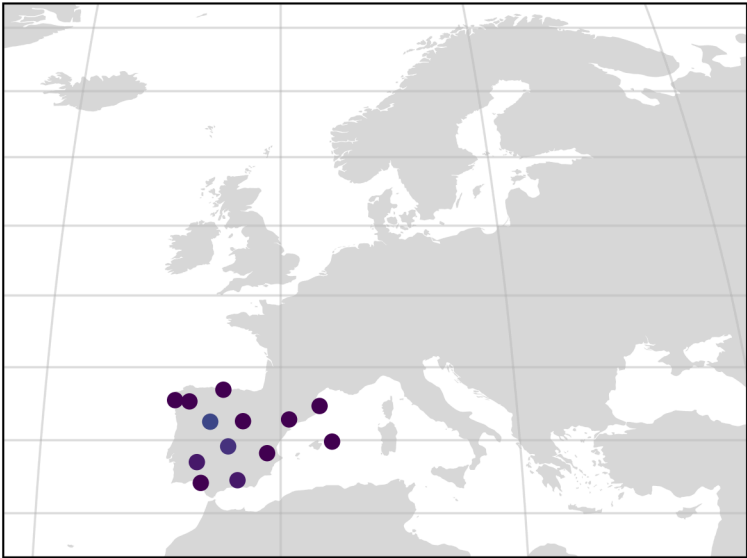
MONARCH
LOCAL·France (1 station)



MONARCH
LOCAL·Germany (6 stations)



MONARCH
LOCAL·Spain (13 stations)

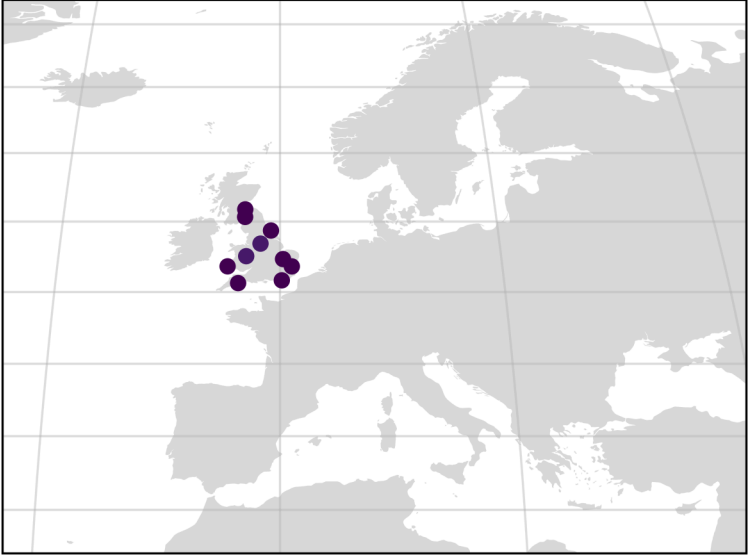


Map NME (Summary)
EBAS|sconchno2

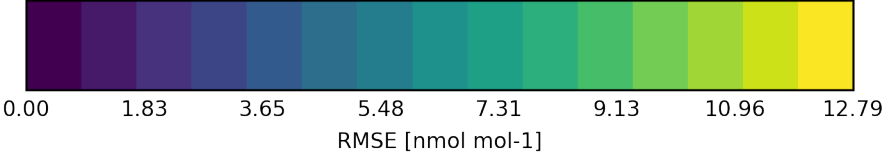


NME [%]

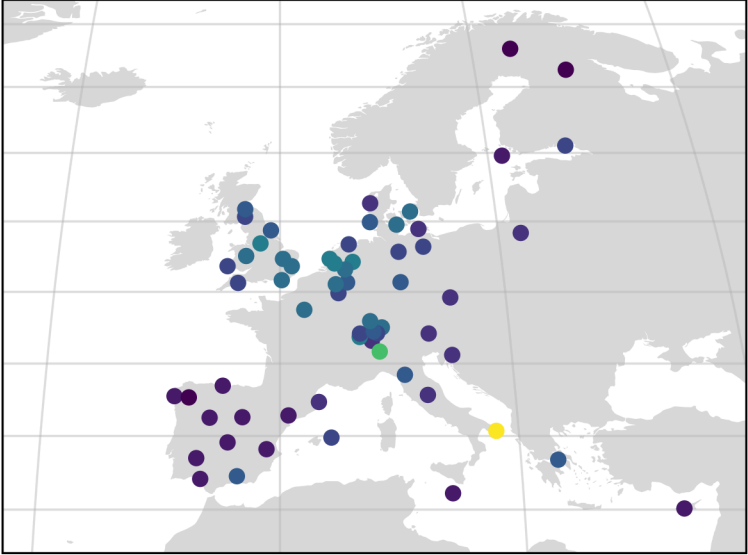
MONARCH
LOCAL-UK (10 stations)



Map RMSE (Summary)
EBAS|sconcnO2



MONARCH
LOCAL·All (62 stations)



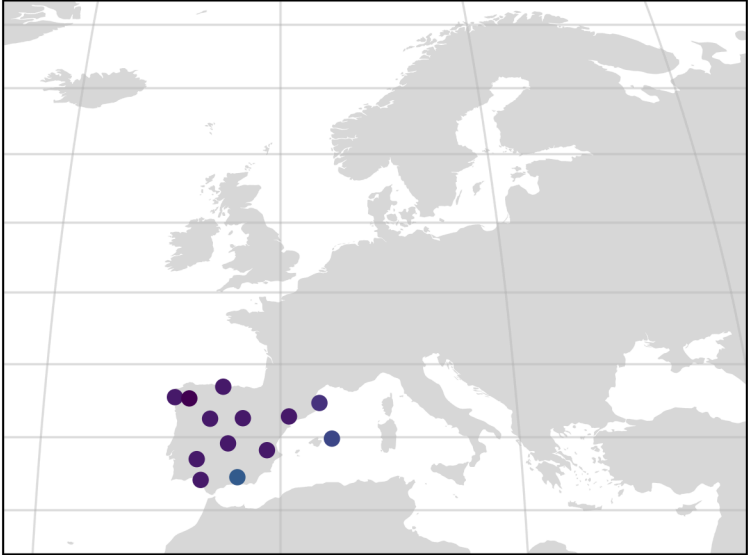
MONARCH
LOCAL·France (1 station)



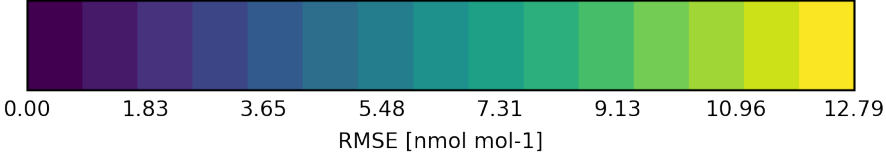
MONARCH
LOCAL·Germany (6 stations)



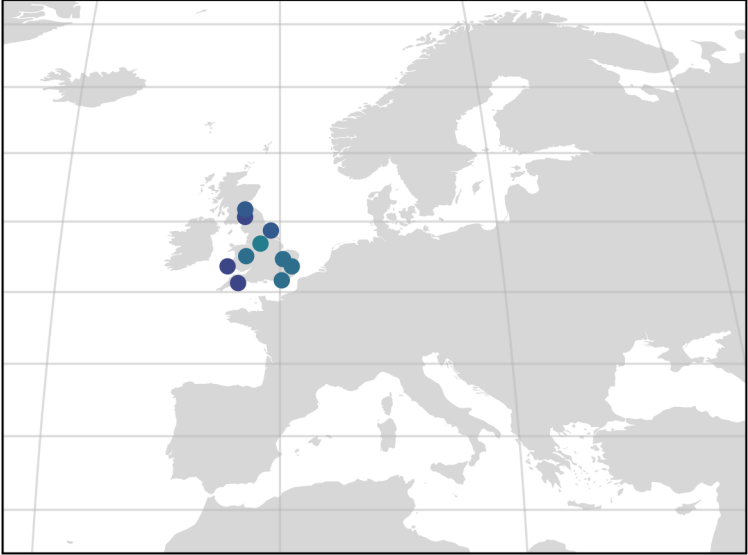
MONARCH
LOCAL·Spain (13 stations)



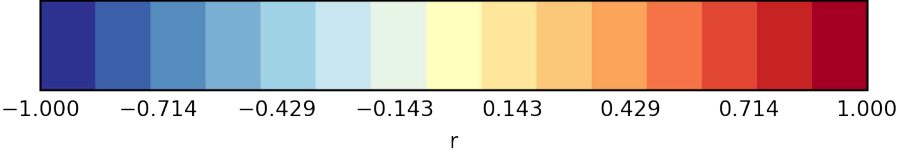
Map RMSE (Summary)
EBAS|sconcn02



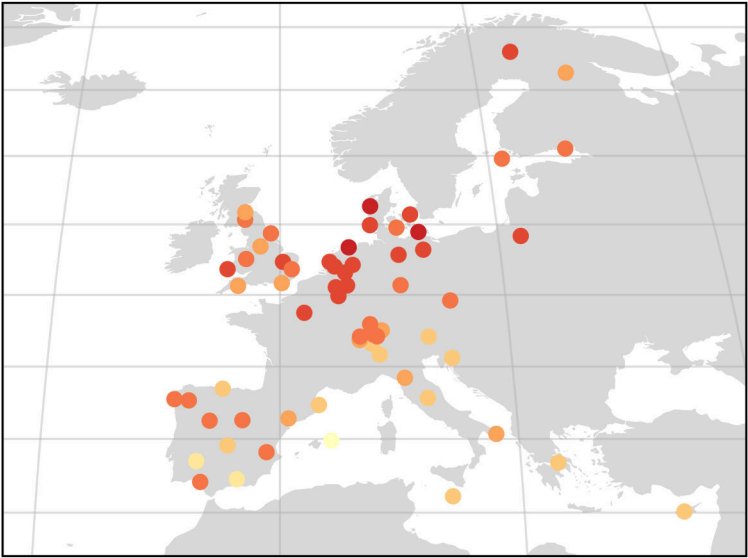
MONARCH
LOCAL-UK (10 stations)



Map r (Summary)
EBAS|sconcn02



MONARCH
LOCAL·All (62 stations)



MONARCH
LOCAL·France (1 station)



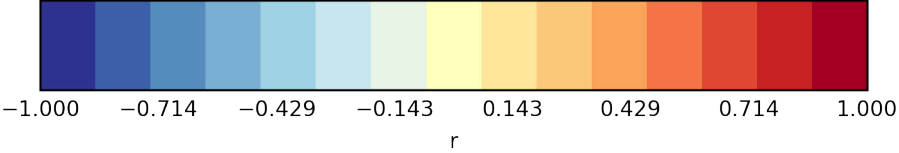
MONARCH
LOCAL·Germany (6 stations)



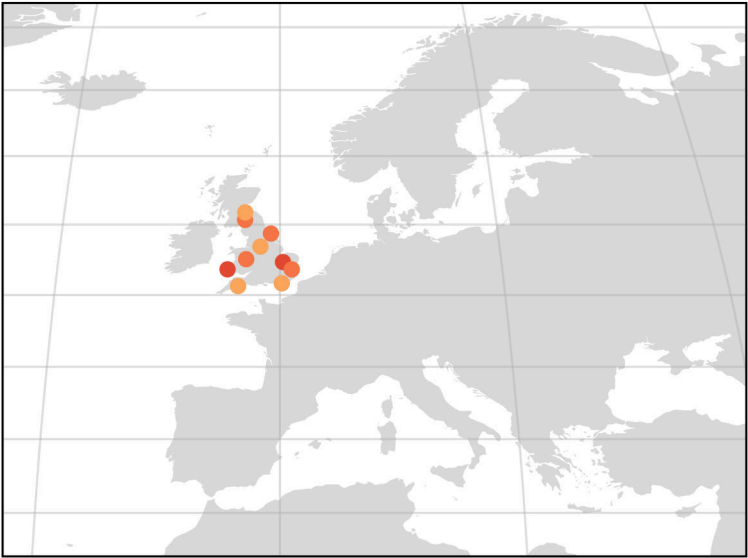
MONARCH
LOCAL·Spain (13 stations)

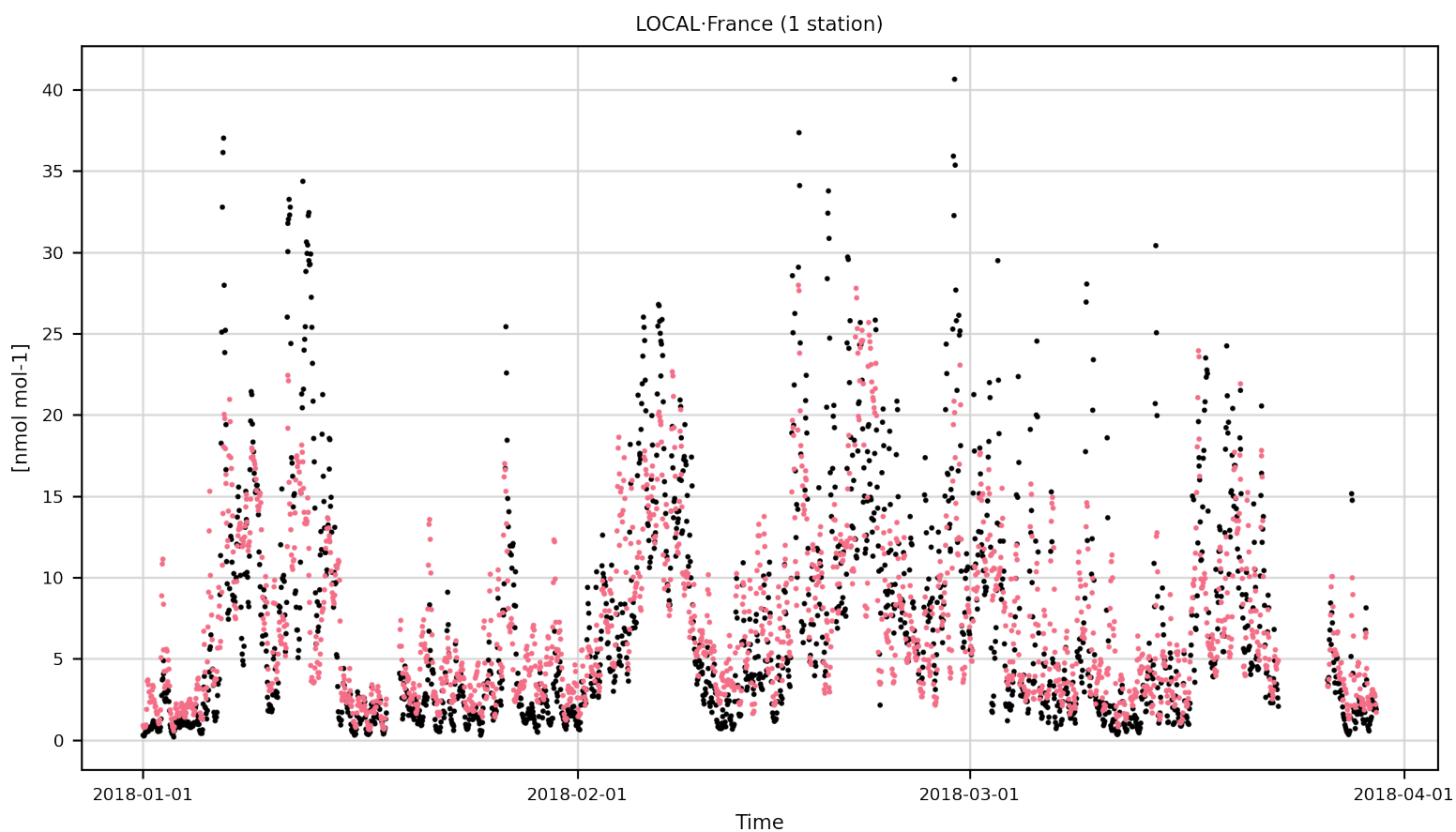
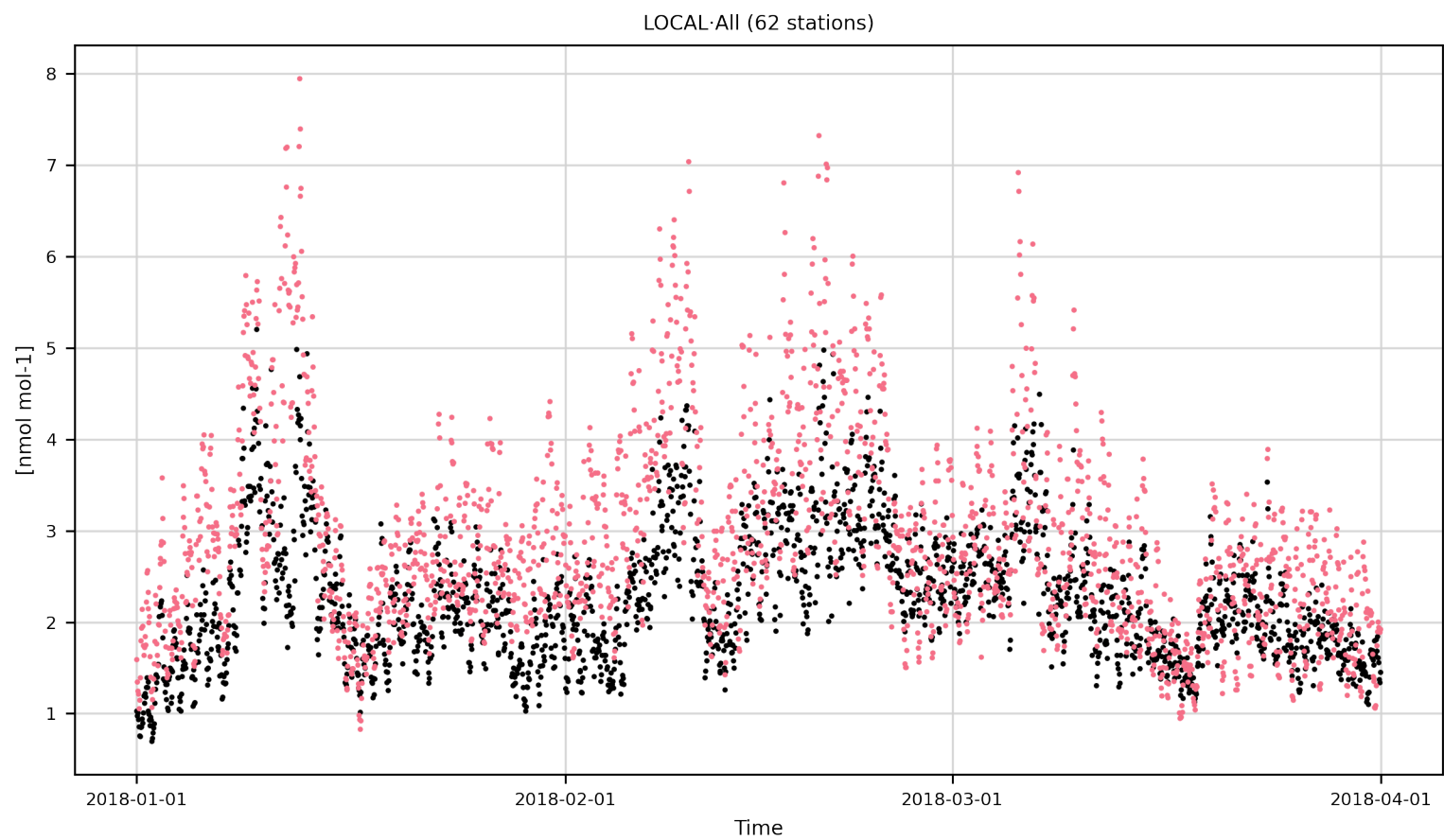


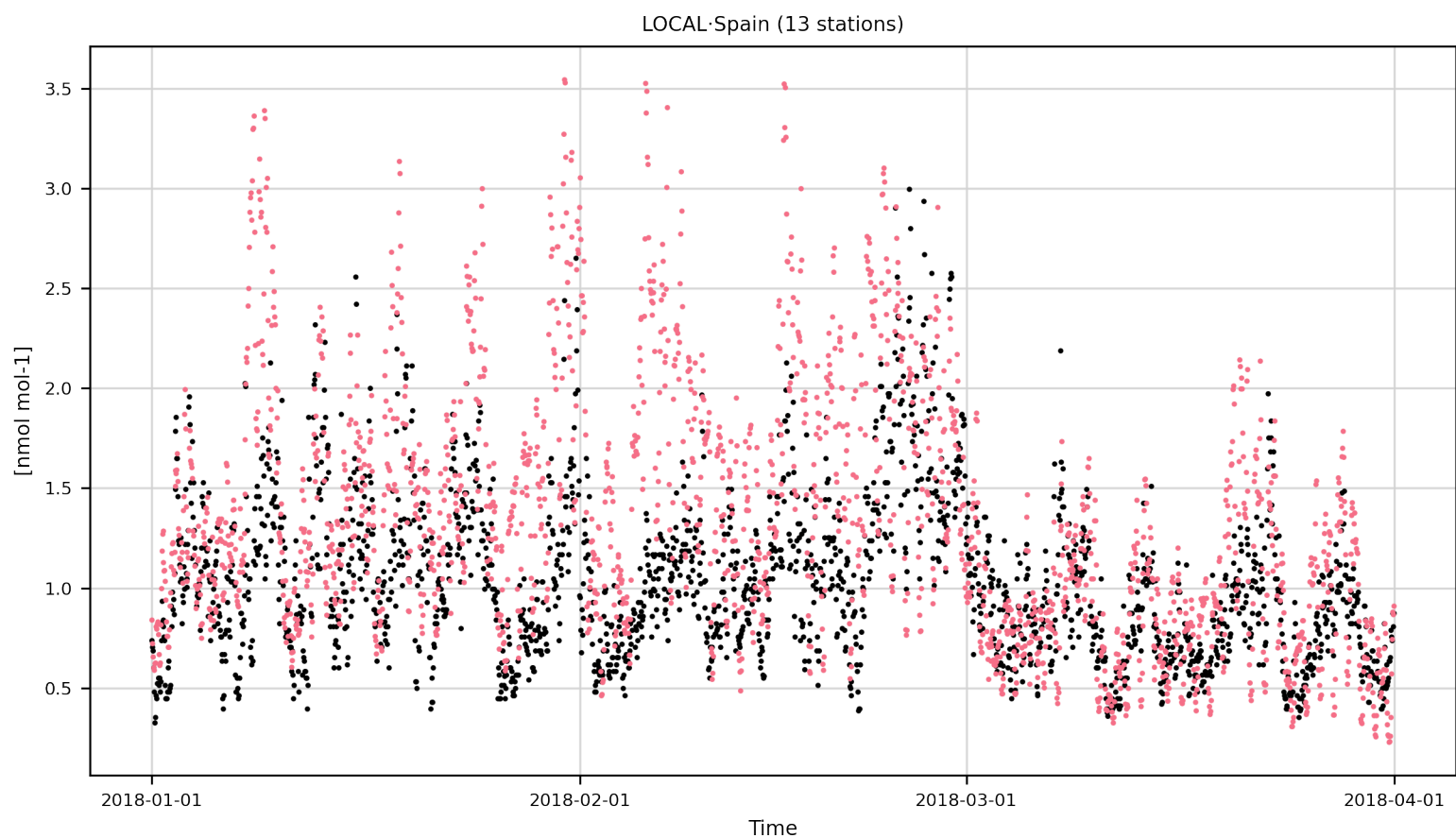
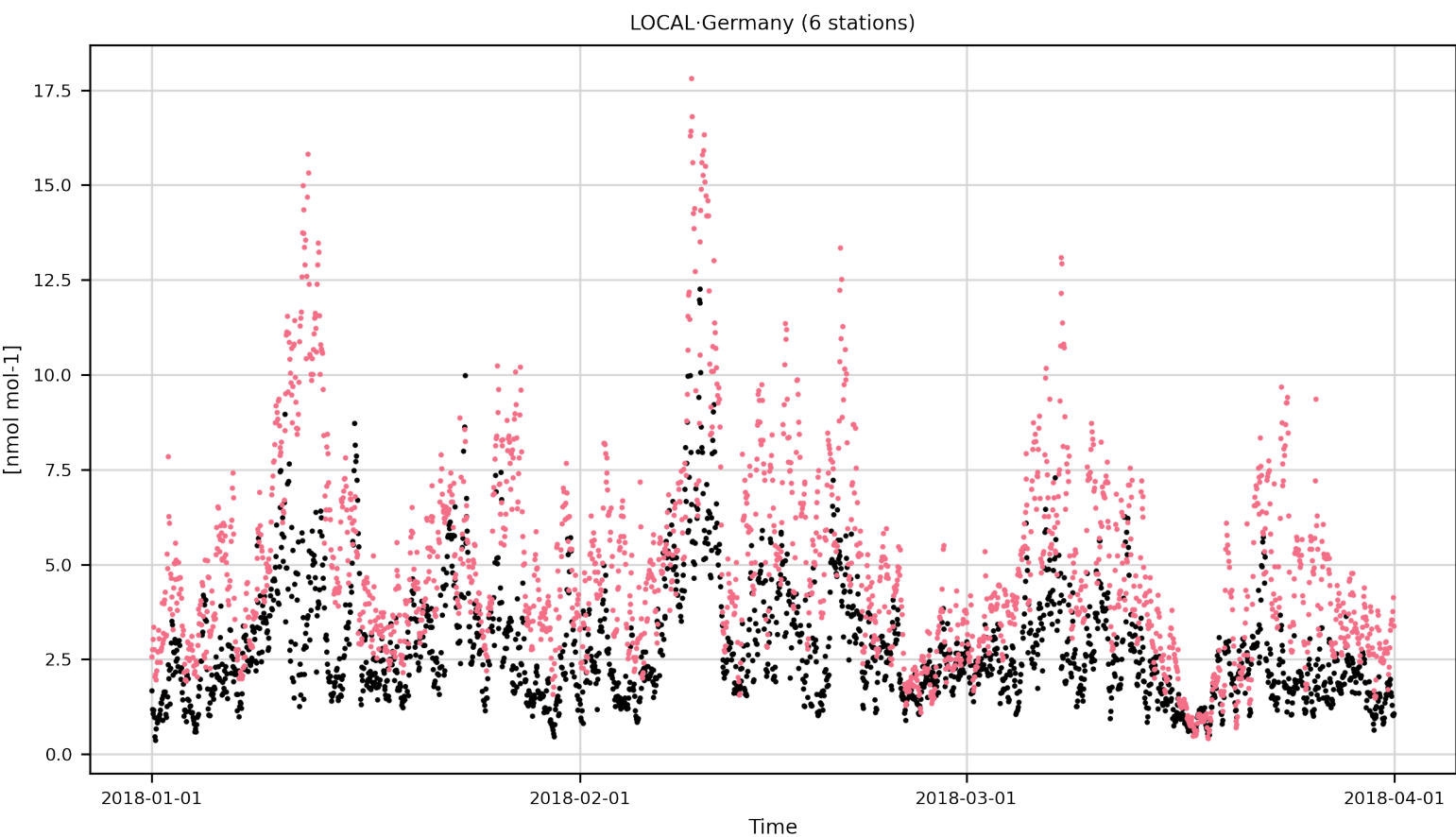
Map r (Summary)
EBAS|sconcn02

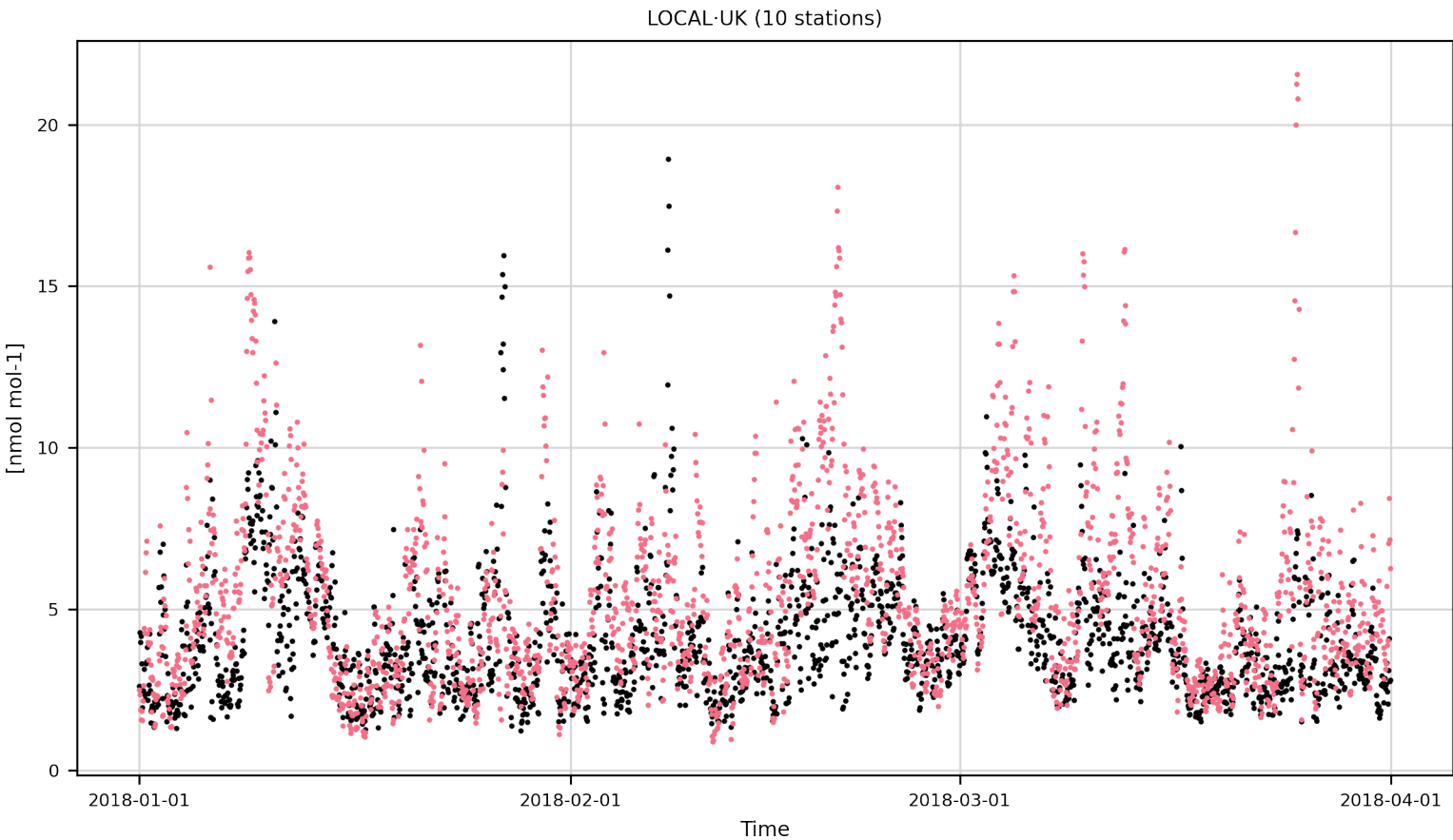


MONARCH
LOCAL·UK (10 stations)





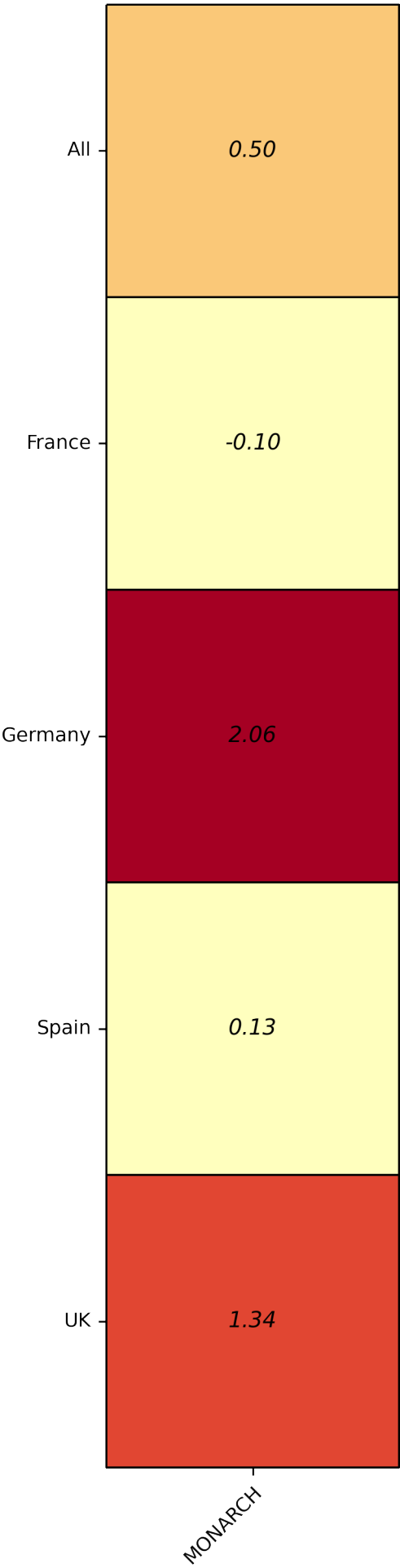
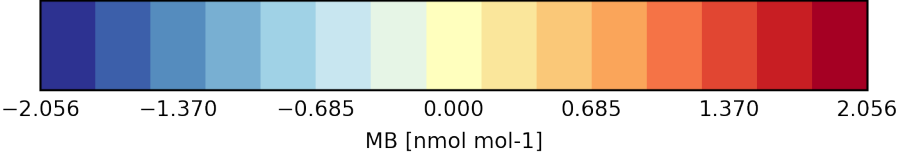




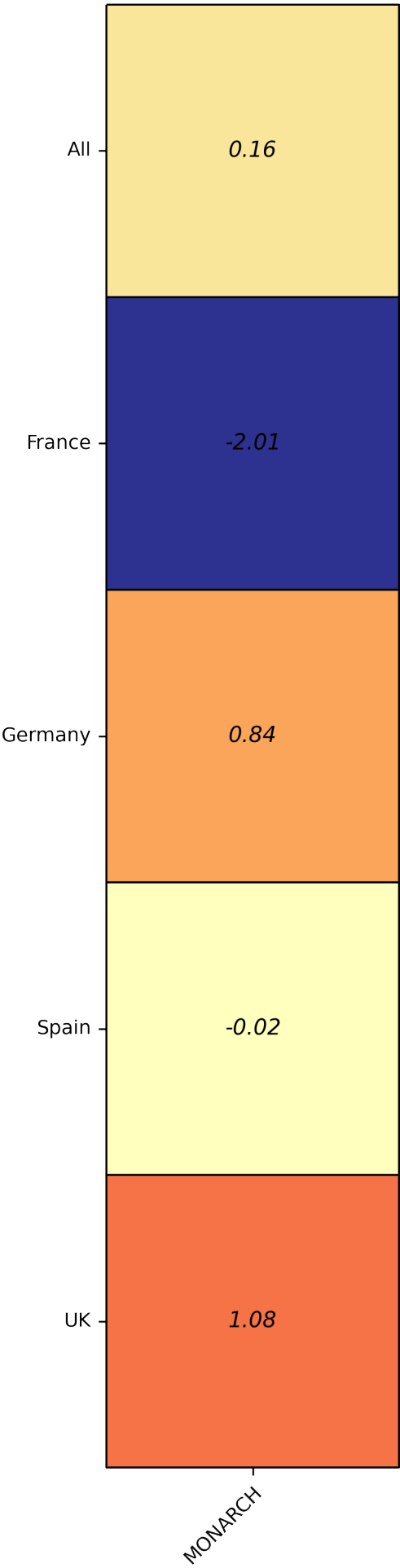
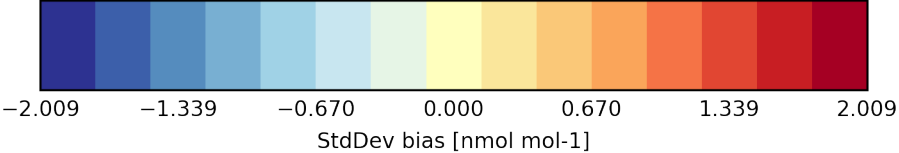
		Mean	StdDev	p5	Median	p95
All	observations	3.16	2.78	0.72	2.39	8.35
	MONARCH	3.71	2.86	0.87	3.00	9.38
France	observations	7.31	7.06	0.76	4.60	22.01
	MONARCH	7.21	5.06	1.67	5.63	17.38
Germany	observations	3.05	2.28	0.55	2.48	7.63
	MONARCH	5.13	3.25	1.33	4.20	11.79
Spain	observations	1.28	1.04	0.33	1.03	3.07
	MONARCH	1.37	1.01	0.38	1.11	3.33
UK	observations	4.39	3.18	1.34	3.26	10.54
	MONARCH	5.92	4.20	1.22	4.71	14.74

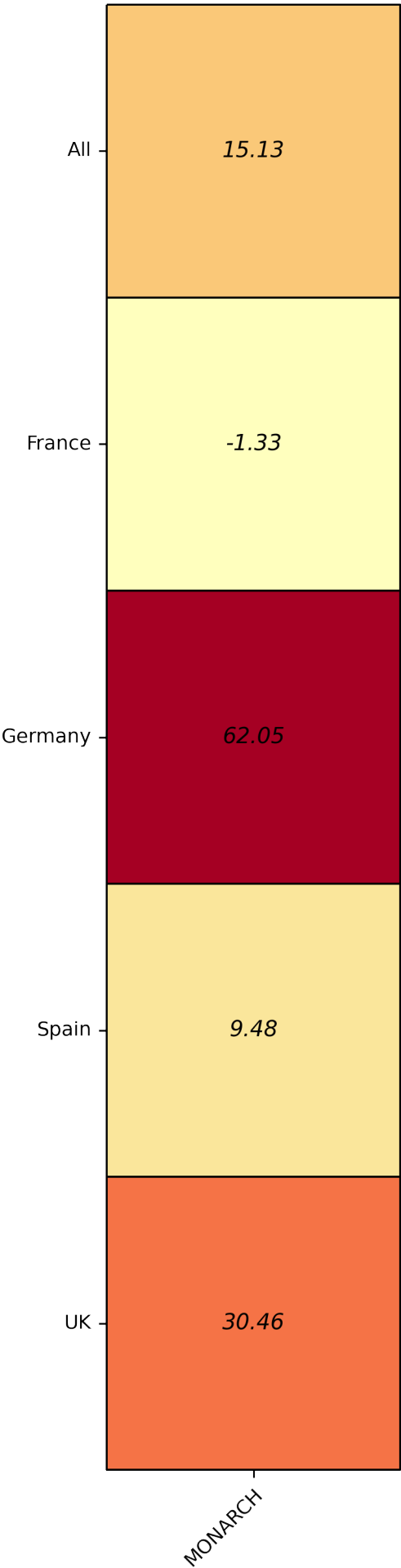
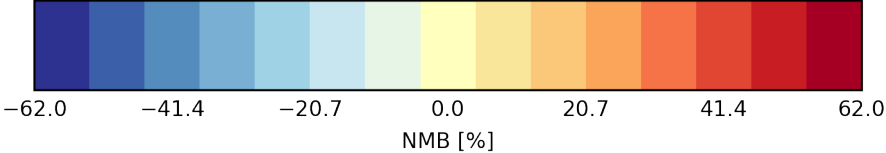
		Mean_bias	StdDev_bias	NMB	NME	RMSE	r
All	MONARCH	0.50	0.16	15.13	57.89	3.03	0.48
France	MONARCH	-0.10	-2.01	-1.33	43.36	4.84	0.73
Germany	MONARCH	2.06	0.84	62.05	71.81	3.35	0.61
Spain	MONARCH	0.13	-0.02	9.48	59.86	1.36	0.42
UK	MONARCH	1.34	1.08	30.46	58.78	4.13	0.48

Heatmap MB (Summary)
EBAS|sconcn02

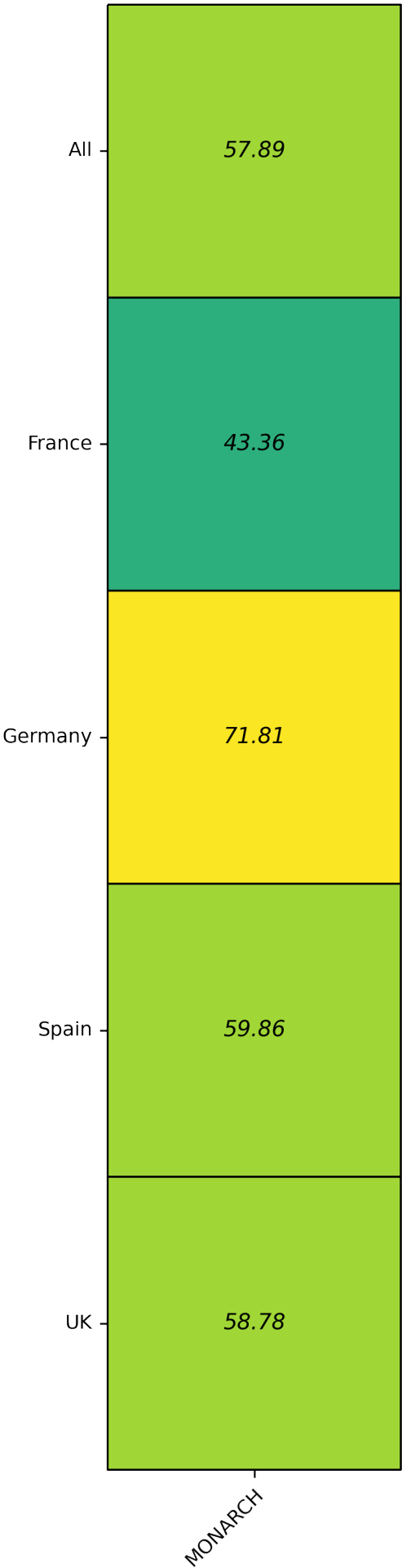
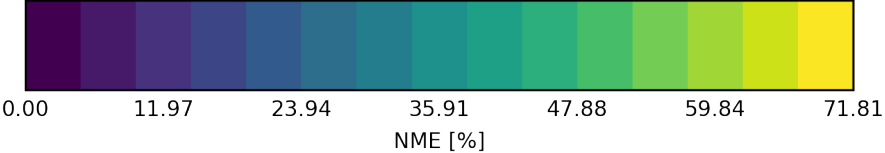


Heatmap StdDev bias (Summary)
EBAS|sconcn02

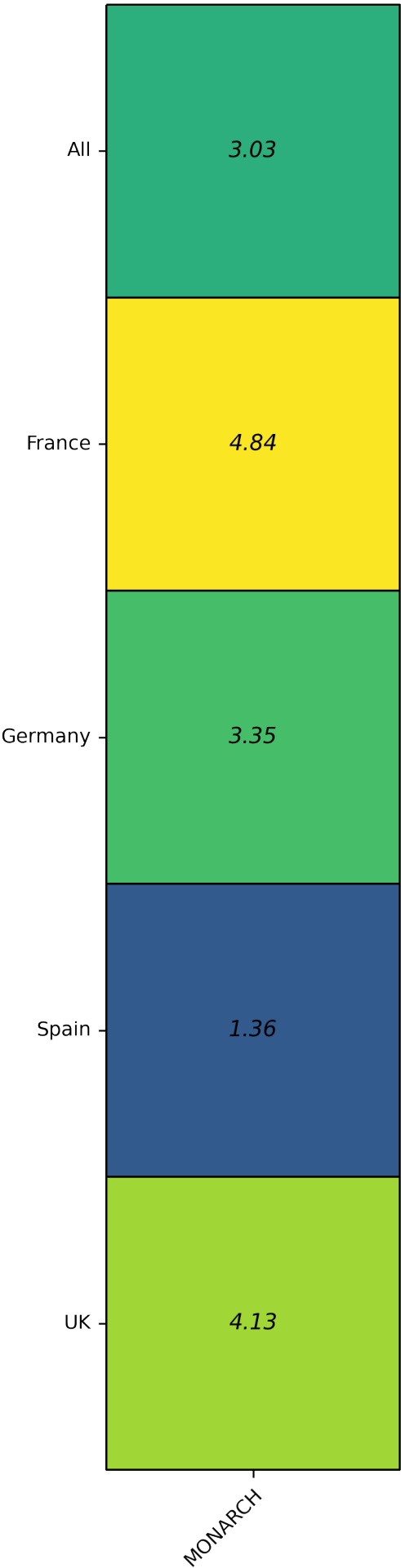
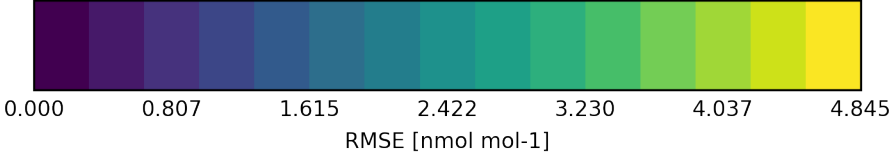




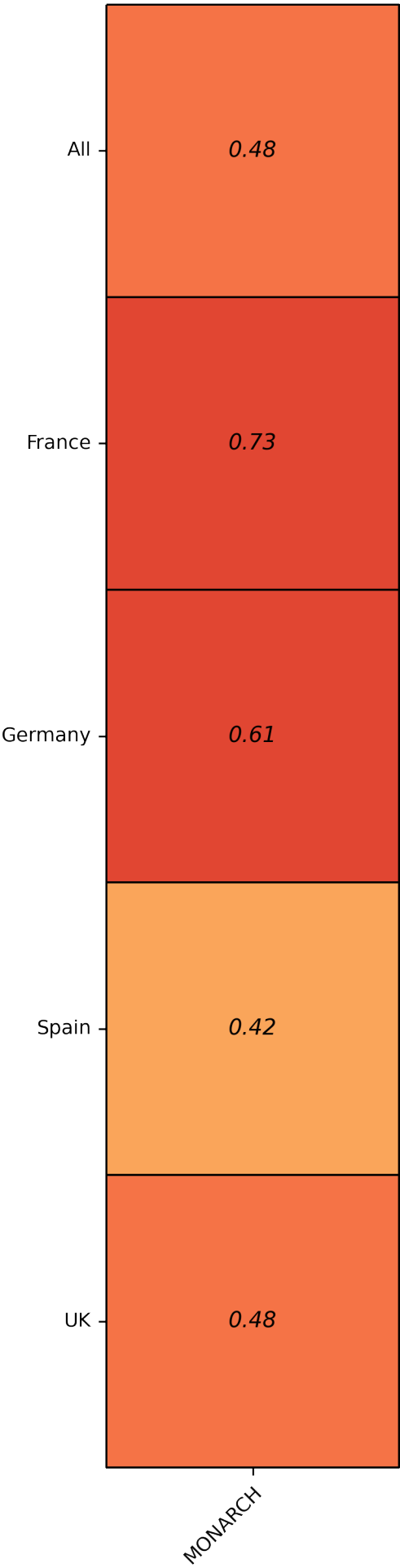
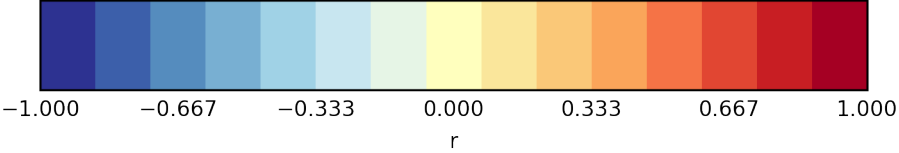
Heatmap NME (Summary)
EBAS|sconcn02



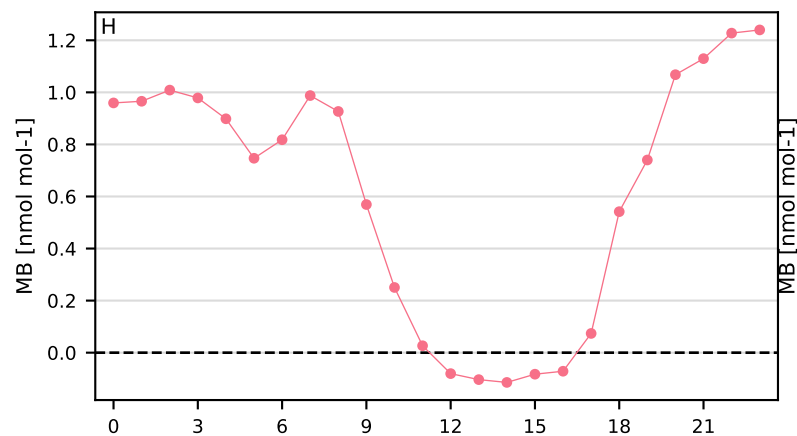
Heatmap RMSE (Summary)
EBAS|sconcn02



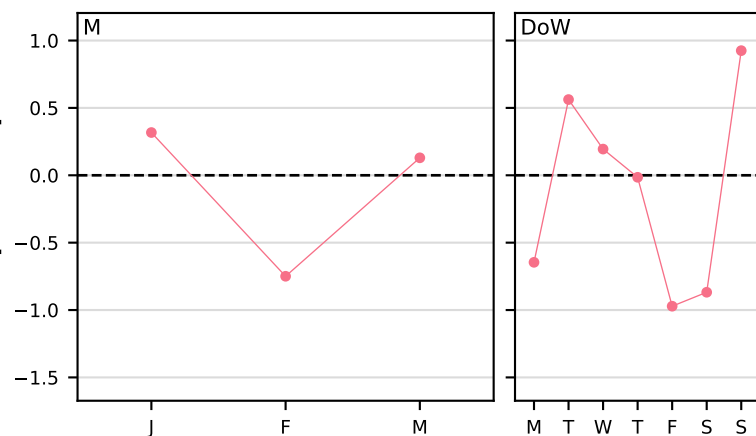
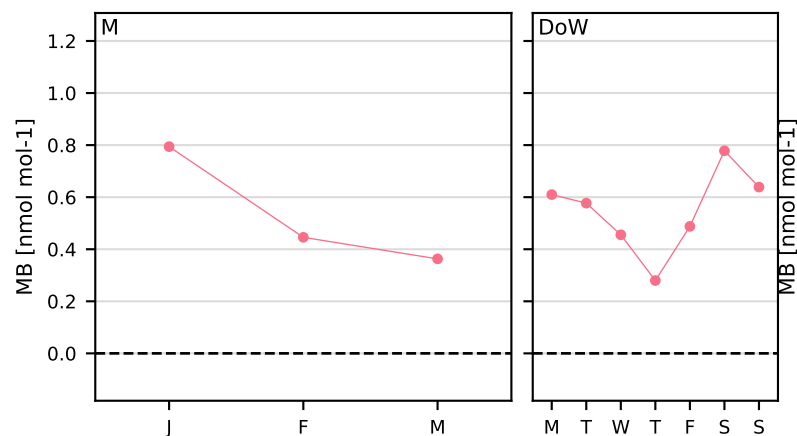
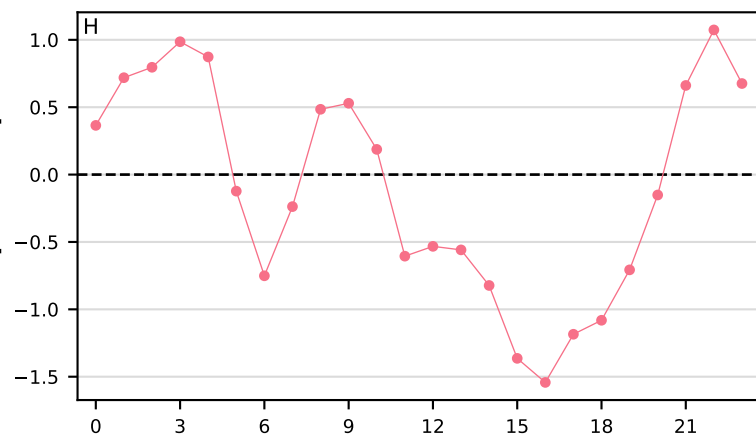
Heatmap r (Summary)
EBAS|sconcno2



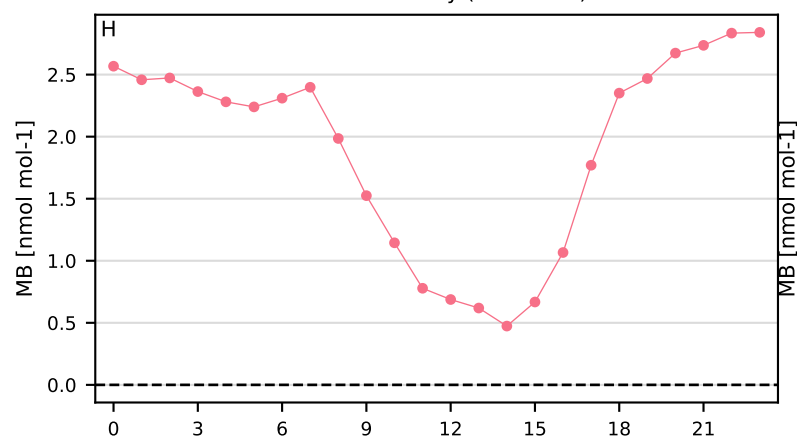
LOCAL·All (62 stations)



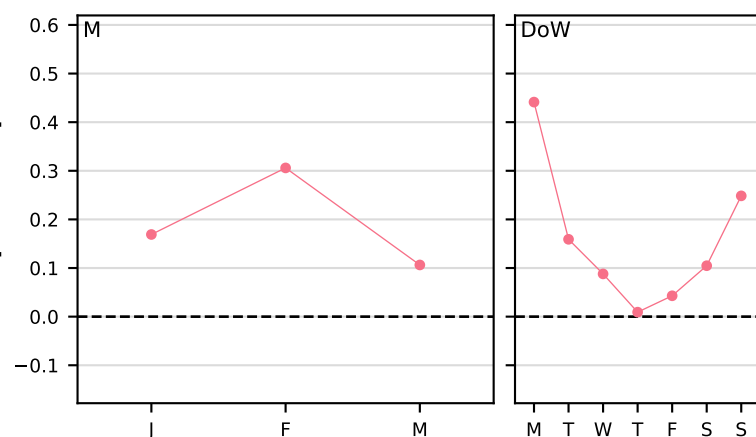
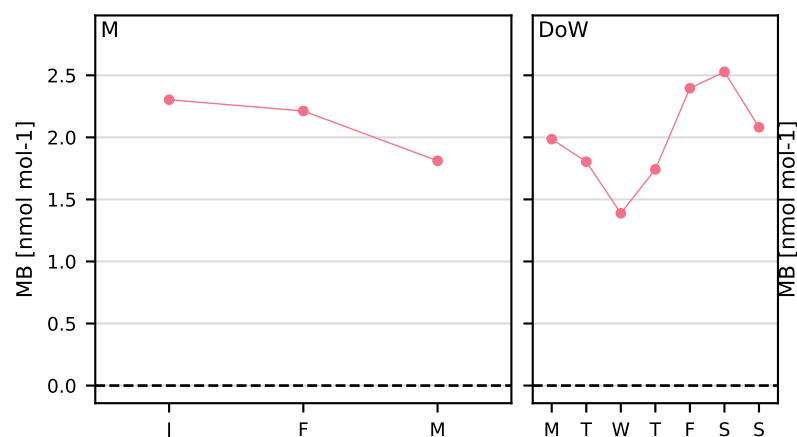
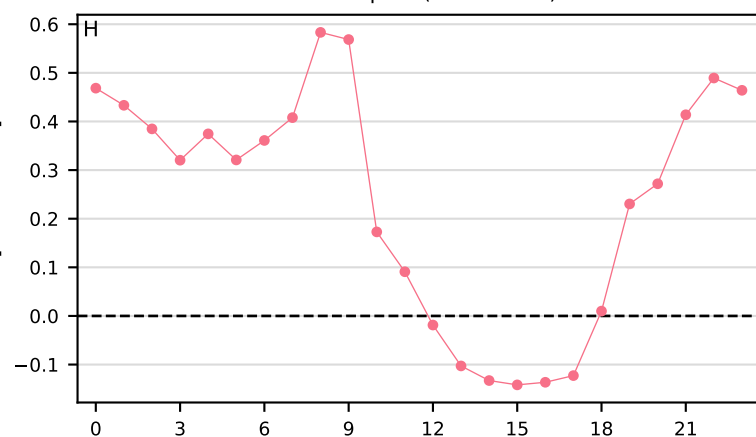
LOCAL·France (1 station)



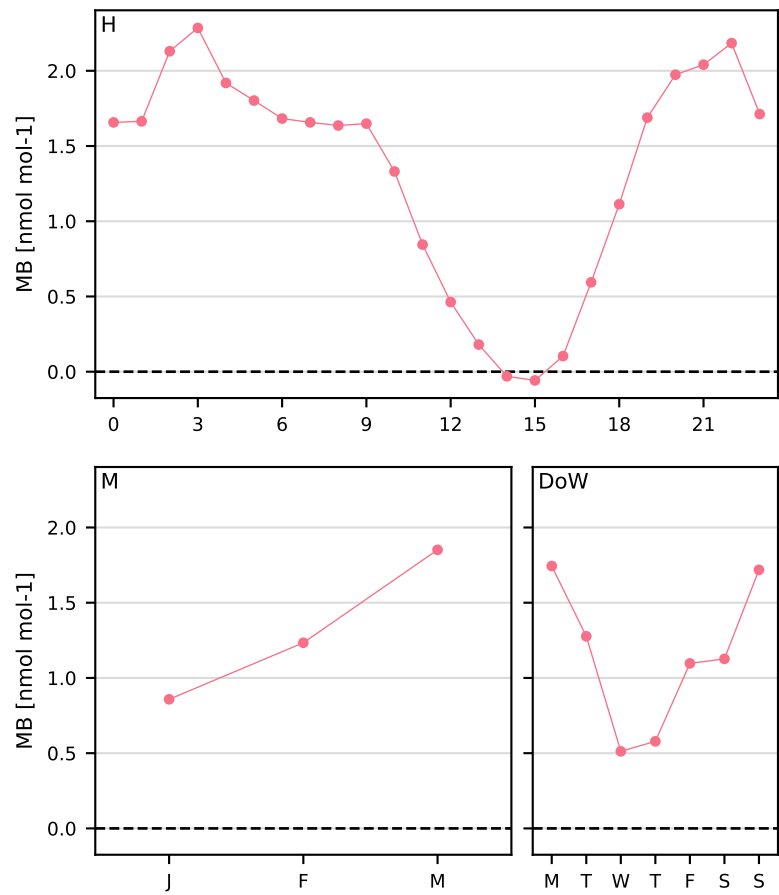
LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)

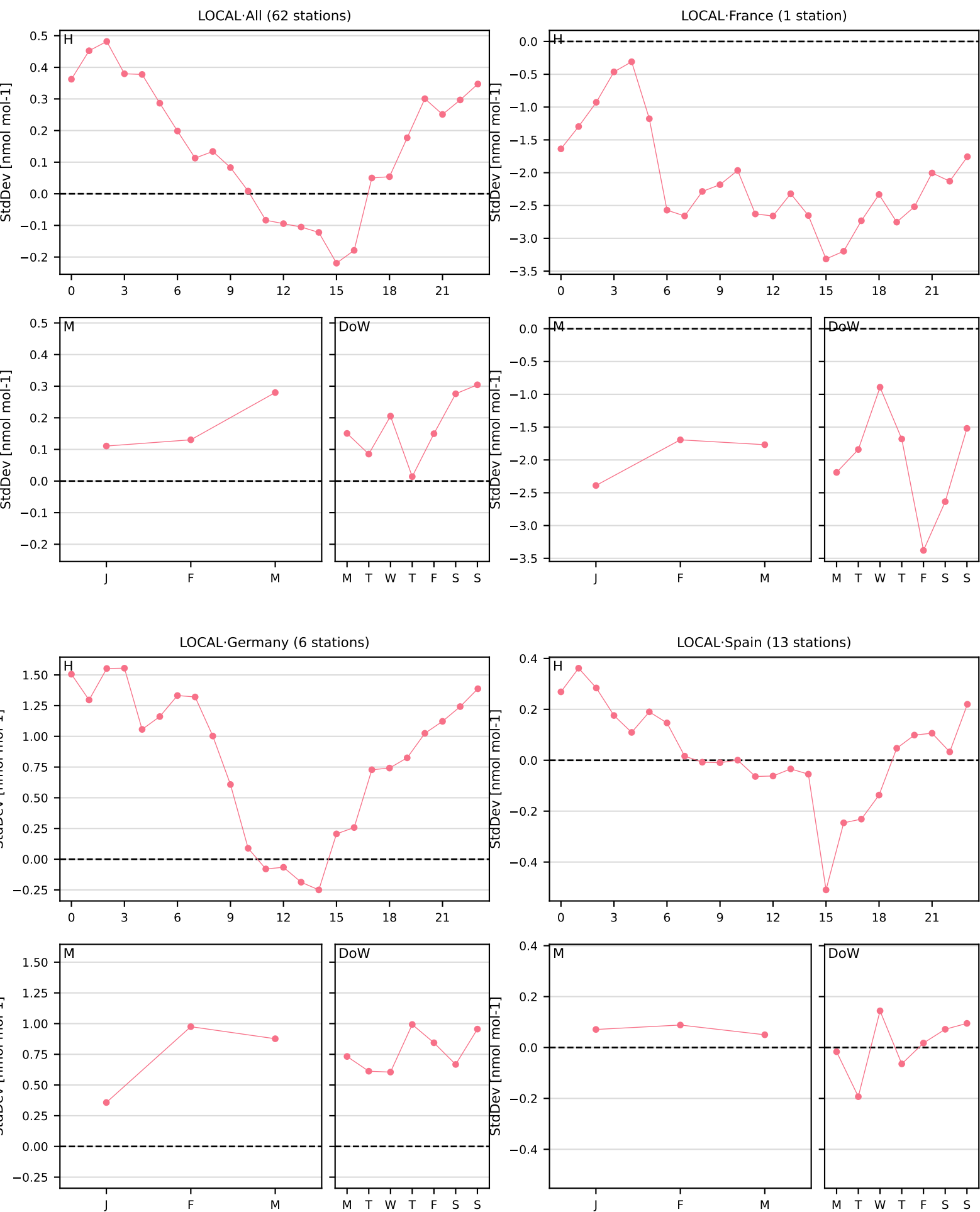


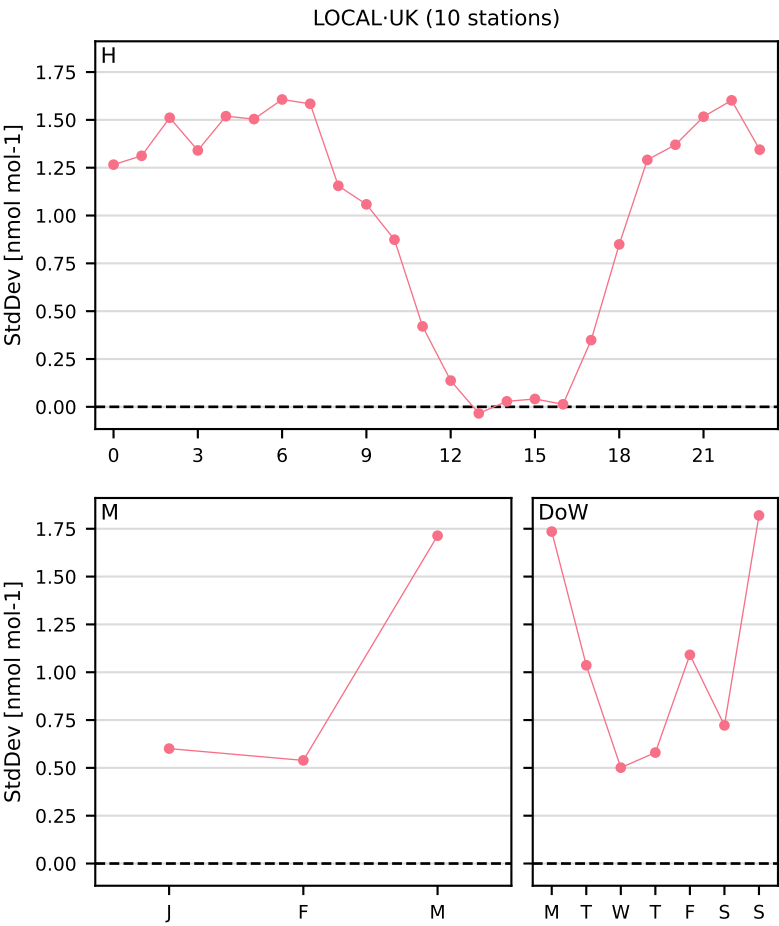
LOCAL·UK (10 stations)



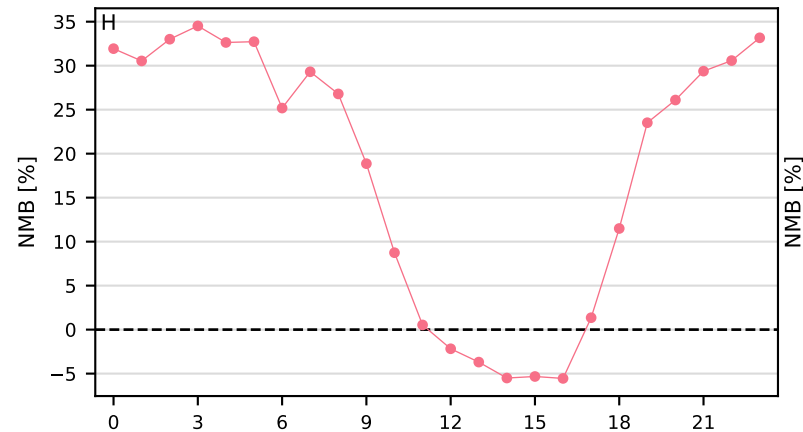
Periodic StdDev bias (Summary)
EBAS|sconcn02

MONARCH

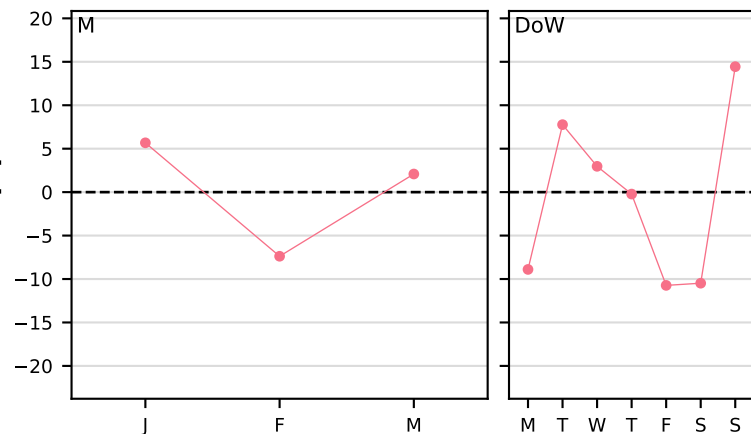
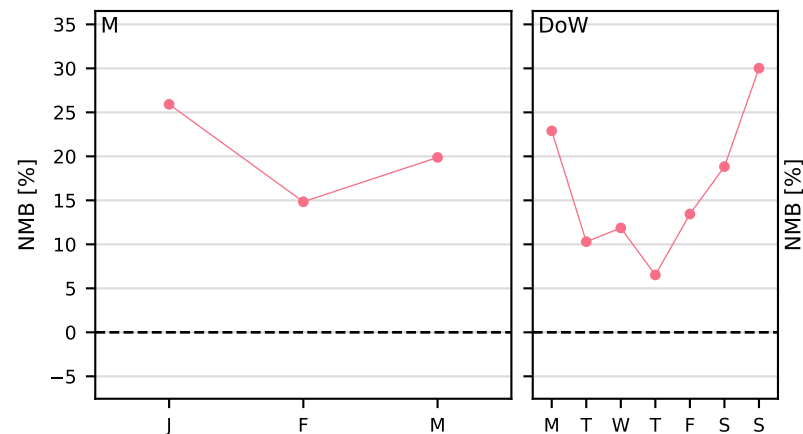
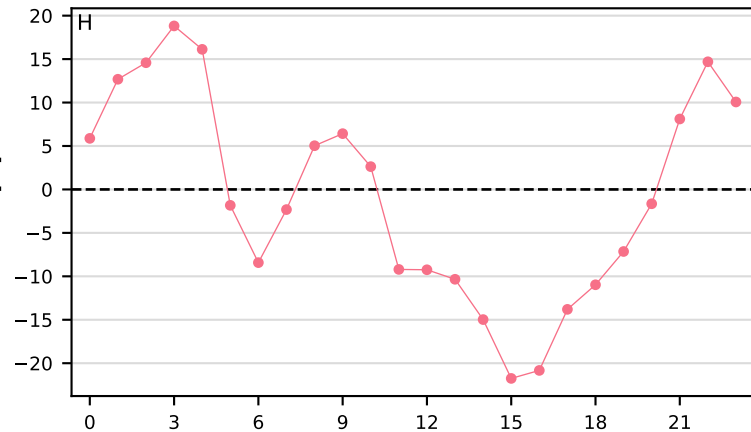




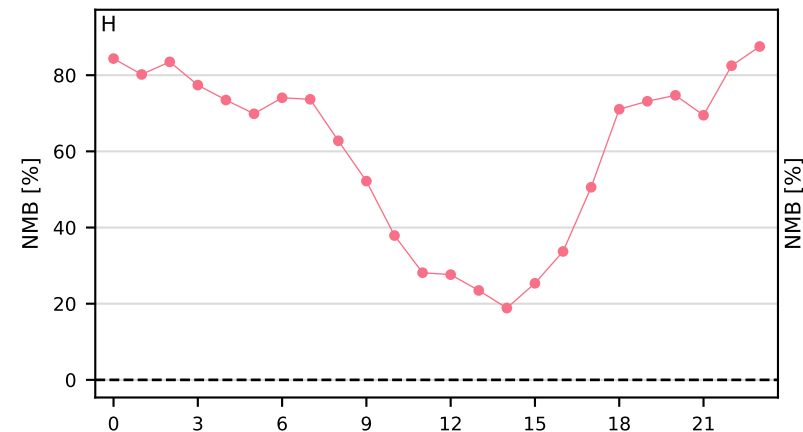
LOCAL·All (62 stations)



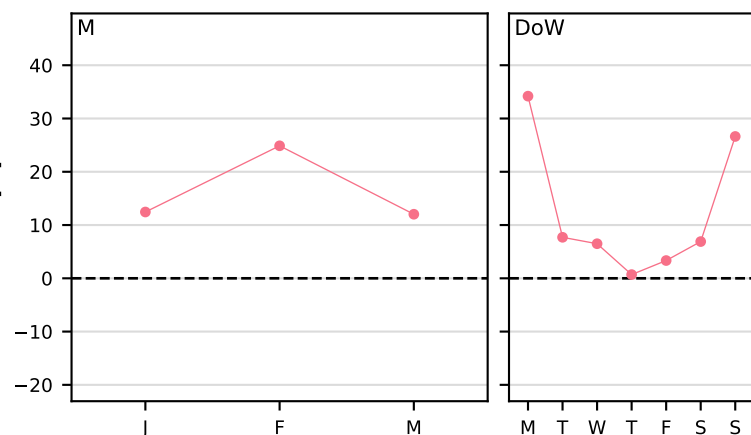
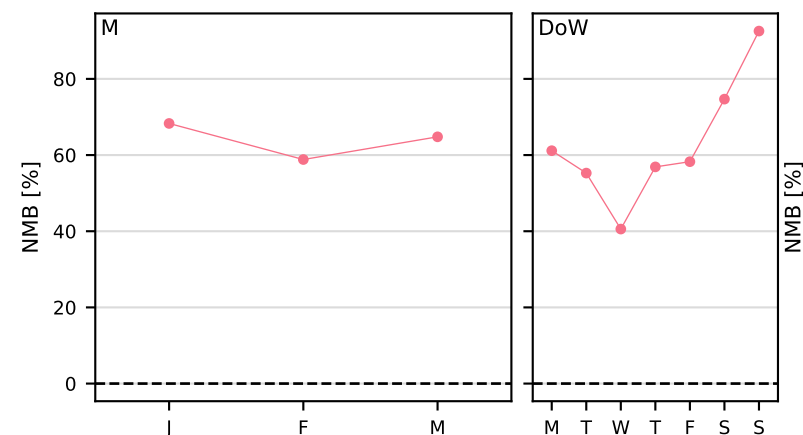
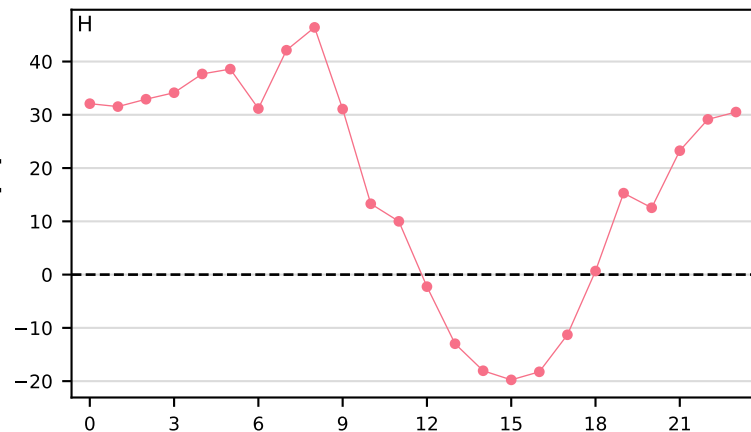
LOCAL·France (1 station)

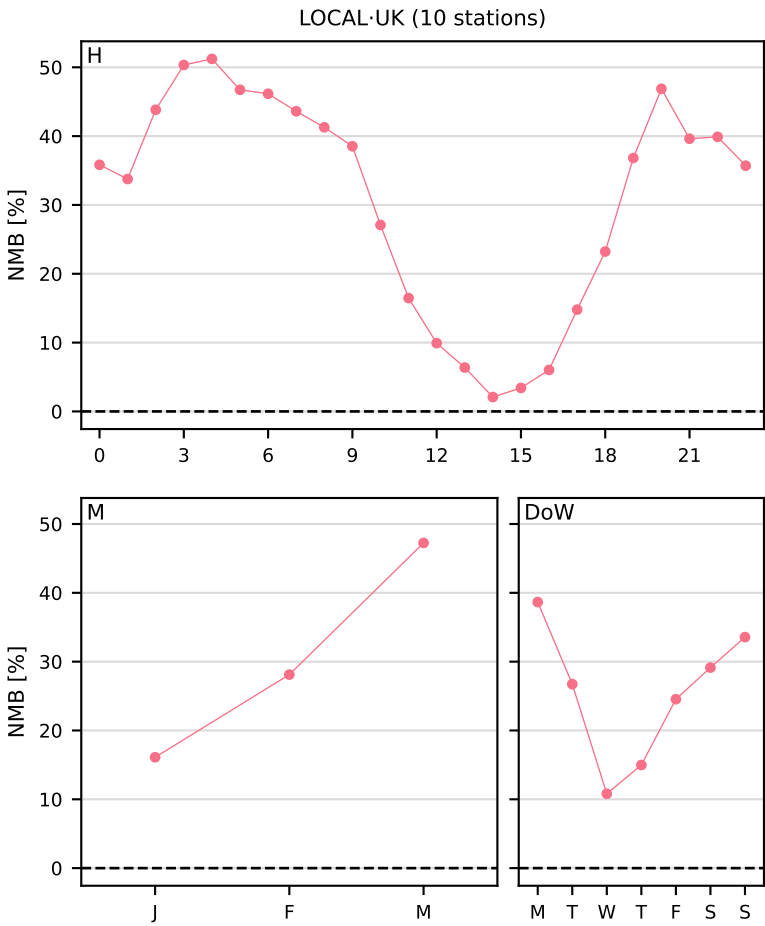


LOCAL·Germany (6 stations)

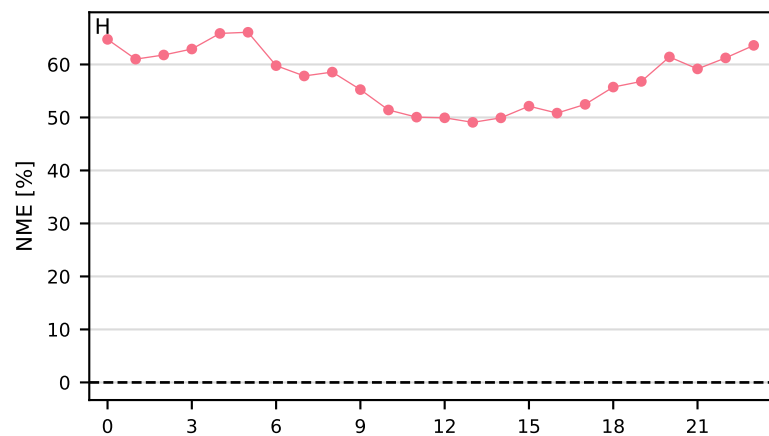


LOCAL·Spain (13 stations)

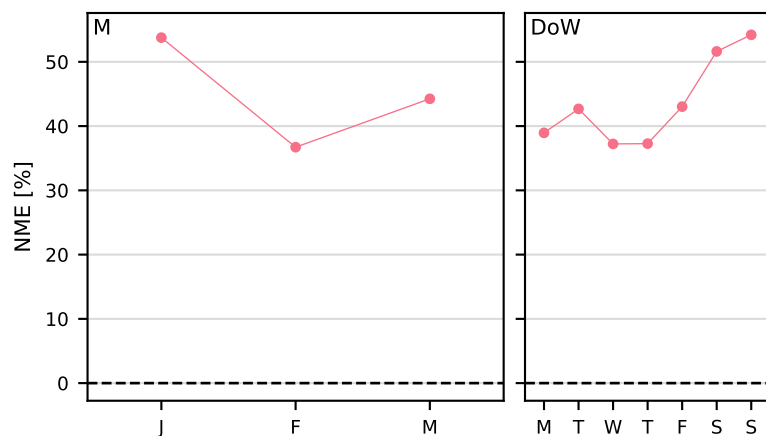
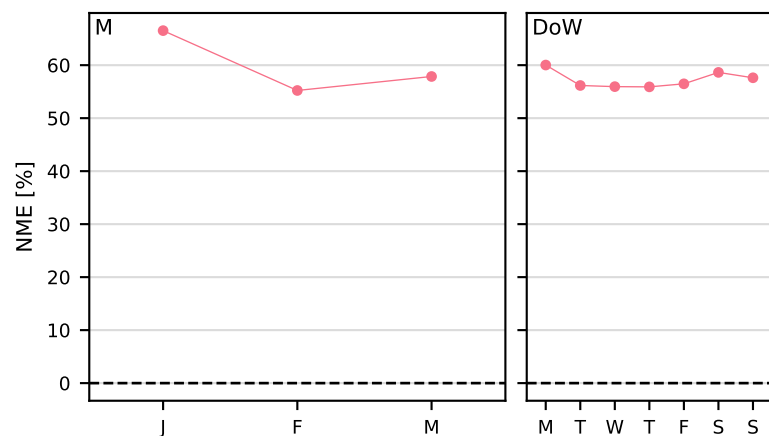
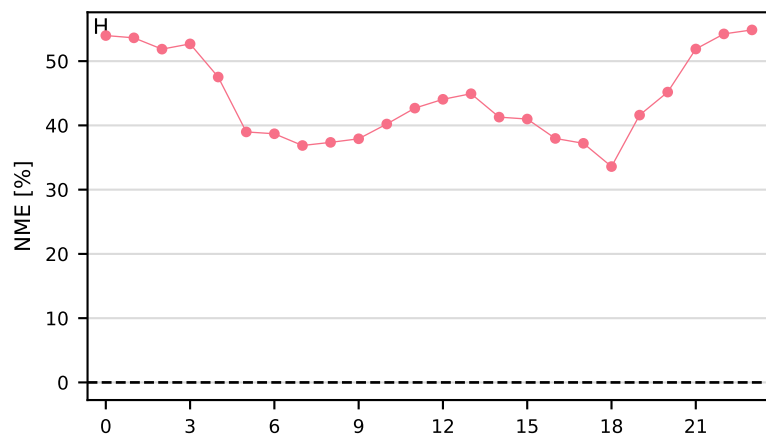




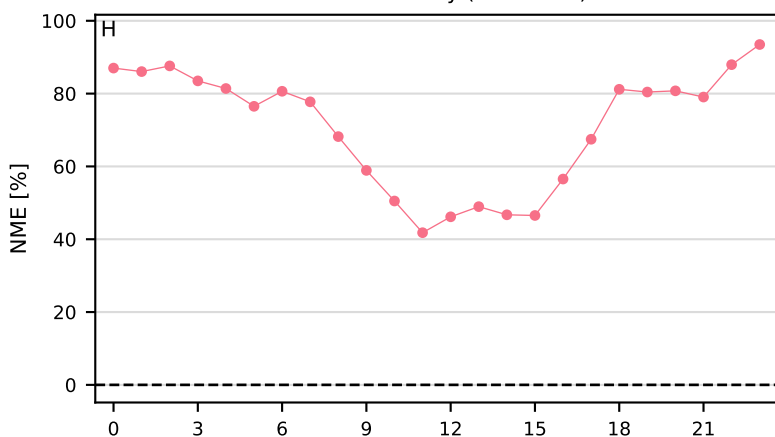
LOCAL·All (62 stations)



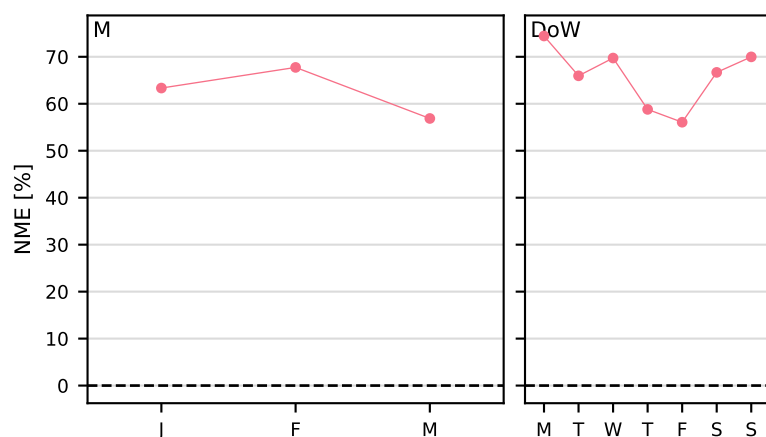
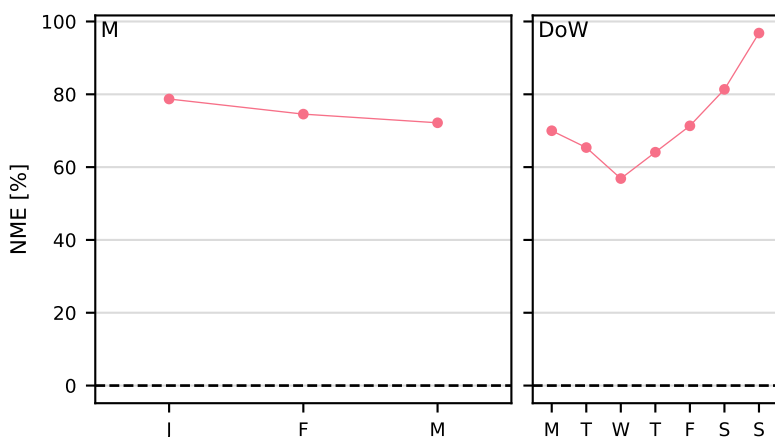
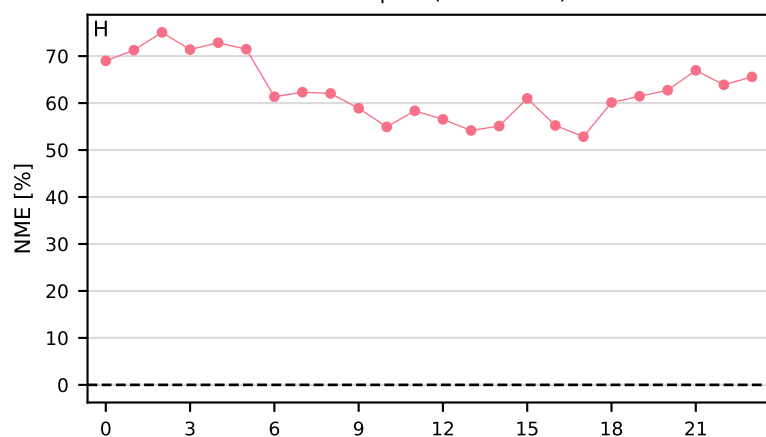
LOCAL·France (1 station)

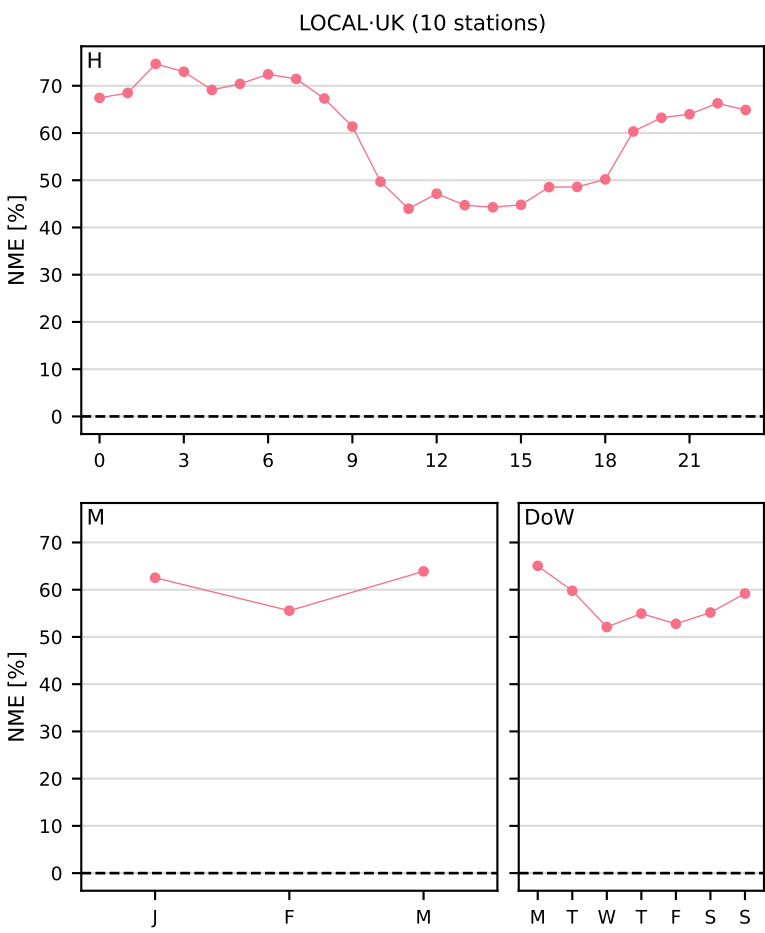


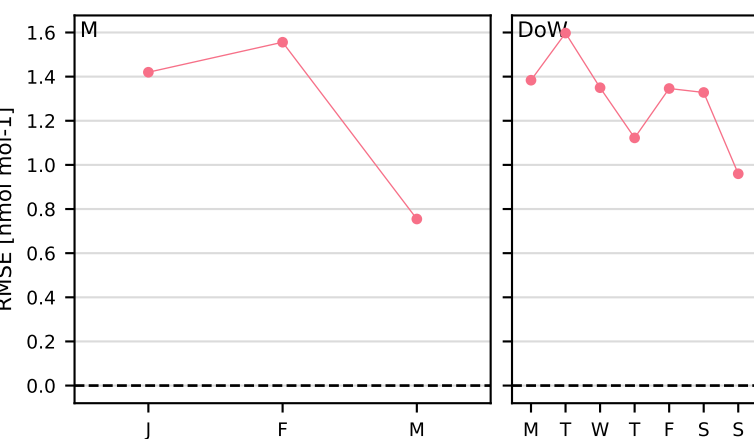
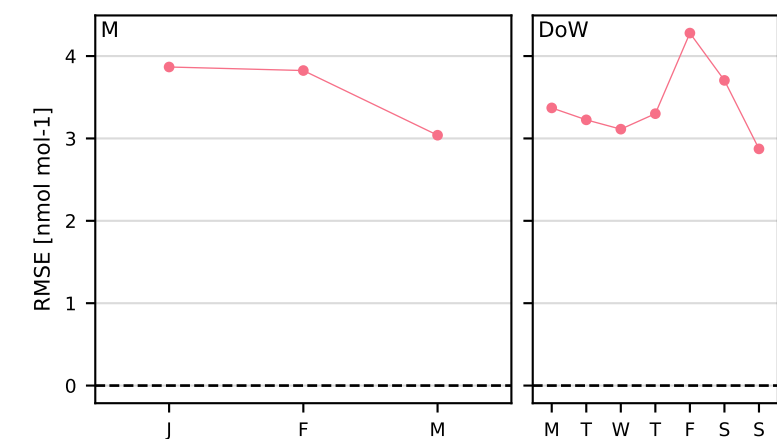
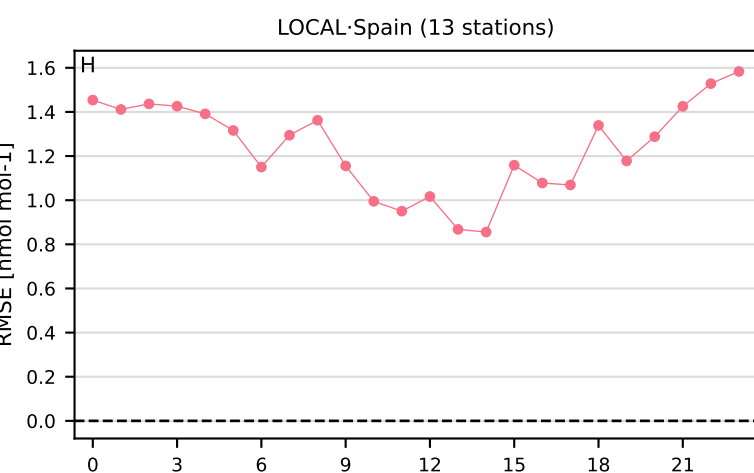
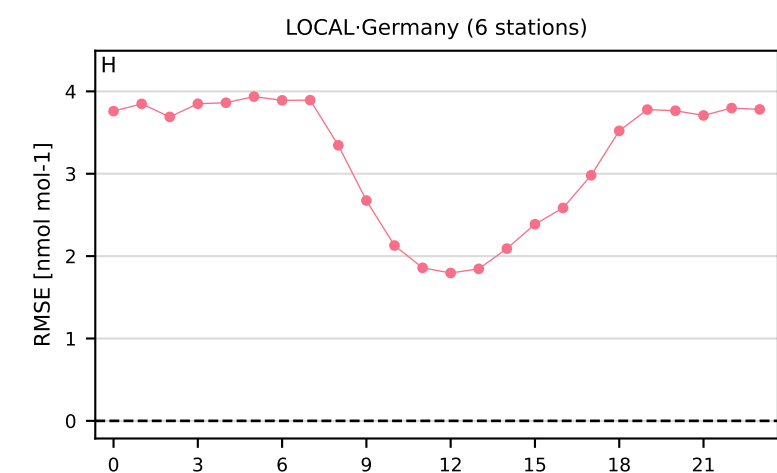
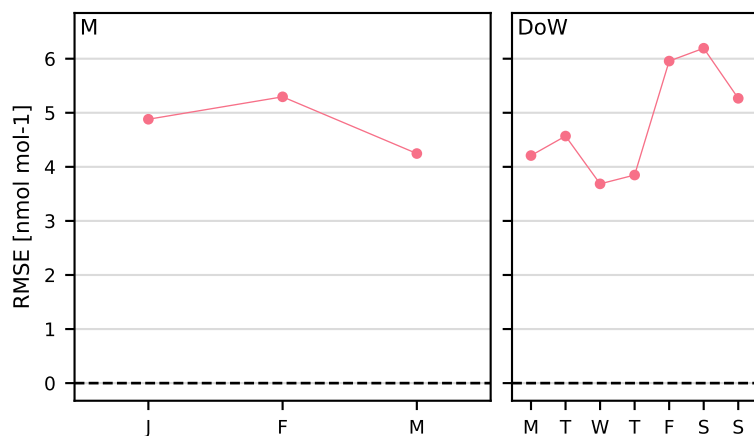
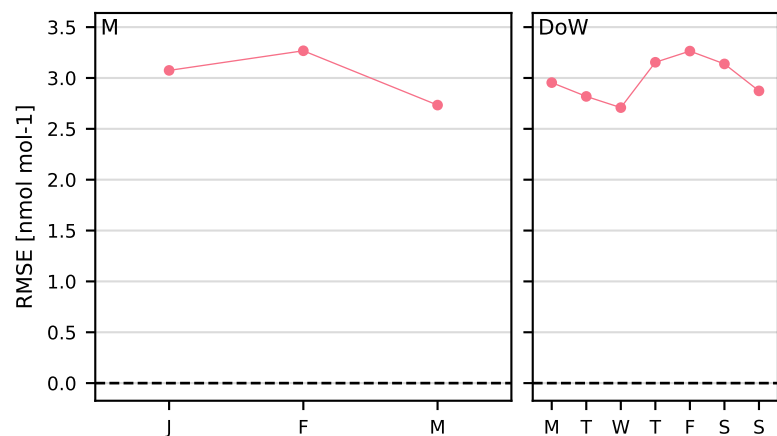
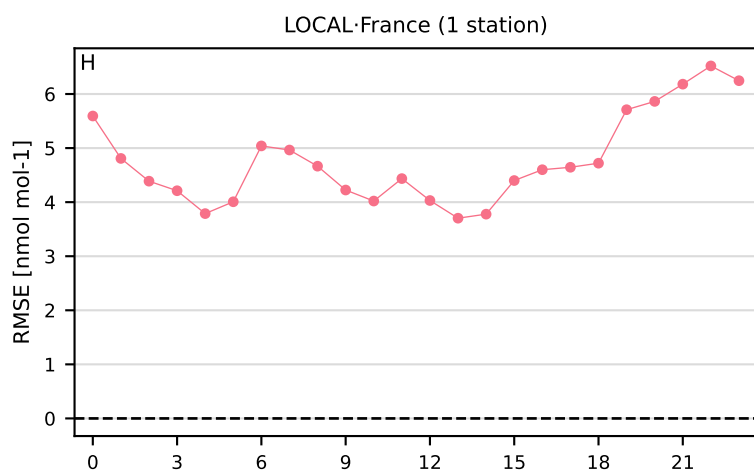
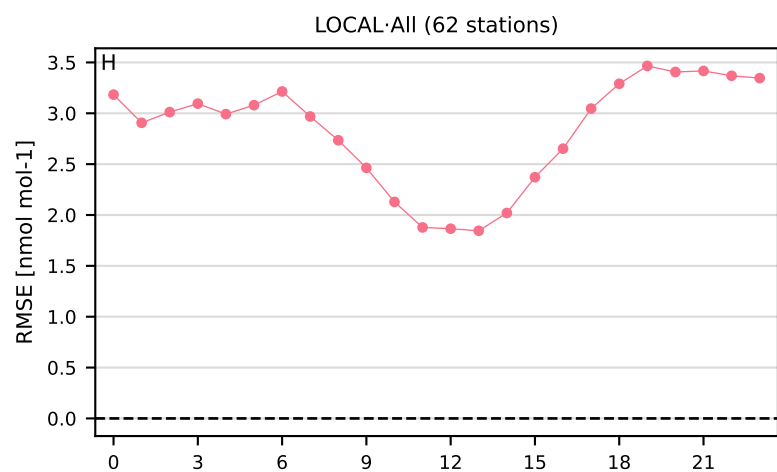
LOCAL·Germany (6 stations)



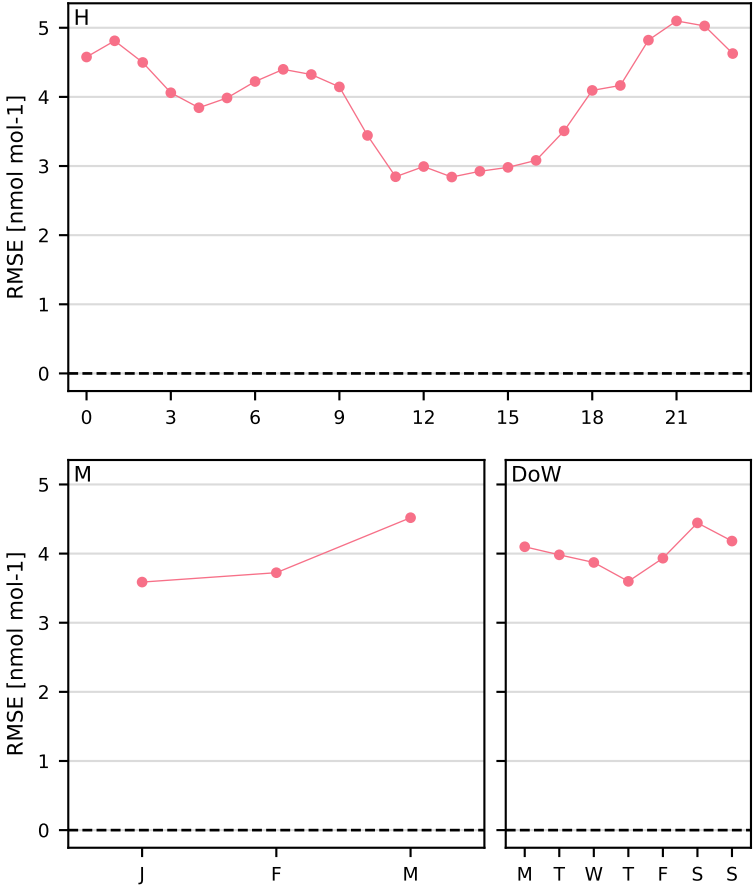
LOCAL·Spain (13 stations)



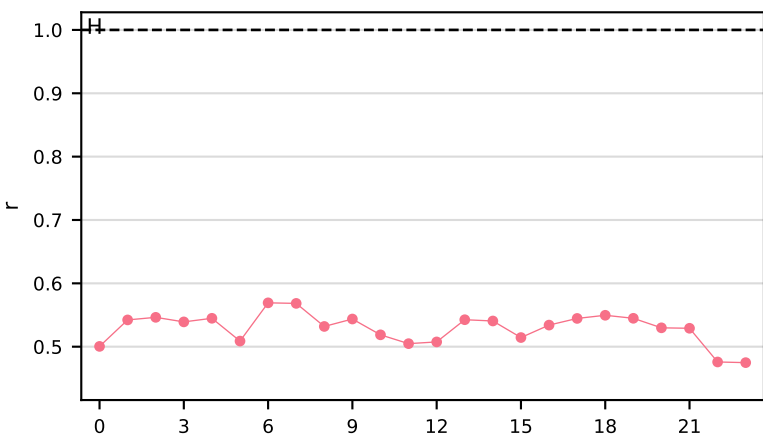




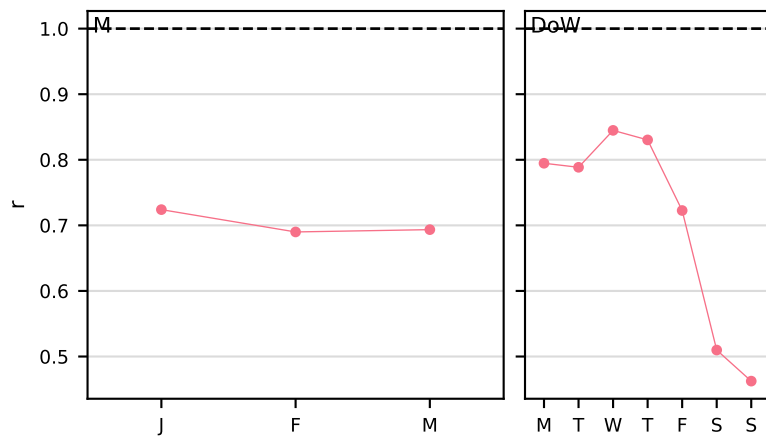
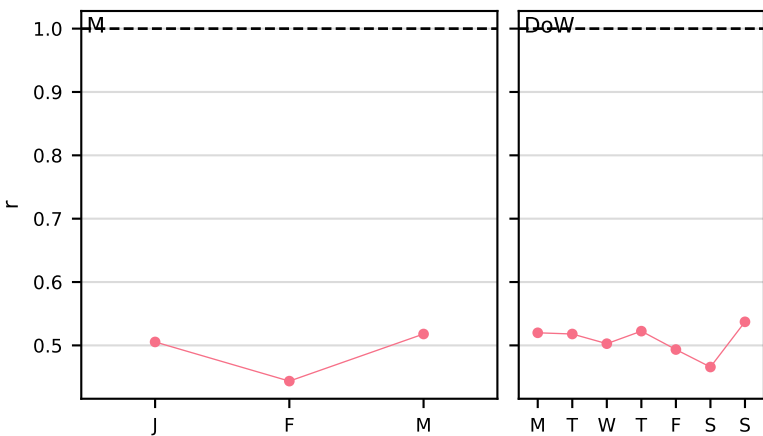
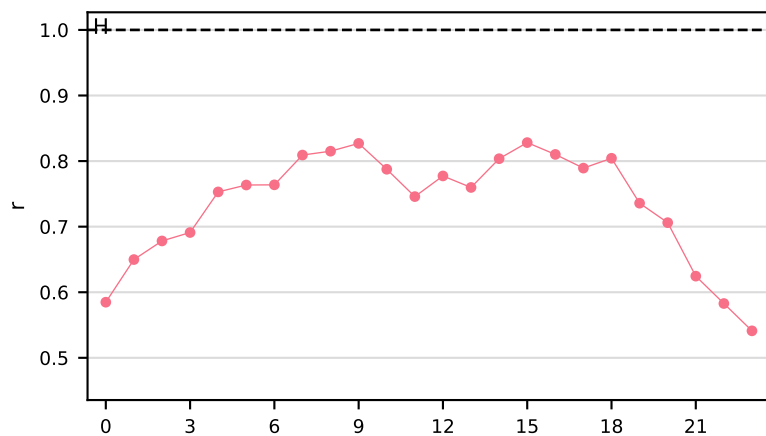
LOCAL·UK (10 stations)



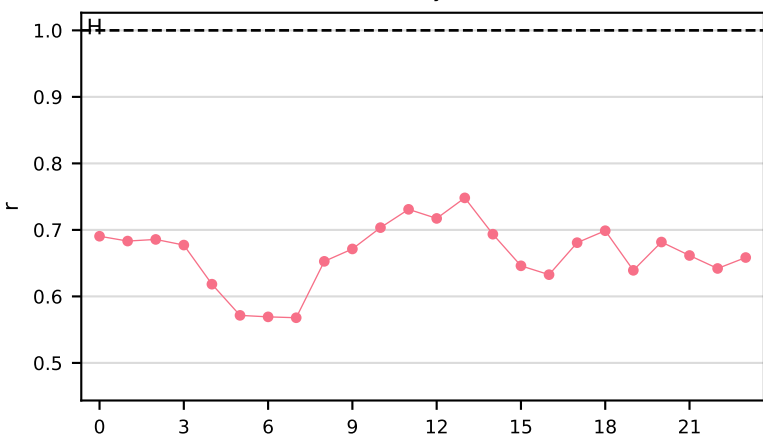
LOCAL·All (62 stations)



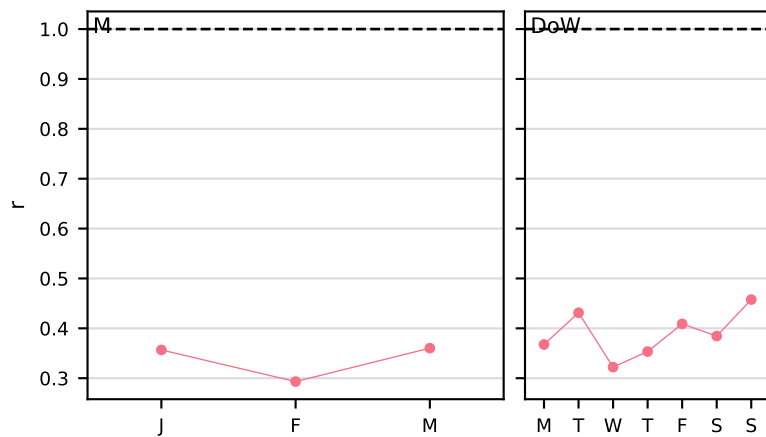
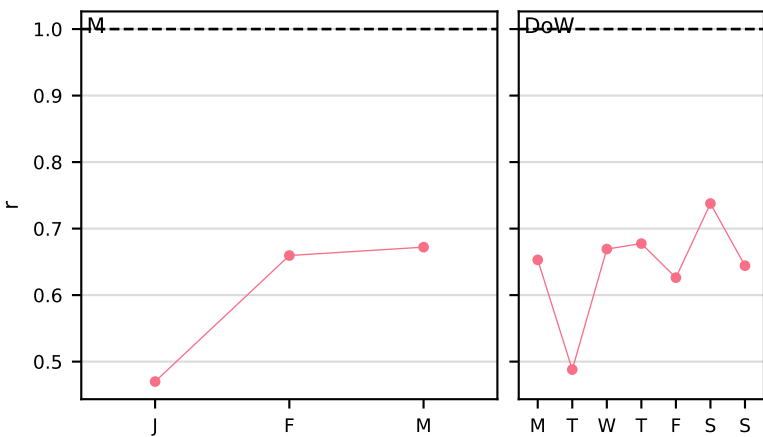
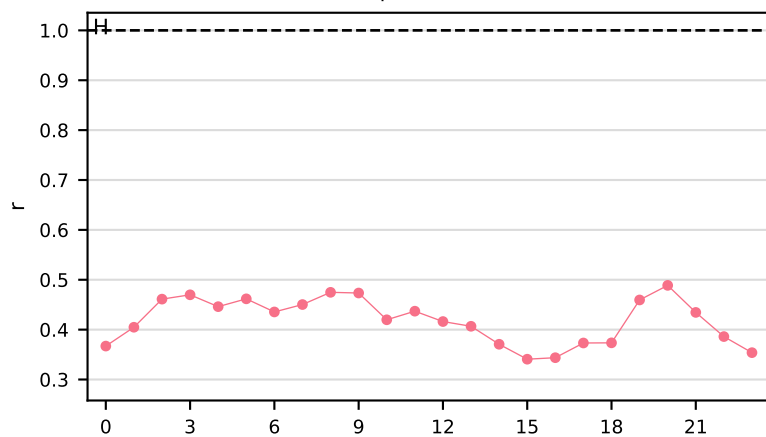
LOCAL·France (1 station)



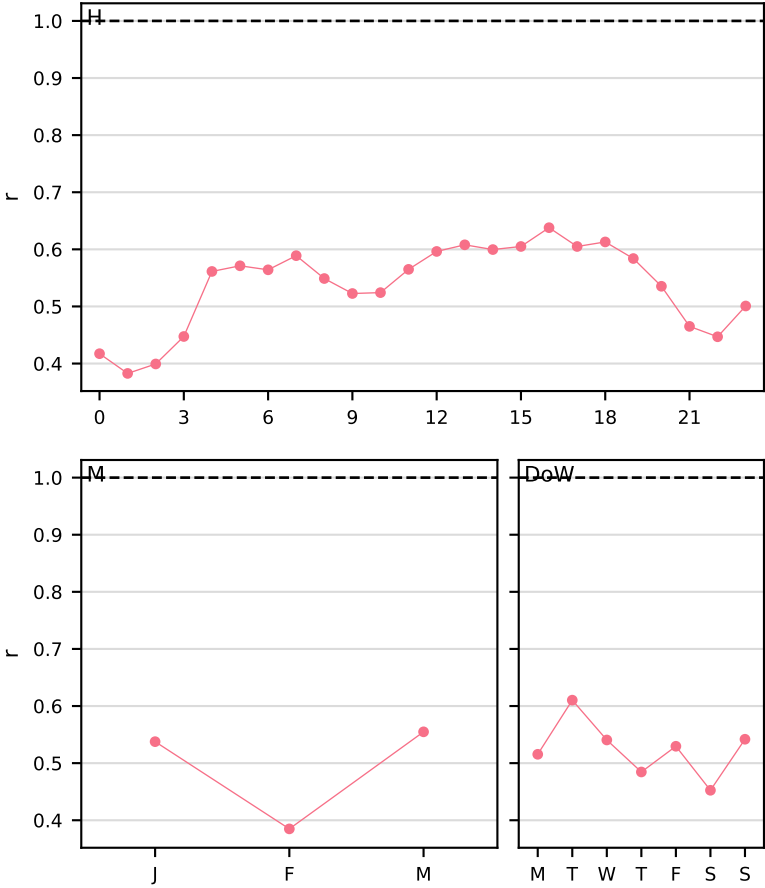
LOCAL·Germany (6 stations)

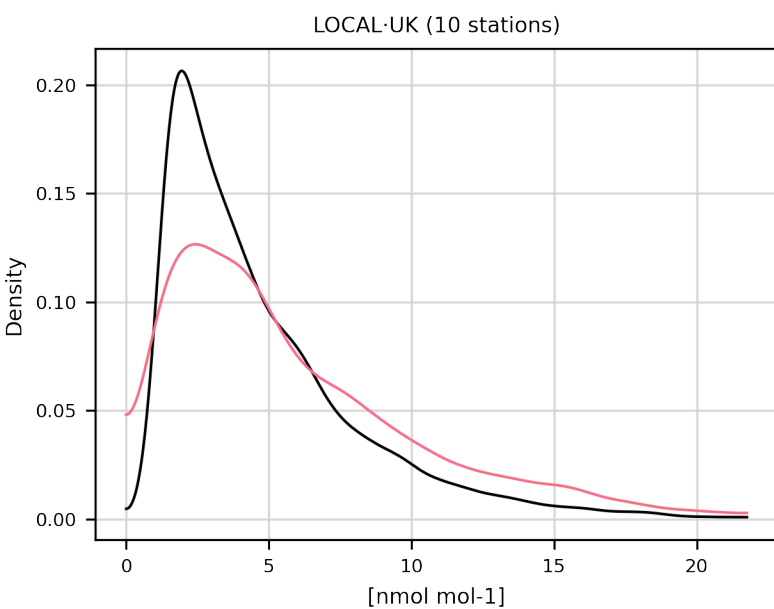
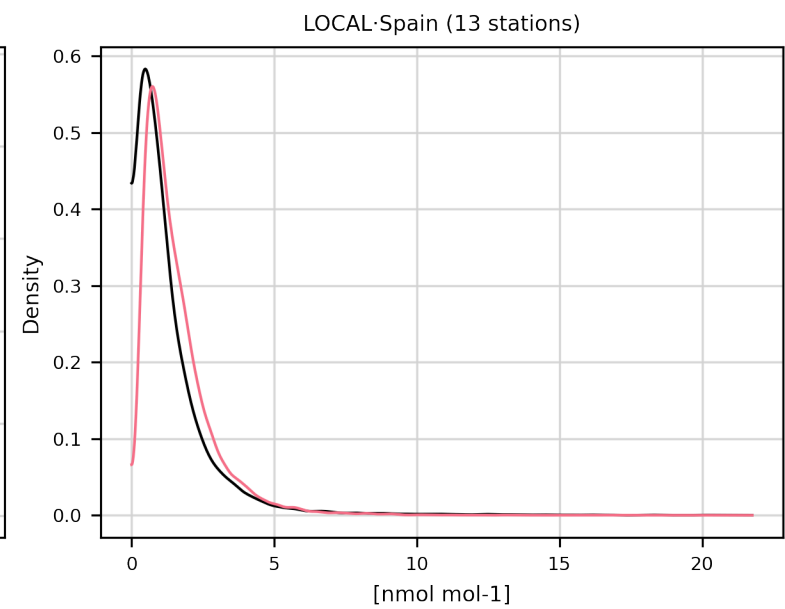
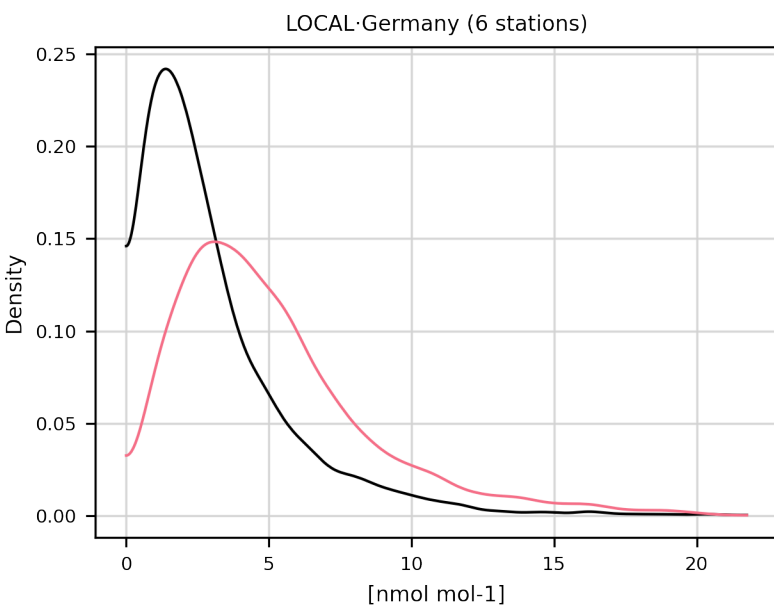
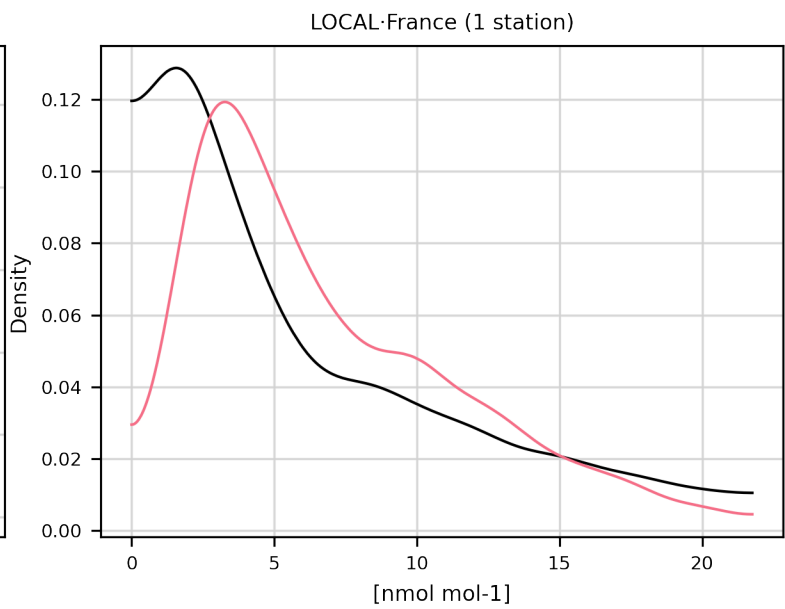
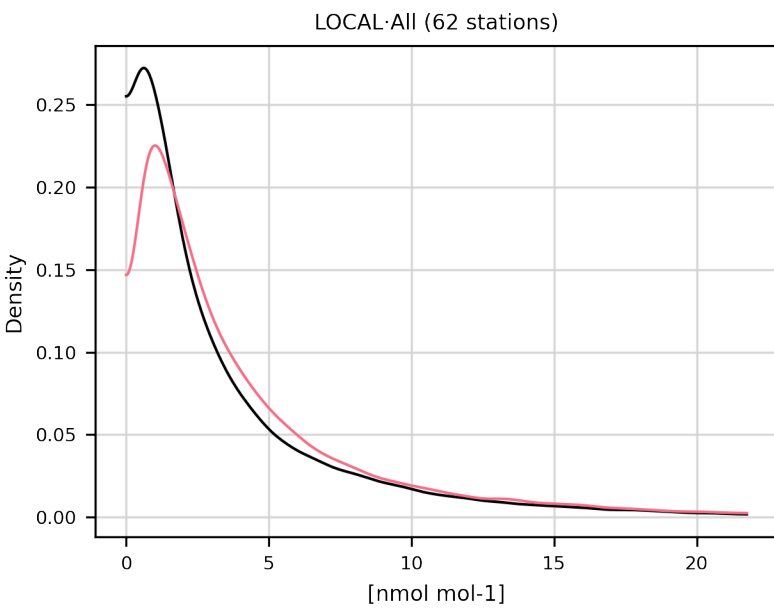


LOCAL·Spain (13 stations)

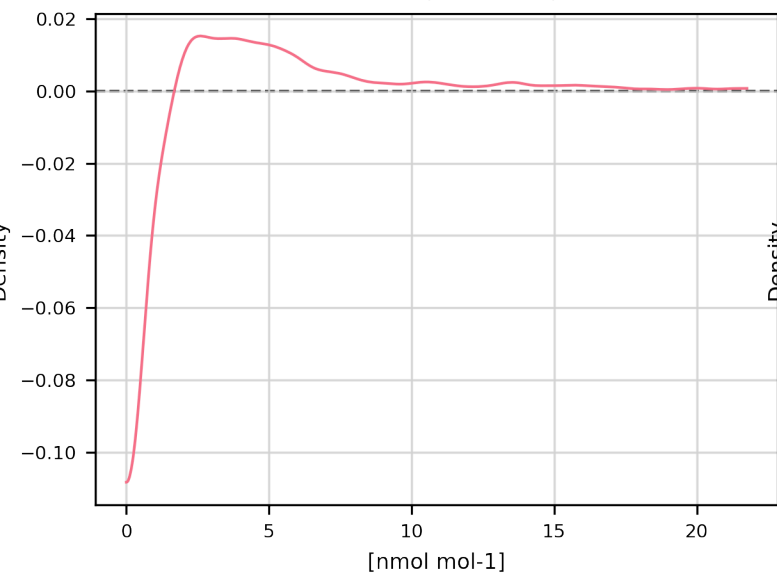


LOCAL·UK (10 stations)

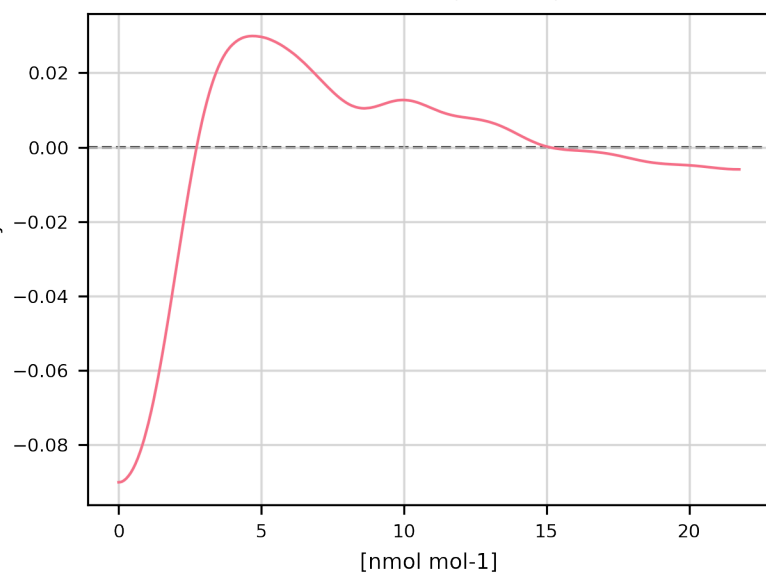




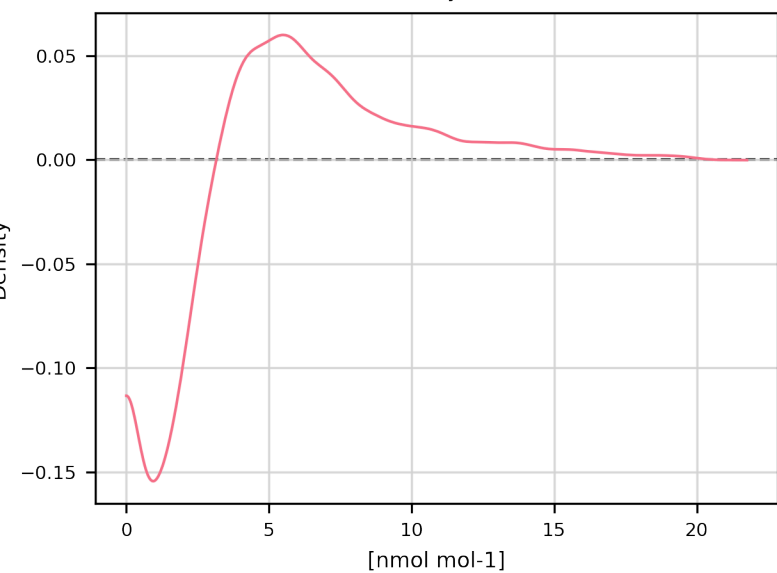
LOCAL·All (62 stations)



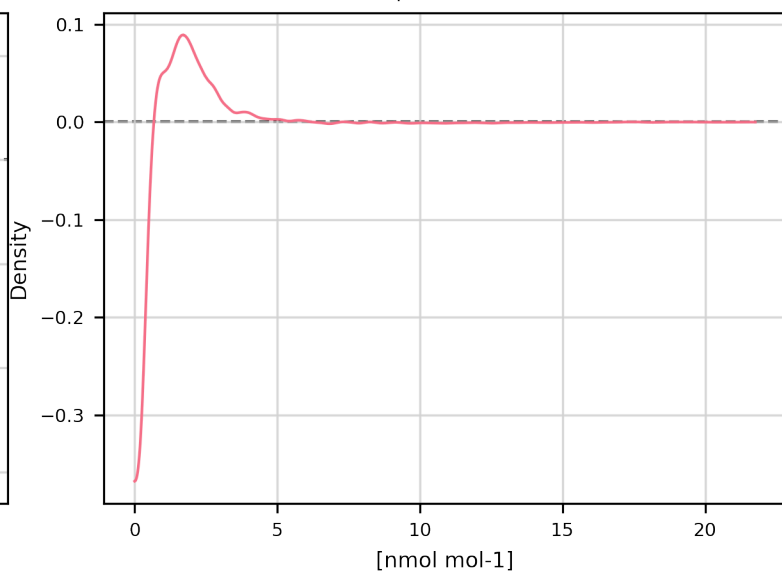
LOCAL·France (1 station)



LOCAL·Germany (6 stations)



LOCAL·Spain (13 stations)



LOCAL·UK (10 stations)

